

CPU Test Platform Manual

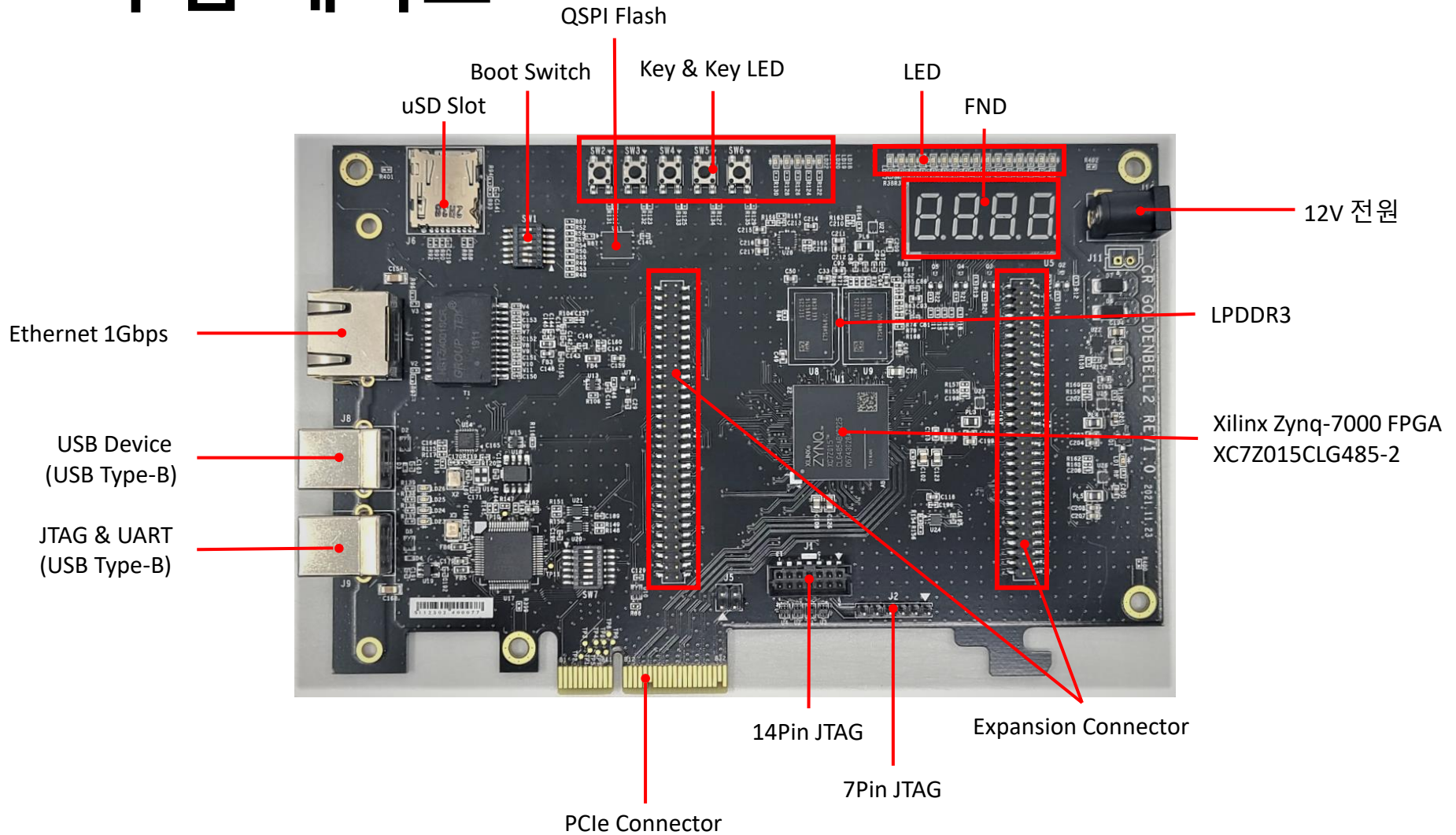
Goldenbell2 Board

CRZ Technology

Goldenbell2 Board(Rev 1.0) Features

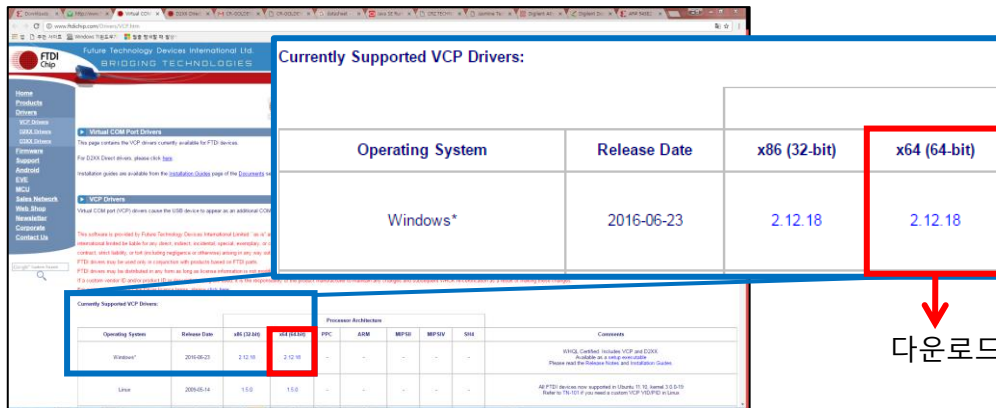
- Xilinx Zynq-7000 FPGA (XC7Z015CLG485-2)
 - Vivado 2022.2
 - 아래 링크에서 Vivado 2022.2 download/install
https://www.xilinx.com/member/forms/download/xef.html?filename=Xilinx_Unified_2022.2_1014_8888_Win64.exe
- Peripheral features
 - ARM Cortex A9 Dual Core 667MHz
 - DRAM : 1GBytes
 - 16MBytes Serial flash memory
 - Power: DC-JACK 12V
 - USB-JTAG and USB-UART connection (FT2232H)
 - 16x LED
 - 1x 7-Segment (4 Digit FND)
 - 5x Key, 5x Key LED
 - 1x USB device
 - 1x 1Gbps Ethernet
 - PCIe Gen2 x4 lane
 - 1x uSD slot

부품 배치도



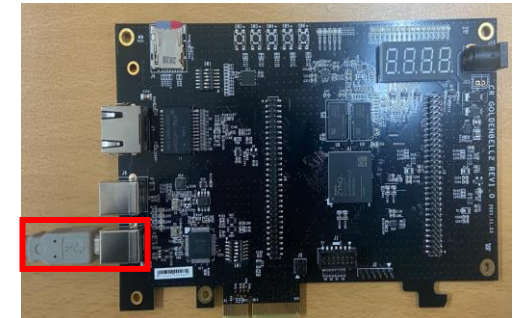
JTAG 연결

- USB 연결은 2가지 목적으로 사용
 - JTAG, UART 연결
 - FT2232H for USB-JTAG and USB-UART connection
- 처음 연결 시 FT2232H Driver를 download해야 함
 - <http://www.ftdichip.com/Drivers/VCP.htm>



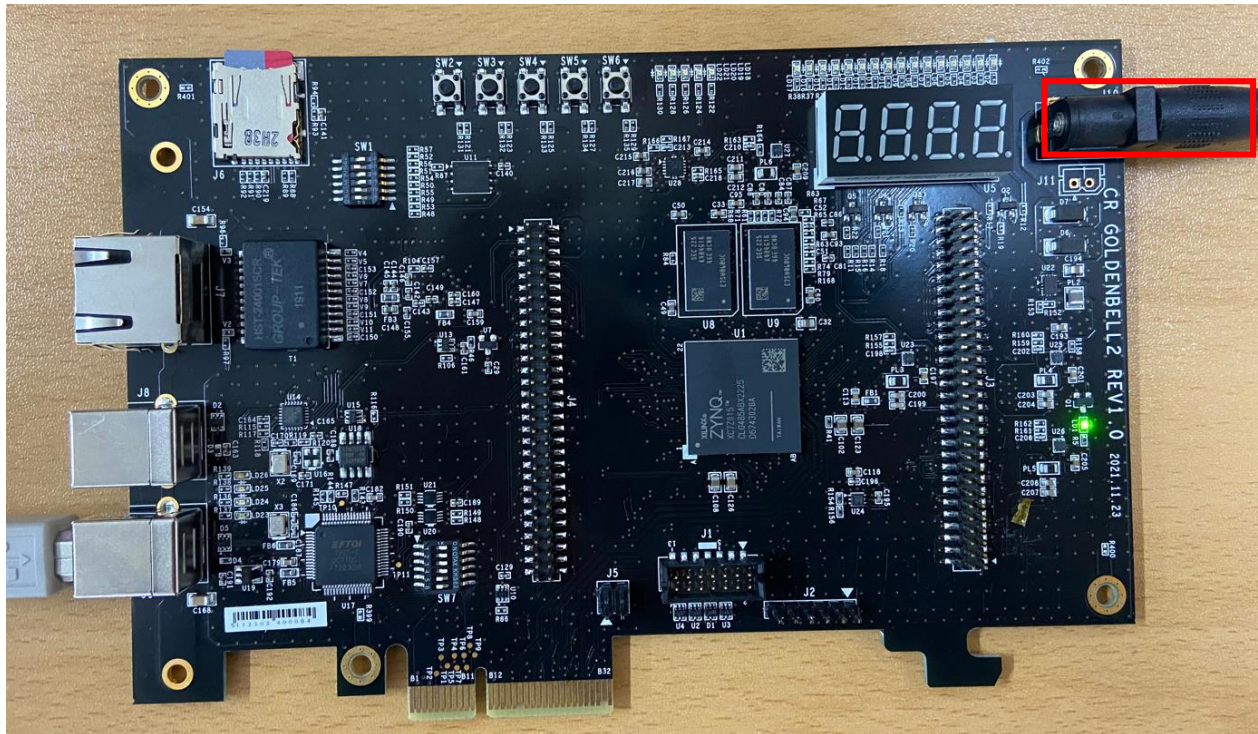
Operating System	Release Date	x86 (32-bit)	x64 (64-bit)
Windows*	2016-06-23	2.12.18	2.12.18

다운로드



보드에 Power 공급하는 방법

- DC 12V를 J10에 공급한다.



전원 Cable 연결

Key Switch를 이용한 reset 구현 (1/2)

- 설명

- Goldenbell2 보드에는 별도의 하드웨어 reset 버튼이 존재하지 않음
- Key Switch(AA19)를 이용하여 program 내부에서 software로 reset을 구현하여 사용

- Pin assignment

- Key Switch(AA19)에 reset 신호를 연결
set_property PACKAGE_PIN AA12 [get_ports {reset}]

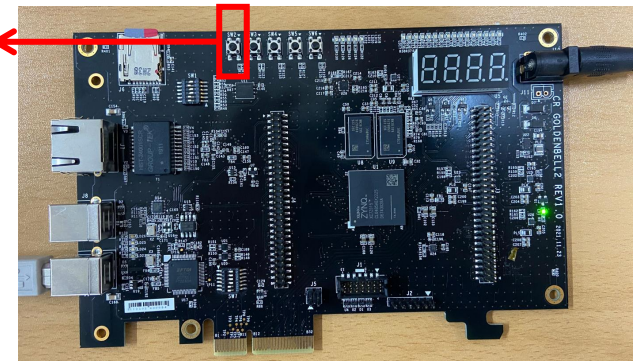
Key Switch를 이용한 reset 구현 (2/2)

- 동작 방법
 - Reset: 1번 key switch를 press / release

- Source code

```
always @(posedge clk_50MHz or posedge reset)
begin
if (reset)
begin
//이 자리에 reset시 동작 code 입력
end
else
begin
//이 자리에 reset이 아닐 시 동작 code 입력
end
end
```

Reset switch



Peripheral Validation

LED 구동 (1/2)

- 방법

- LED에 1을 인가하면 on상태, 0을 인가하면 off상태가 됨

- Example 동작 방법

- Reset시, 모든 LED off
 - 모든 LED에 0을 인가
 - Reset이 아닐 시, led[7:4] & led[15:12] on
 - on 시킬 led (led[7:4] & led[15:12])에 1을 인가

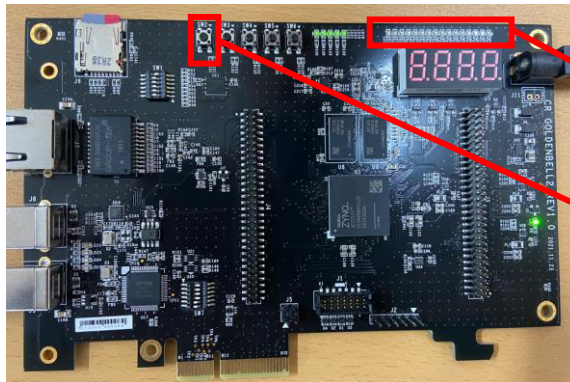
- Source code

- led_test.v

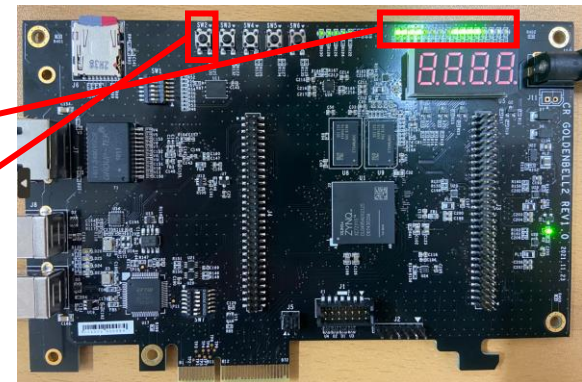
```
begin
if (reset)
begin
led <= 16'b0;
end
else
begin
led <= 16'b1111_0000_1111_0000;
end
end
```

LED 구동 (2/2)

- Result
 - Reset 시, 모든 LED off
 - Reset 아닐 시, led[7:4] & led[15:12] on



Reset 상태



Non-reset 상태

led[15]-led[0]

Reset 스위치

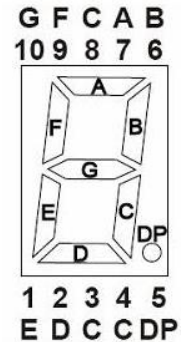
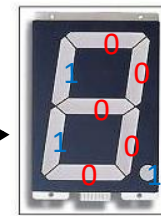
FND 구동 (1/3)

- FND 제어 방법

- FND는 0을 인가하면 on상태, 1을 인가하면 off상태

- 예) digit[3], digit[1]에 동시에 3을 출력하는 경우

```
digit<=4'b0101; fnd<=8'b00001101;
```



- 동작 방법

- Reset시, 모든 digit에 0 출력

- Reset이 아닐 시, digit[3]=3 digit[2]=2 digit[1]=1 digit[0]=0 출력

- 4개의 FND에 서로 다른 숫자 출력

- digit_sel은 제어할 digit을 선택해주는 register

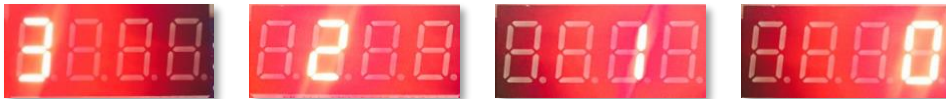
- Counter가 증가함에 따라서 digit_sel 값을 0,1,2,3 바뀌가며 각 digit_sel 값에 한 digit의 값을 출력 (시간 차를 두고 FND가 켜지는 효과 이용)

FND 구동 (2/3)

- Source code

- fnd_test.v

```
begin
  digit_sel<=digit_sel+1'b1;
  if(digit_sel==2'b11)
    digit_sel<=0;
    case(digit_sel)
      0: begin digit<=4'b1110; number<=0; end
      1: begin digit<=4'b1101; number<=1; end
      2: begin digit<=4'b1011; number<=2; end
      3: begin digit<=4'b0111; number<=3; end
    endcase
  end
end
```



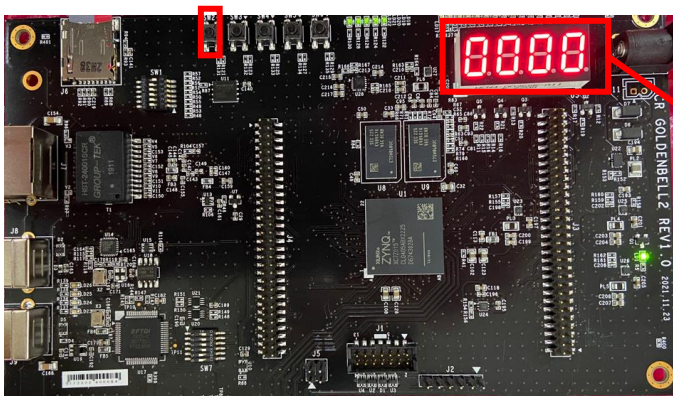
의 빠른 출력으로 →  이 동시에 출력되는 것처럼 보임

FND	a	b	c	d	e	f	g	.	2진수
0	0	0	0	0	0	0	1	1	8'b00000011
1	1	0	0	1	1	1	1	1	8'b10011111
2	0	0	1	0	0	1	0	1	8'b00100101
3	0	0	0	0	1	1	0	1	8'b00001101
4	1	0	0	1	1	0	0	1	8'b10011001
5	0	1	0	0	1	0	0	1	8'b01001001
6	0	1	0	0	0	0	0	1	8'b01000001
7	0	0	0	1	1	0	1	1	8'b00011011
8	0	0	0	0	0	0	0	1	8'b00000001
9	0	0	0	1	1	0	0	1	8'b00011001
A	1	0	0	0	1	0	0	1	8'b10001001
B	1	0	0	0	0	0	1	1	8'b10000011
C	0	1	1	0	0	0	1	1	8'b01100011
D	1	0	0	0	0	1	0	1	8'b10000101
E	0	1	1	0	0	0	0	1	8'b01100001
F	0	1	1	1	0	0	0	1	8'b01110001

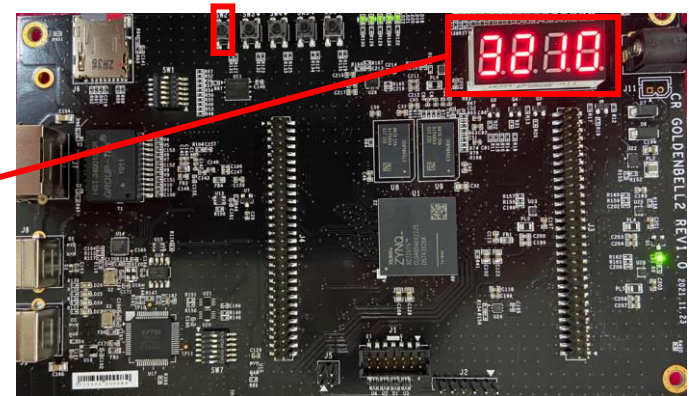
FND 구동 (3/3)

- Result

- Reset시, 모든 digit에 0 출력
- Reset이 아닐 시, digit[3]=3 digit[2]=2 digit[1]=1 digit[0]=0 출력



Reset 상태



Non-reset 상태

digit[3]-digit[0]

Key & Key LED 구동 (1/2)

- Key 입력 제어
 - 눌렀을 때 1, Normal 0 인 상태
- Key LED 출력 제어
 - LED에 0을 인가하면 LED on, 1을 인가하면 LED off
- Example 동작 방법
 - Reset시, 모든 keyLed<=5'b11111 을 해주어 off 상태를 만들어 줌
 - Reset이 아닐 시, 각 key를 누르면 key의 상태가 1이므로 keyLed에 0을 인가해 주어 켜주기 위해 ~key를 keyLed에 대입

Key & Key LED 구동 (2/2)

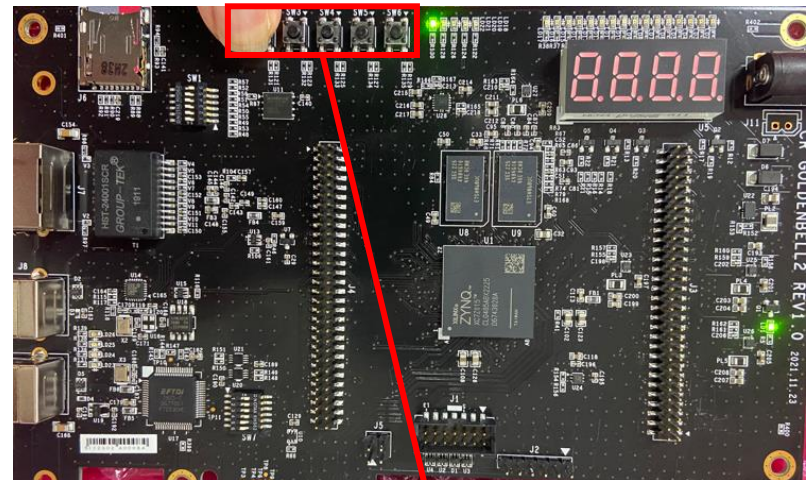
- Source code

- Key_test.v

```
begin
  if (reset)
    begin
      keyLed <=5'b11111;
    end
  else
    begin
      keyLed <=~key;
    end
  end
end
```

- Result

- Reset시, 모든 LED off
 - Reset이 아닐 시, Key를 push했을 때 각각의 key에 해당하는 key LED on



key[0] key[1] key[2] key[3] key[4]

Vivado Project 생성 방법

LED project를 생성하고 이를 FPGA에 프로그램

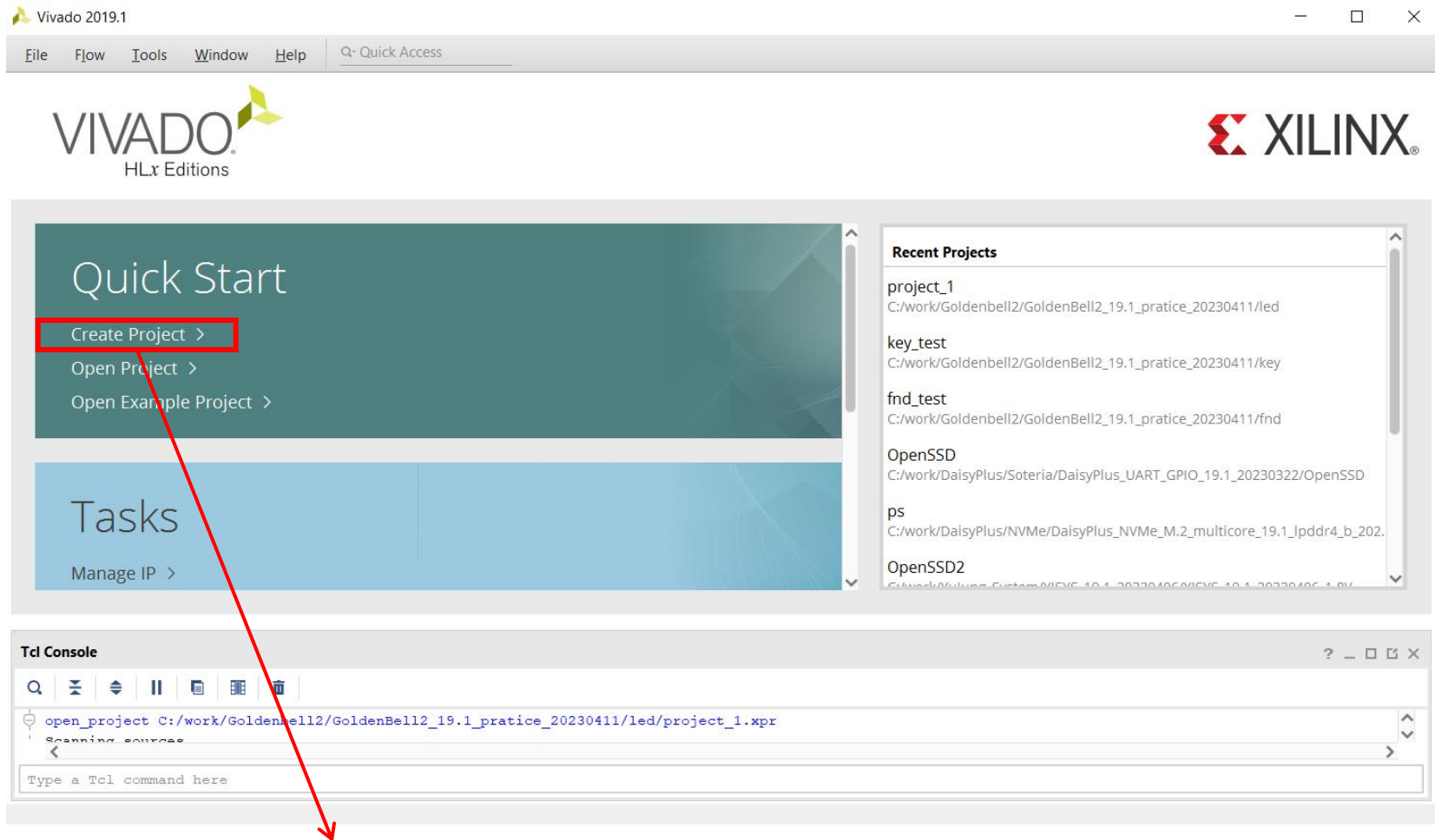
Step 1. Creating a new project

Step 2. Creating constraint file

Step 3. Creating bit file

Step 4. Downloading a bit file to FPGA

Step 1. Creating new project



① Create Project 클릭

② 파일이름 지정

New Project

Project Name
Enter a name for your project and specify a directory where the project data files will be stored.

Project name: led_test

Project location: C:/work/Goldenbell2/GoldenBell2_19.1_pratice_20230411/led

☐ Create project subdirectory

Project will be created at: C:/work/Goldenbell2/GoldenBell2_19.1_pratice_20230411/led

③ 파일 경로지정

④ Next 클릭

< Back Next > Finish Cancel

New Project
 ✕

Default Part

Choose a default Xilinx part or board for your project.

Parts | Boards

[Reset All Filters](#)

Category: All

Package: All

Temperature: All

Family: All

Speed: All

Static power: All

Search: xc7z015
 ✕

(4 matches)

Part	I/O Pin Count	Available IOBs	LUT Elements	FlipFlops	Block RAMs	Ultra RAMs	DSPs	Gb T
xc7z015clg485-3	485	150	46200	92400	95	0	160	4
xc7z015clg485-2	485	150	46200	92400	95	0	160	4
xc7z015clg485-1	485	150	46200	92400	95	0	160	4
xc7z015iclg485-1L	485	150	46200	92400	95	0	160	4

⑤ Goldenbell2보드의 FPGA 설정
Zynq-7000, XC7Z015CLG485-2

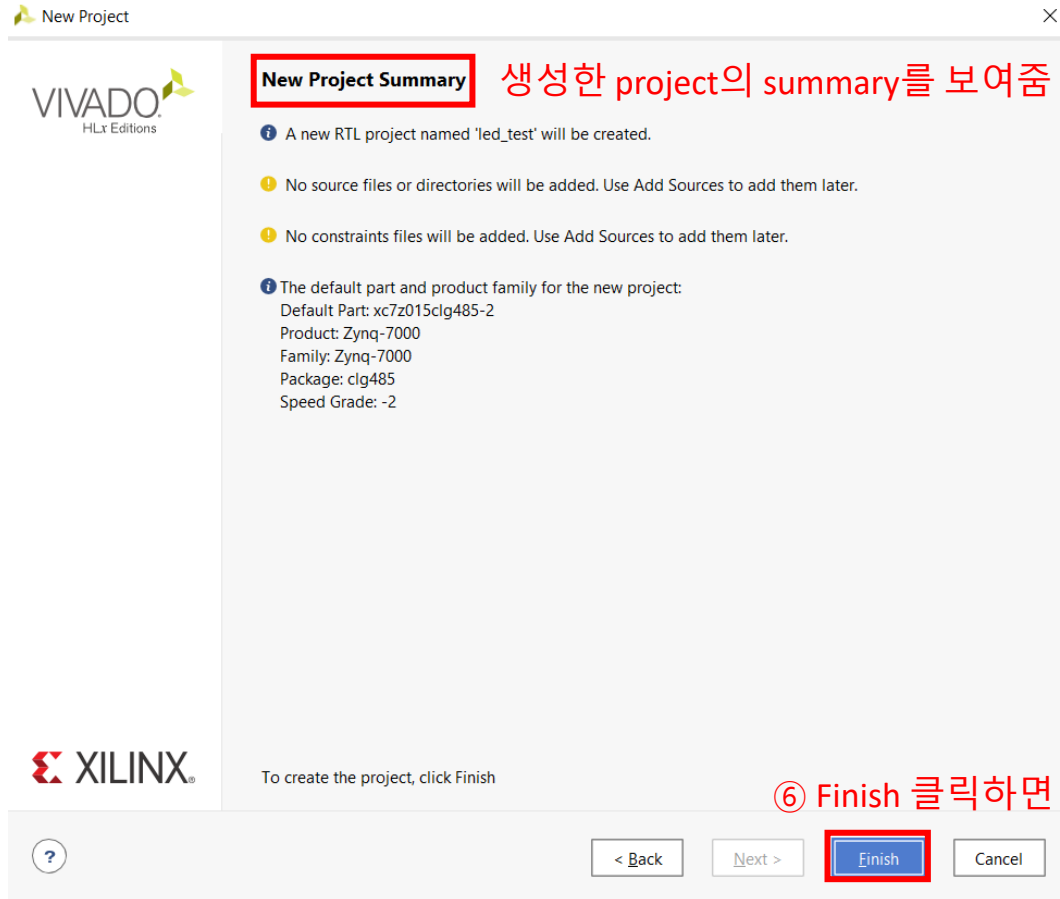
?

< Back

Next >

Finish

Cancel



led_test - [C:/work/Goldenbell2/GoldenBell2_19.1_pratice_20230411/led/led_test.xpr] - Vivado 2019.1

File Edit Flow Tools Reports Window Layout View Help Quick Access

Flow Navigator PROJECT MANAGER - led_test

PROJECT MANAGER

- Settings
 - Add Sources
 - Language Templates
 - IP Catalog
- IP INTEGRATOR
 - Create Block Design
 - Open Block Design
 - Generate Block Design
- SIMULATION
 - Run Simulation
- RTL ANALYSIS
 - Open Elaborated Design
- SYNTHESIS
 - Run Synthesis
 - Open Synthesized Design
- IMPLEMENTATION
 - Run Implementation

Sources

Design Sources

- Constr
- Simul

Hierarchy

Properties

Properties...

Hierarchy Update

Refresh Hierarchy

IP Hierarchy

Edit Constraints Sets...

Edit Simulation Sets...

+ Add Sources... Alt+A

Select an object to see properties

Project Summary

Overview | Dashboard

Settings Edit

Project name: led_test

Project location: C:/work/Goldenbell2/GoldenBell2_19.1_pratice_20230411/led

Product family: Zynq-7000

Project part: xc7z015clg485-2

Top module name: Not defined

Target language: Verilog

Simulator language: Mixed

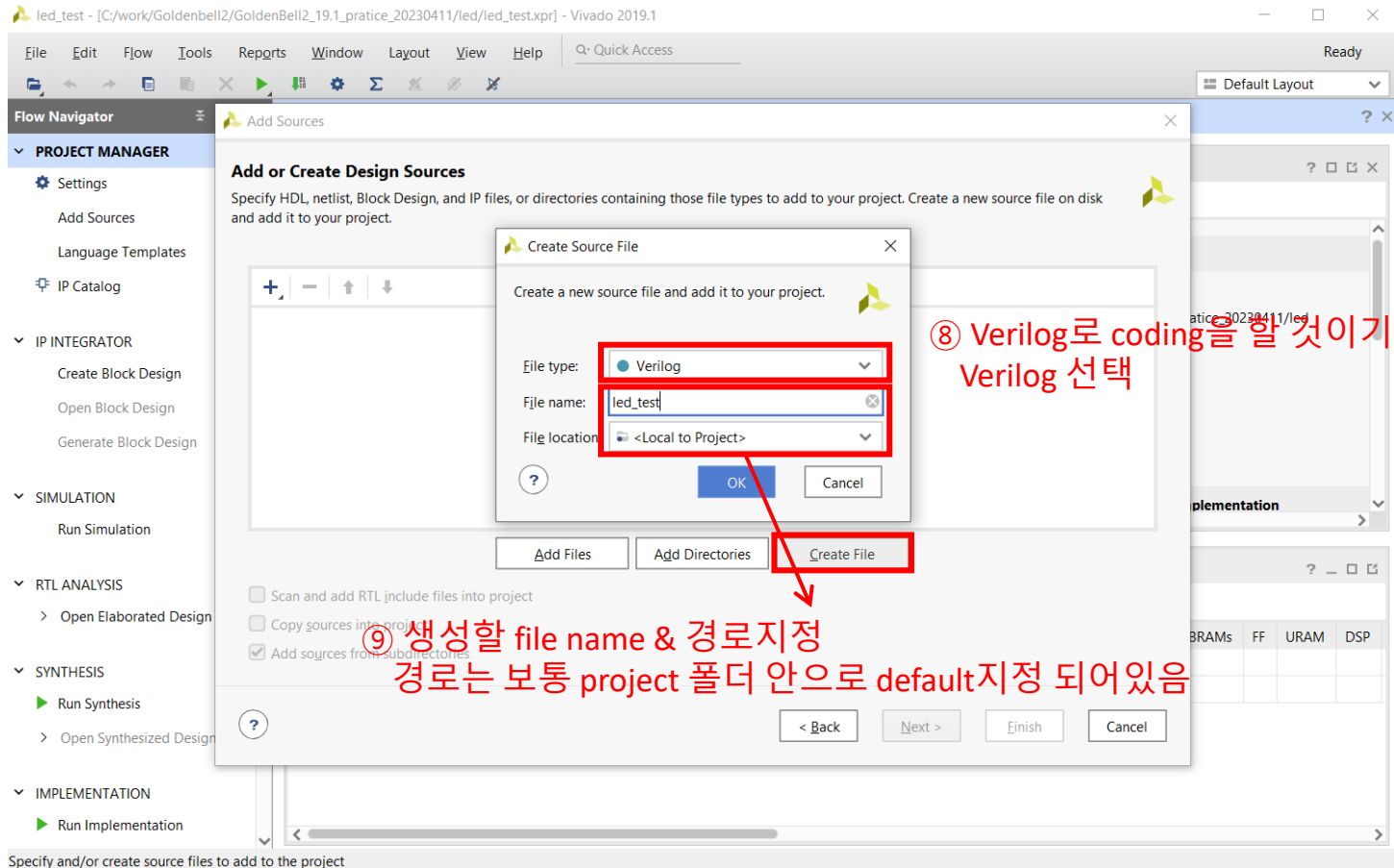
Synthesis **Implementation**

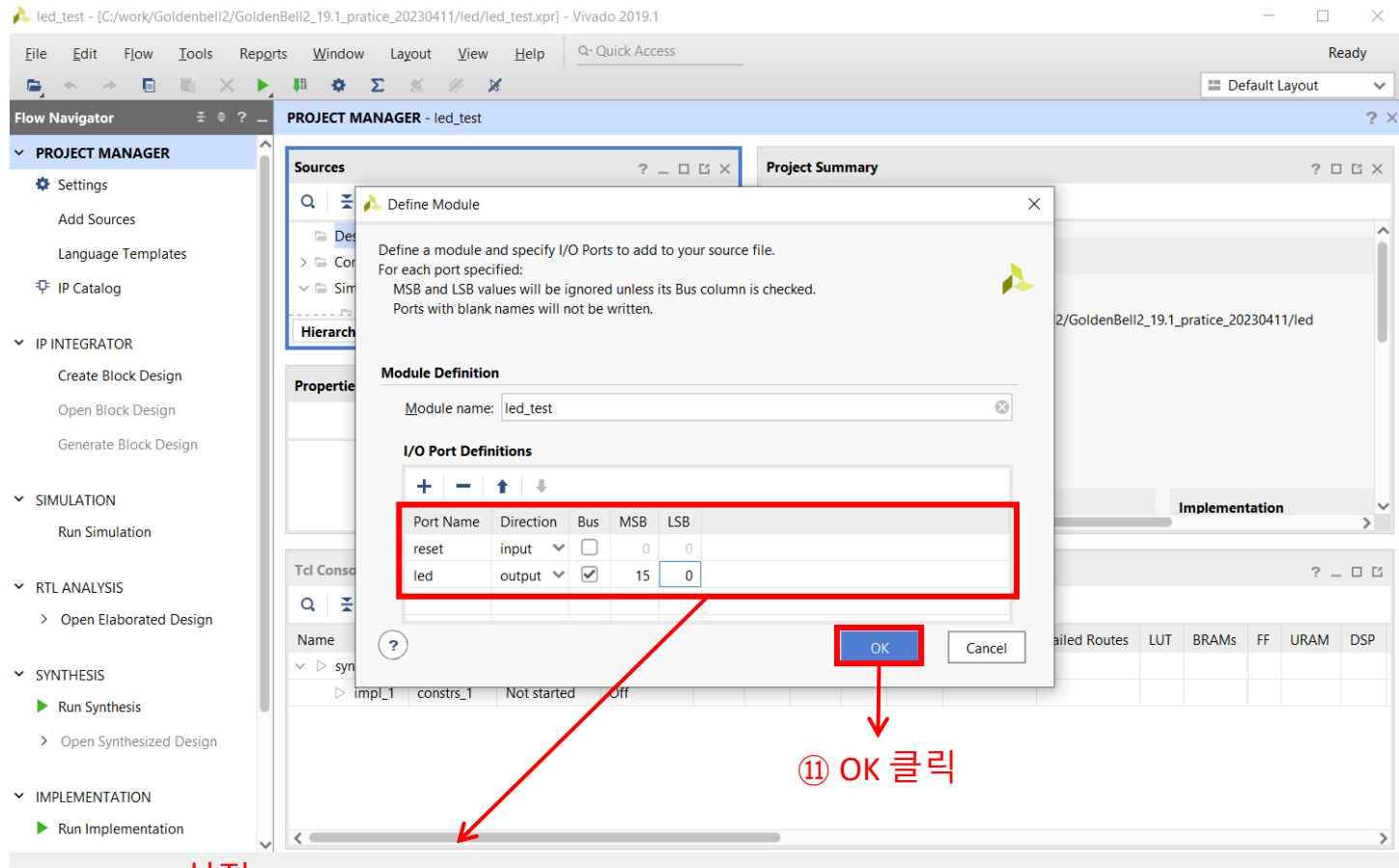
Tcl Console Messages Log Reports **Design Runs**

Name	Constraints	Status	Incremental	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	BRAMs	FF	URAM	DSP
synth_1	constrs_1	Not started	Off												
impl_1	constrs_1	Not started	Off												

⑦ Sources에 Design Sources 우클릭 뒤 Add Source 선택
만든 project에 Source file 생성

Specify and/or create source files to add to the project





⑩ input, output 설정
여기에서 설정해 주지 않아도 file생성 뒤에도 code상 수정 가능
(그냥 OK해도 상관 없음)

led_test - [C:/work/GoldenBell2/GoldenBell2_19.1_practice_20230411/led/led_test.xpr] - Vivado 2019.1

File Edit Flow Tools Repgrts Window Layout View Help Quick Access

Ready

Default Layout

Flow Navigator

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog

IP INTEGRATOR

- Create Block Design
- Open Block Design
- Generate Block Design

SIMULATION

- Run Simulation

RTL ANALYSIS

- Open Elaborated Design

SYNTHESIS

- Run Synthesis
- Open Synthesized Design

IMPLEMENTATION

- Run Implementation

PROJECT MANAGER - led_test

Sources

- Design Sources (1)
 - led_test (led_test.v)
- Constraints
- Simulation Sources (1)
 - sim_1 (1)

Hierarchy Libraries Compile Order

Source File Properties

led_test.v

Enabled

Location: C:/work/GoldenBell2/GoldenBell2_19.1_practice_20

General Properties

Project Summary

led_test.v

ork/GoldenBell2/GoldenBell2_19.1_practice_20230411/led/led_test.srscs/sources_1/new/led_test.v

```

12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module led_test(
24     input reset,
25     output [15:0] led
26 );
27 endmodule
28

```

Source file 생성 끝 !!!

Tcl Console Messages Log Reports Design Runs

Name	Constraints	Status	Incremental	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	BRAMs	FF	URAM	DSP
synth_1	constrs_1	Not started	Off												
impl_1	constrs_1	Not started	Off												

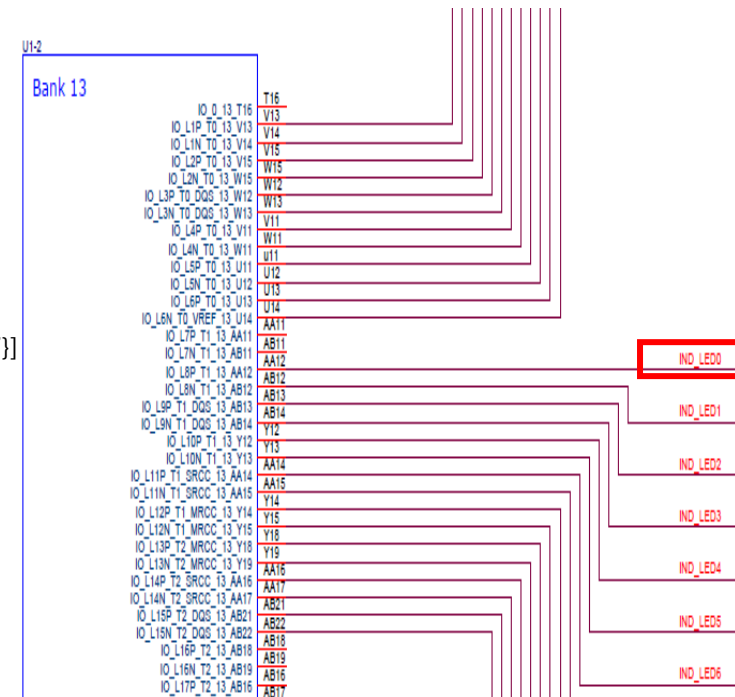
23:0 Insert Verilog

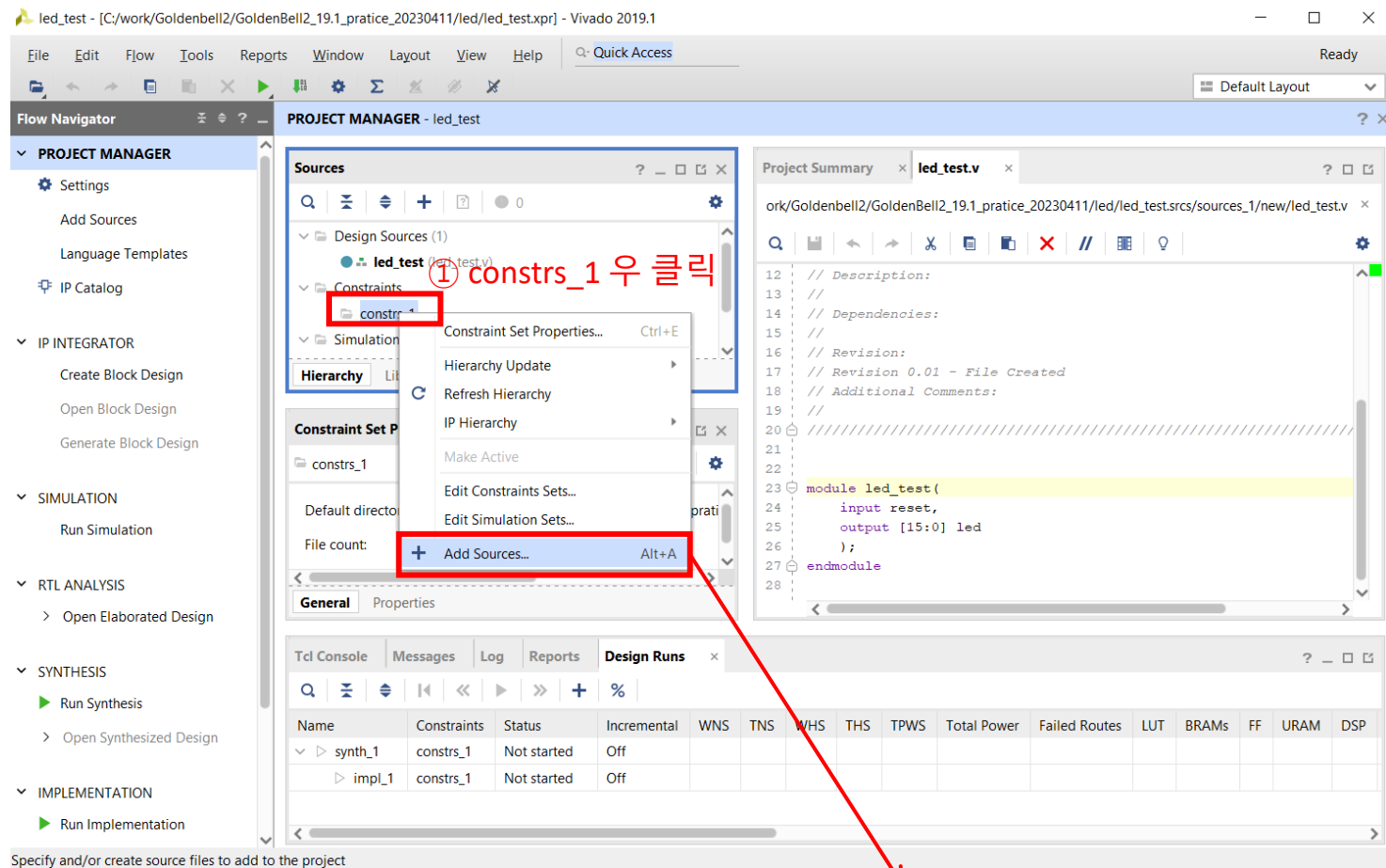
Step 2. Creating constraint file

Pin map을 위한 Constraint파일 생성 방법

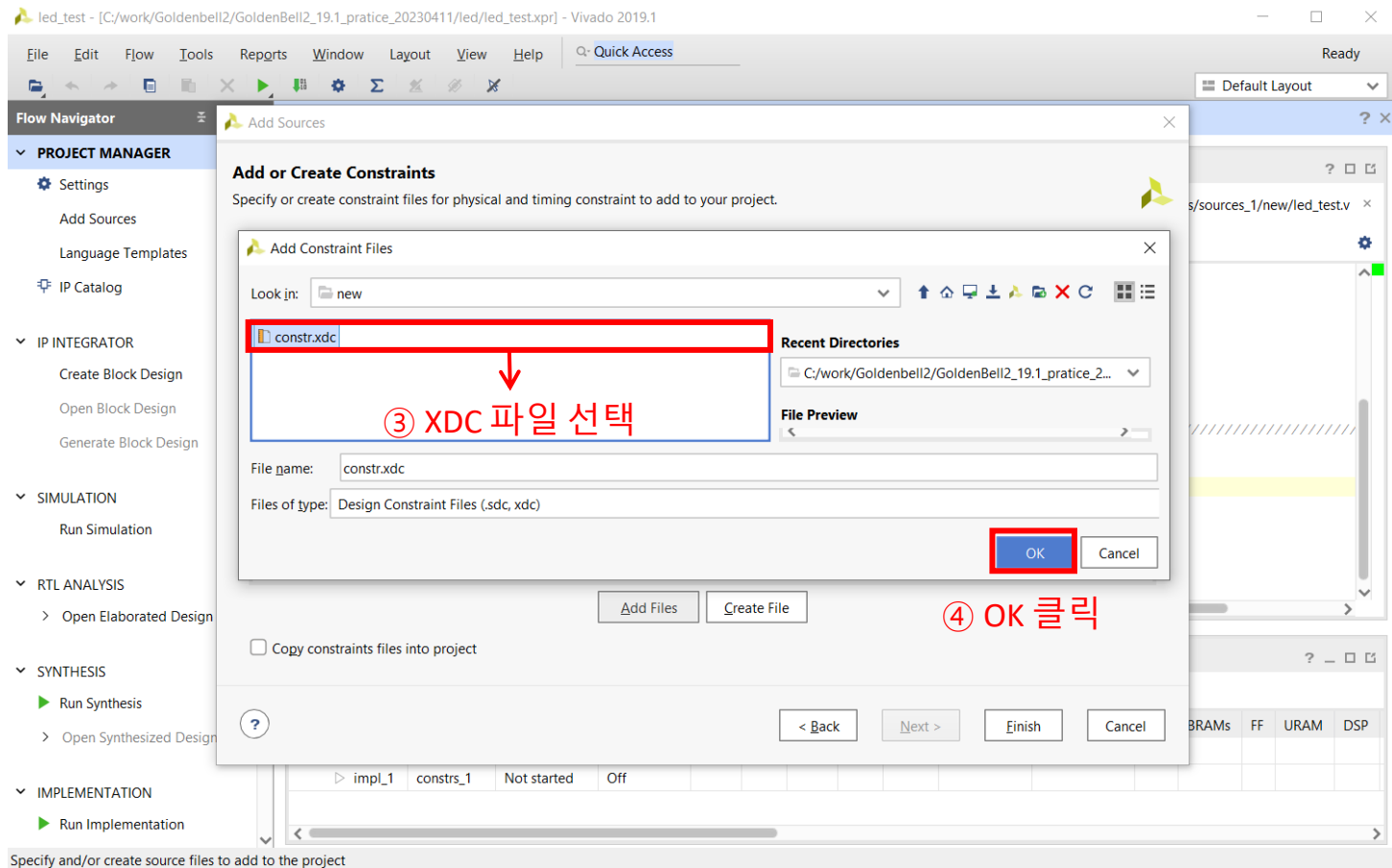
Constraint File Generation Example

- constr.xdc 파일을 프로젝트에 추가한다
 - constr.xdc 파일은 peripheral_test_code 폴더에 존재
 - 현재 working directory에 .xdc file을 옮김
- 보드의 회로도를 확인하고 원하는 input과 output의 번호를 확인
 - 예) Output으로 선언한 led_output으로
Led0[AA12 pin]를 제어하고 싶을 경우
 - Led0가 AA12에 mapping되어있음을 확인
 - XDC 파일에 아래 문법에 맞춰 입력
 - 문법: `set_property PACKAGE_PIN “pin 번호” [get_ports {“output 이름”}]`
 - ex) `set_property PACKAGE_PIN AA12 [get_ports {led[0]}]`





② XDC 파일을 추가하기 위해 Add Sources를 클릭



led_test - [C:/work/Goldenbell2/GoldenBell2_19.1_practice_20230411/led/led_test.xpr] - Vivado 2019.1

File Edit Flow Tools Reports Window Layout View Help Quick Access

Flow Navigator PROJECT MANAGER - led_test

PROJECT MANAGER

- Settings
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- SYNTHESIS
 - Run Synthesis
 - Open Synthesized Design
- IMPLEMENTATION
 - Run Implementation

Sources

- Design Sources (1)
 - led_test (led_test.v)
- Constraints (1)
 - constrs_1 (1)
 - constr.xdc
- Simulation Sources (1)

Hierarchy Libraries Compile Order

Source File Properties

constr.xdc

Enabled

Location: C:/work/Goldenbell2/GoldenBell2_19.1_practice_20230411/led/led_test.xpr

General Properties

Project Summary led_test.v

ork/Goldenbell2/GoldenBell2_19.1_practice_20230411/led/led_test.srscs/sources_1/new/led_test.v

```

12 // Description:
13 // Dependencies:
14 // Revision:
15 // Revision 0.01 - File Created
16 // Additional Comments:
17 //
18 //
19 //
20 //
21 //
22 //
23 module led_test(
24     input reset,
25     output [15:0] led
26 );
27 endmodule
28

```

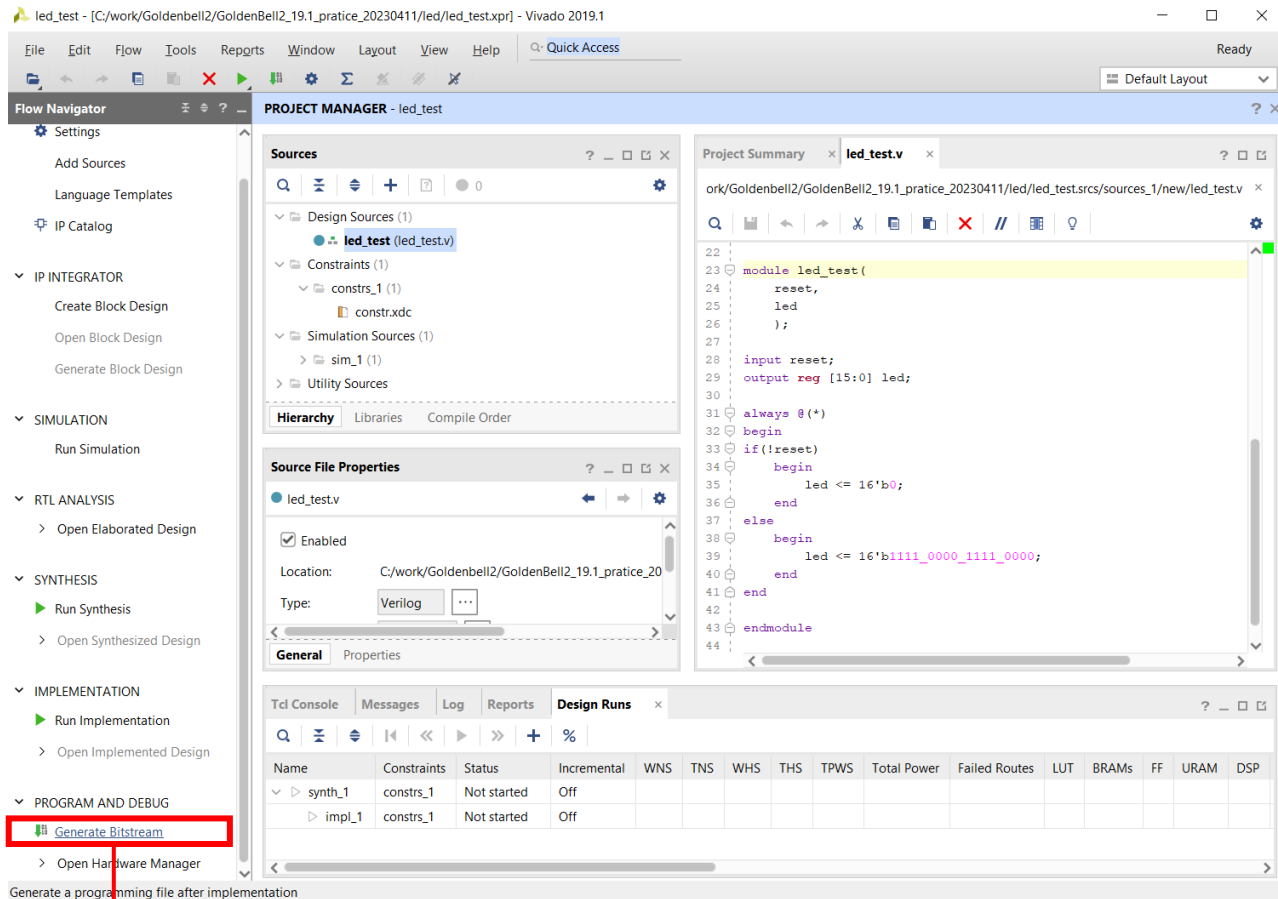
Tcl Console Messages Log Reports Design Runs

Name	Constraints	Status	Incremental	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	BRAMs	FF	URAM	DSP
synth_1	constrs_1	Not started	Off												
impl_1	constrs_1	Not started	Off												

constr.xdc 파일 프로젝트 안에 넣기 끝!!!
이제.... xdc 파일을 수정하여 쓰시면 됩니다

Step 3. Creating a Bit File

보드에 programming하기 위한 bit-file을 생성하는
방법



- ① .xdc 파일까지 수정을 완료 하였으면, bit파일을 생성함
 Generate Bitstream을 클릭하면 bit파일이 생성됨
 Synthesize → Implement design → Generate Bitstream 순서로 진행

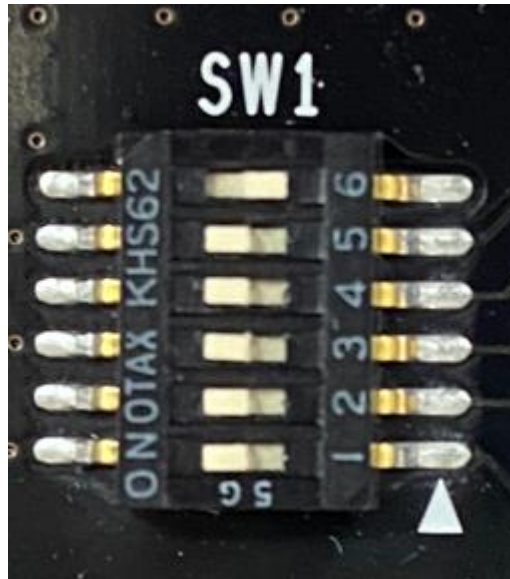
FPGA programming

- FPGA board에 programming하는 방법 3가지 존재
 - 1. bit file FPGA downloading
 - .bit file을 직접 FPGA에 다운로드 하여 바로 실행
 - 2. SPI Flash programming
 - SPI Flash에 programming하여 booting시(전원 인가 시) 바로 실행
 - 이를 위해 먼저 .bin file을 생성해야 함
 - 3. SD card 부트
 - 생성된 .bin 파일을 SD card에 복사하여 booting시(전원 인가 시) 바로 실행

Step 4. Downloading a Bit File to FPGA

생성한 bit-file을 보드에 download하는 방법

JTAG Boot Switch(SW1) 설정



1-2-3-4-5-6
ON-ON-ON-ON-ON-OFF

led_test - [C:/work/Goldenbell2/GoldenBell2_19.1_pratice_20230411/led/led_test.xpr] - Vivado 2019.1

File Edit Flow Tools Reports Window Layout View Help Quick Access write_bitstream Complete ✓ Default Layout

Flow Navigator

- IP INTEGRATOR
 - Create Block Design
 - Open Block Design
 - Generate Block Design
- SIMULATION
 - Run Simulation
- RTL ANALYSIS
 - Open Elaborated Design
- SYNTHESIS
 - Run Synthesis
 - Open Synthesized Design
- IMPLEMENTATION
 - Run Implementation
 - Open Implemented Design
- PROGRAM AND DEBUG
 - Generate Bitstream
 - Open Hardware Manager**

Open the hardware program and debug manager

PROJECT MANAGER - led_test

Sources

- Design Sources (1)
 - led_test (led_test.v)
- Constraints (1)
 - constrs_1 (1)
 - constr.xdc

Hierarchy Libraries Compile Order

Properties

led_test.v

Enabled

Location: C:/work/Goldenbell2/GoldenBell2_19.1_pratice_20230411/led/led_test.sr

General Properties

Project Summary

led_test.v

C:/work/Goldenbell2/GoldenBell2_19.1_pratice_20230411/led/led_test.sr

```

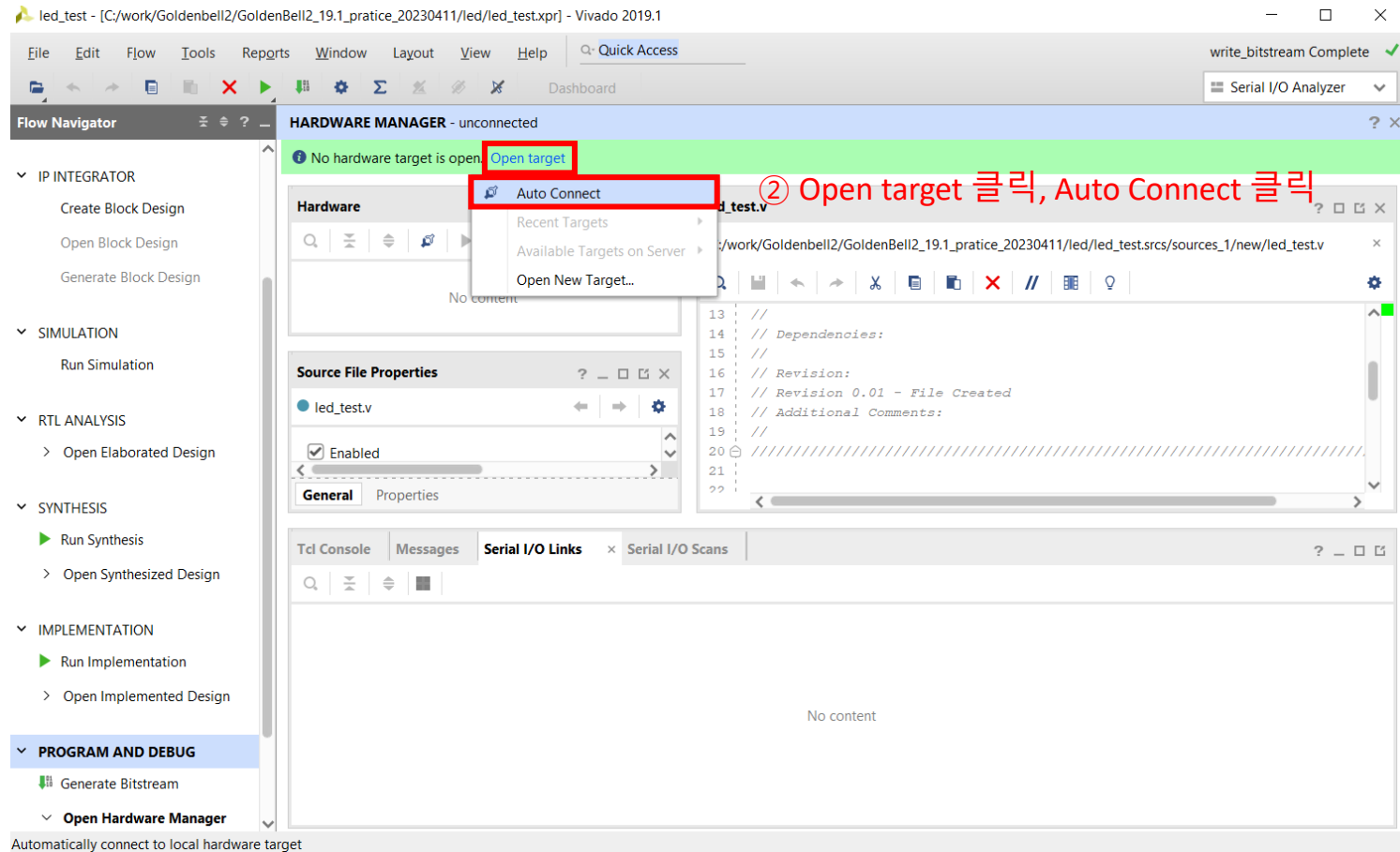
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module led_test(
24     reset,
25     led
26 );
27
28 input reset;
  
```

Tcl Console Messages Log Reports **Design Runs**

Name	Constraints	Status	Incremental	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	BRAMs	FF
✓ synth_1	constrs_1	synth_design Complete!	Off								0	0.0	0
✓ impl_1	constrs_1	write_bitstream Complete!	Off	NA	NA	NA	NA	NA	6.006	0	0	0.0	0

*이 flow는 bit file을 생성 했음을 가정으로 함

① Open Hardware Manager를 클릭



led_test - [C:/work/GoldenBell2/GoldenBell2_19.1_pratice_20230411/led/led_test.xpr] - Vivado 2019.1

File Edit Flow Tools Repgrts Window Layout View Help Quick Access

Dashboard

Flow Navigator

- IP INTEGRATOR
 - Create Block Design
 - Open Block Design
 - Generate Block Design
- SIMULATION
 - Run Simulation
- RTL ANALYSIS
 - Open Elaborated Design
- SYNTHESIS
 - Run Synthesis
 - Open Synthesized Design
- IMPLEMENTATION
 - Run Implementation
 - Open Implemented Design
- PROGRAM AND DEBUG**
 - Generate Bitstream
 - Open Hardware Manager

HARDWARE MANAGER - localhost/xilinx_tcf/Digilent/210205329746A

There are no debug cores. Program device Refresh device

Hardware

Name	Status
localhost (1)	Connected
xilinx_tcf/Digilent/210205329746A (2)	Open
arm_dap_0 (0)	N/A
xc7z015_1 (1)	Programmed
XADC (System Monitor)	

③ Jtag 연결이 정상적이면 device가 창에 뜸

Properties

Select an object to see properties

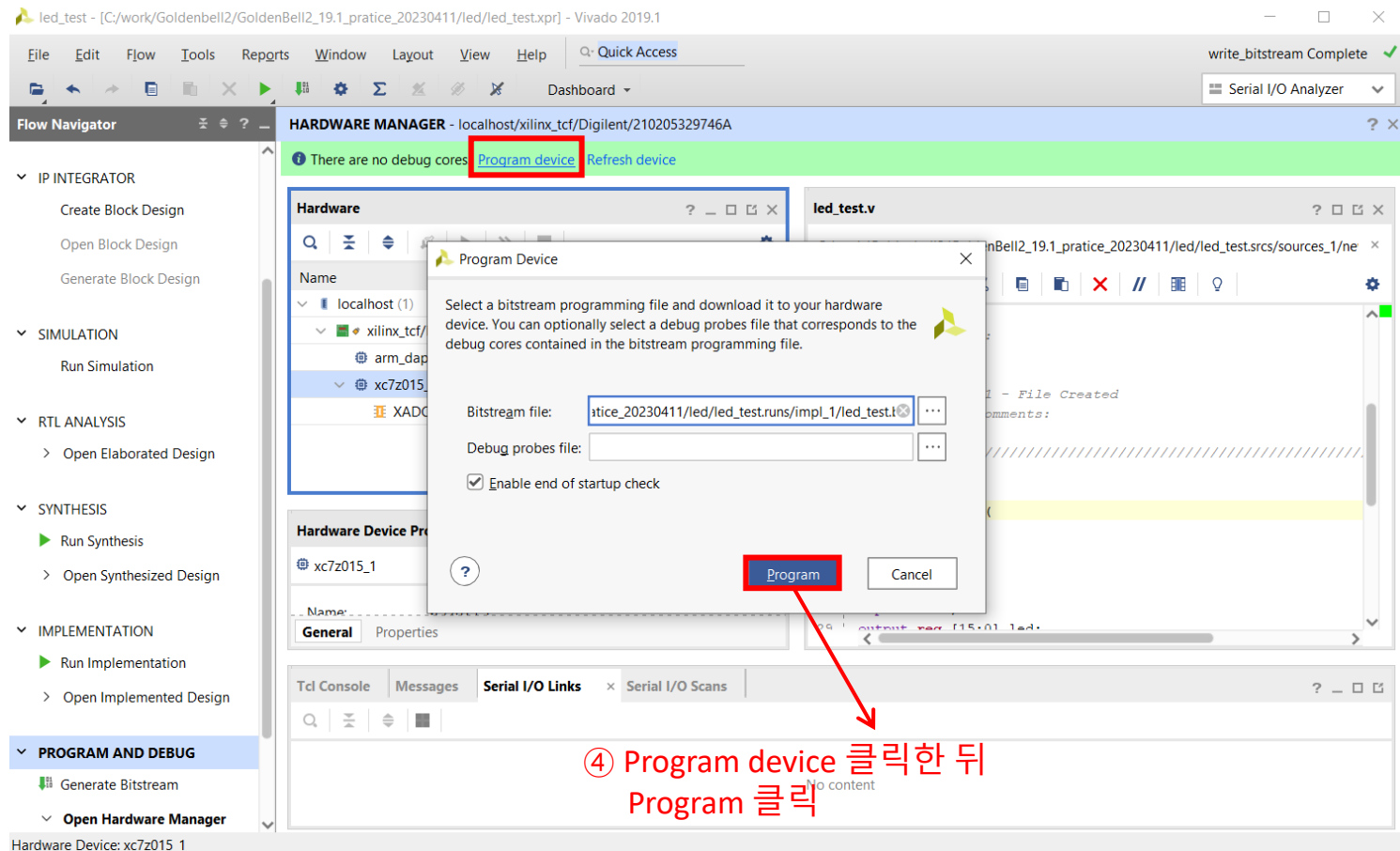
led_test.v

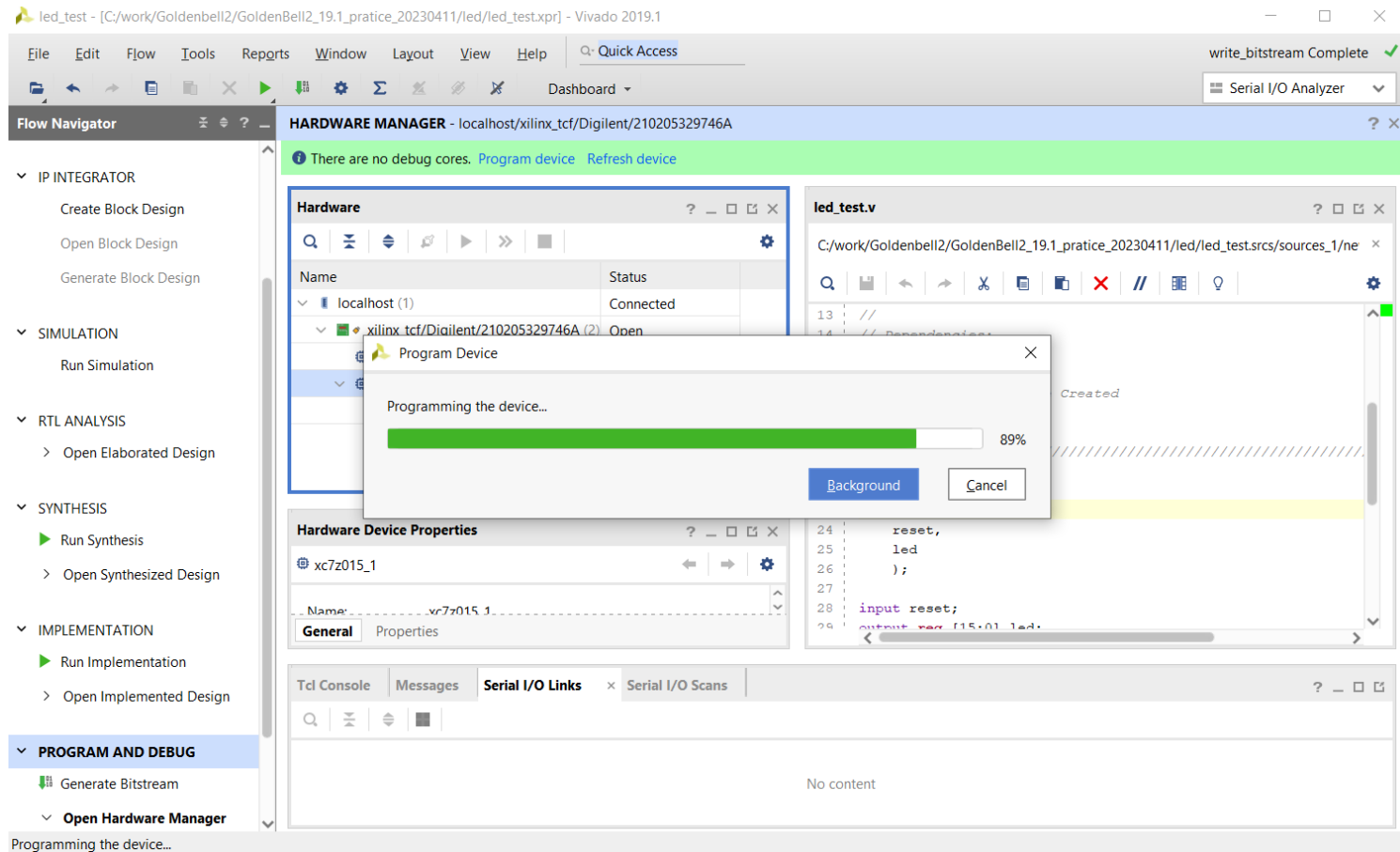
```

13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //
21 //
22 //
23 module led_test(
24     reset,
25     led
26 );
27
28 input reset;
29 output reg [15:0] led;
  
```

Tcl Console Messages Serial I/O Links Serial I/O Scans

No content



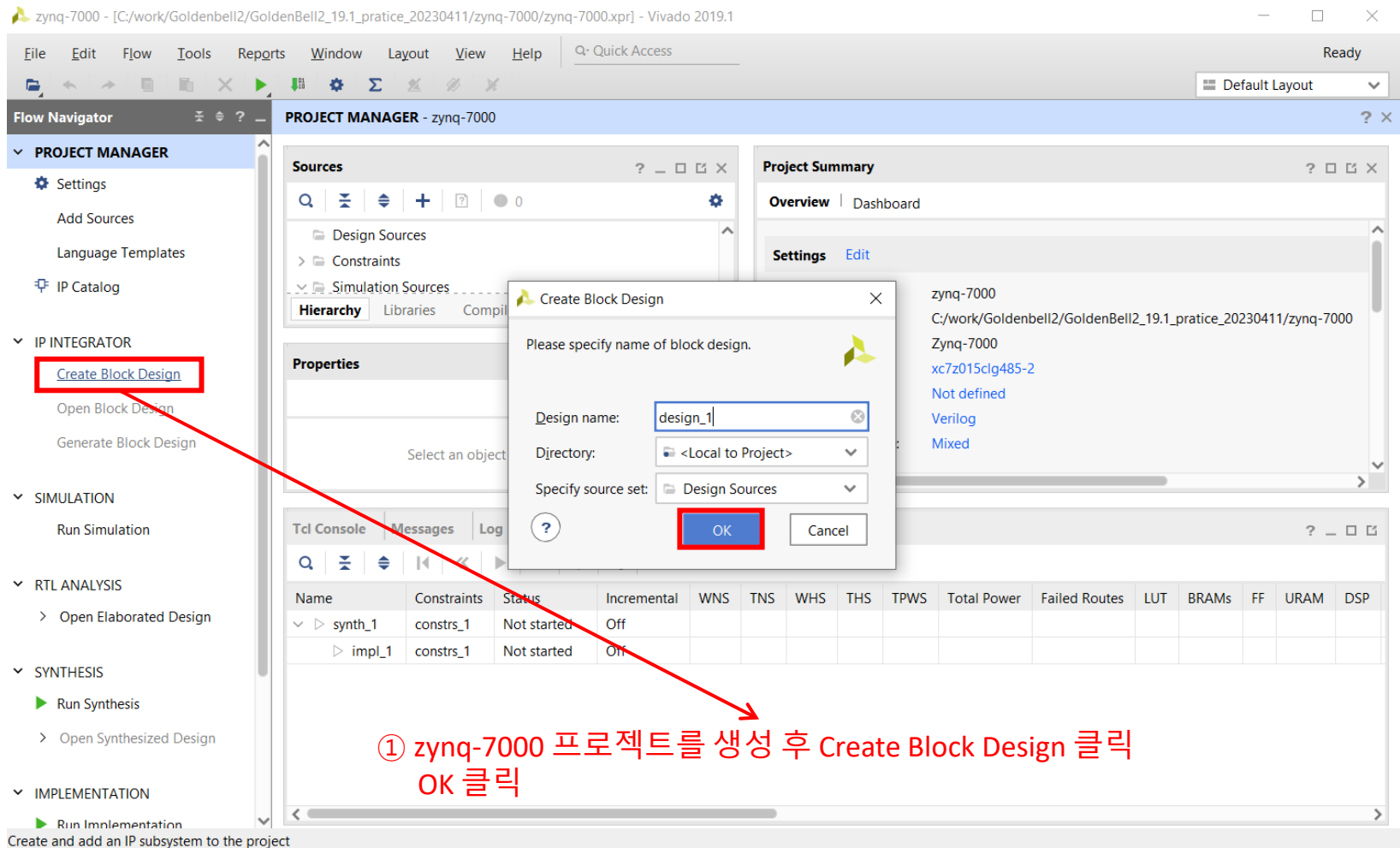


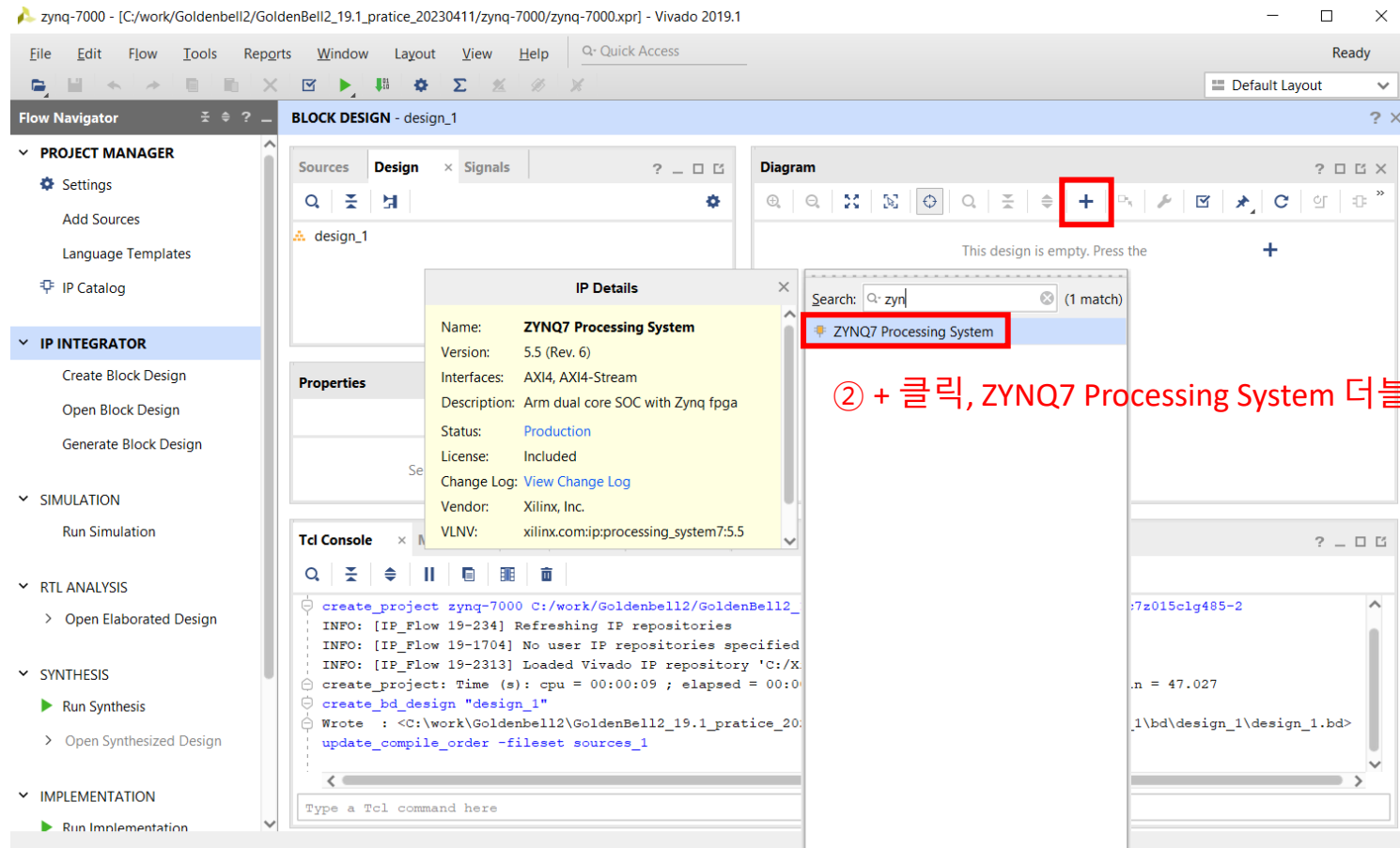
⑤ Bit 파일을 프로그램함

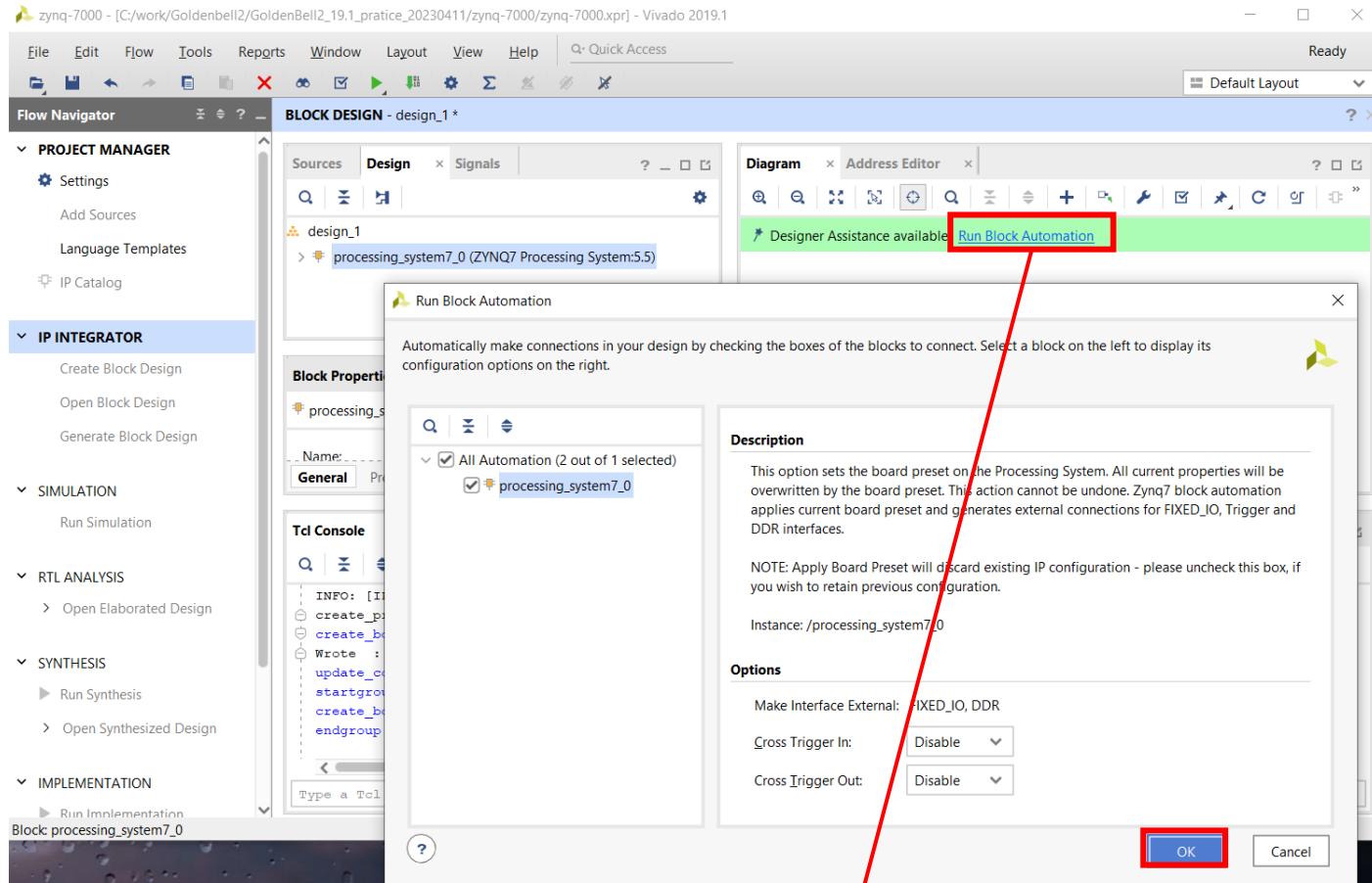
SPI Flash에 FPGA Bit File 프로그램

SPI Flash

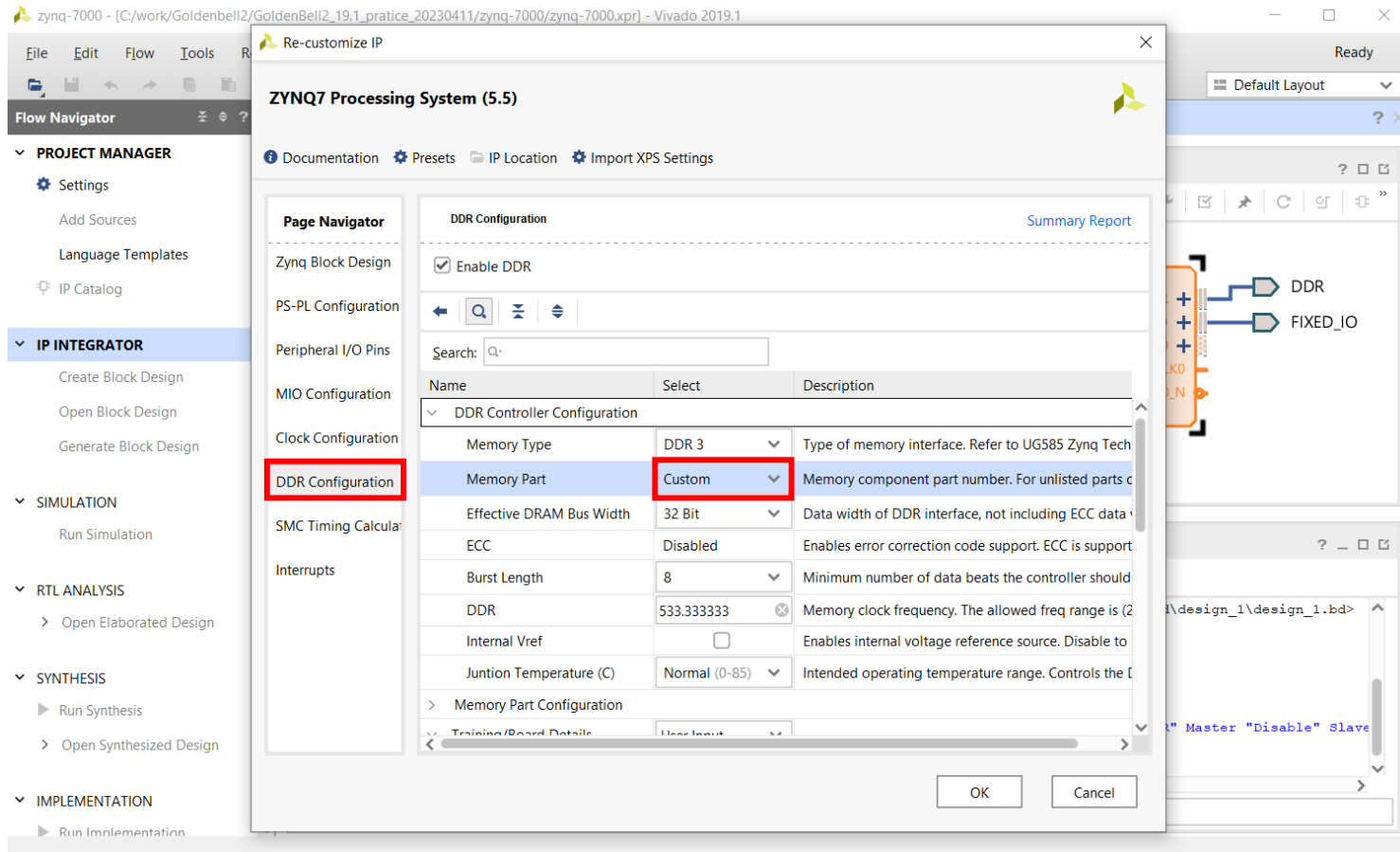
- 최초 전원 인가 시 바로 SPI flash에 저장된 bit file 실행
- SPI flash spec
 - Winbond 128Mb NOR Flash memory (25Q128F)
 - Datasheet는 <Goldenbell2_CPU_test_platform_manual/datasheet> 폴더에 존재
- SPI flash에 program하는 방법
 - Bit file-BIN file로 변환
 - Bit file를 프로그램해 주는 FSBL(First Stage Boot Loader)가 필요
 - FSBL 이미지 + Bit file -> BIN file 생성
 - BIN file programming
 - FSBL 이미지 빌드를 위해서 Zynq-7000 IP를 포함하는 디자인 필요







③ Run Block Automation 클릭, OK 클릭



④ ZYNQ7 Processing System IP 더블 클릭,
DDR Configuration 클릭,
Memory Part에서 Custom 선택

zynq-7000 - [C:/work/Goldenbell2/GoldenBell2] Re-customize IP

File Edit Flow Tools Reports

Flow Navigator

- PROJECT MANAGER
 - Settings
 - Add Sources
 - Language Templates
 - IP Catalog
- IP INTEGRATOR
 - Create Block Design
 - Open Block Design
 - Generate Block Design
- SIMULATION
 - Run Simulation
- RTL ANALYSIS
 - Open Elaborated Design
- SYNTHESIS
 - Run Synthesis
 - Open Synthesized Design
- IMPLEMENTATION
 - Run Implementation

ZYNQ7 Processing System (5.5)

Documentation Presets IP Location Import XPS Settings

Page Navigator

- Zynq Block Design
- PS-PL Configuration
- Peripheral I/O Pins
- MIO Configuration
- Clock Configuration
- DDR Configuration
- SMC Timing Calculator
- Interrupts

DDR Configuration

Enable DDR

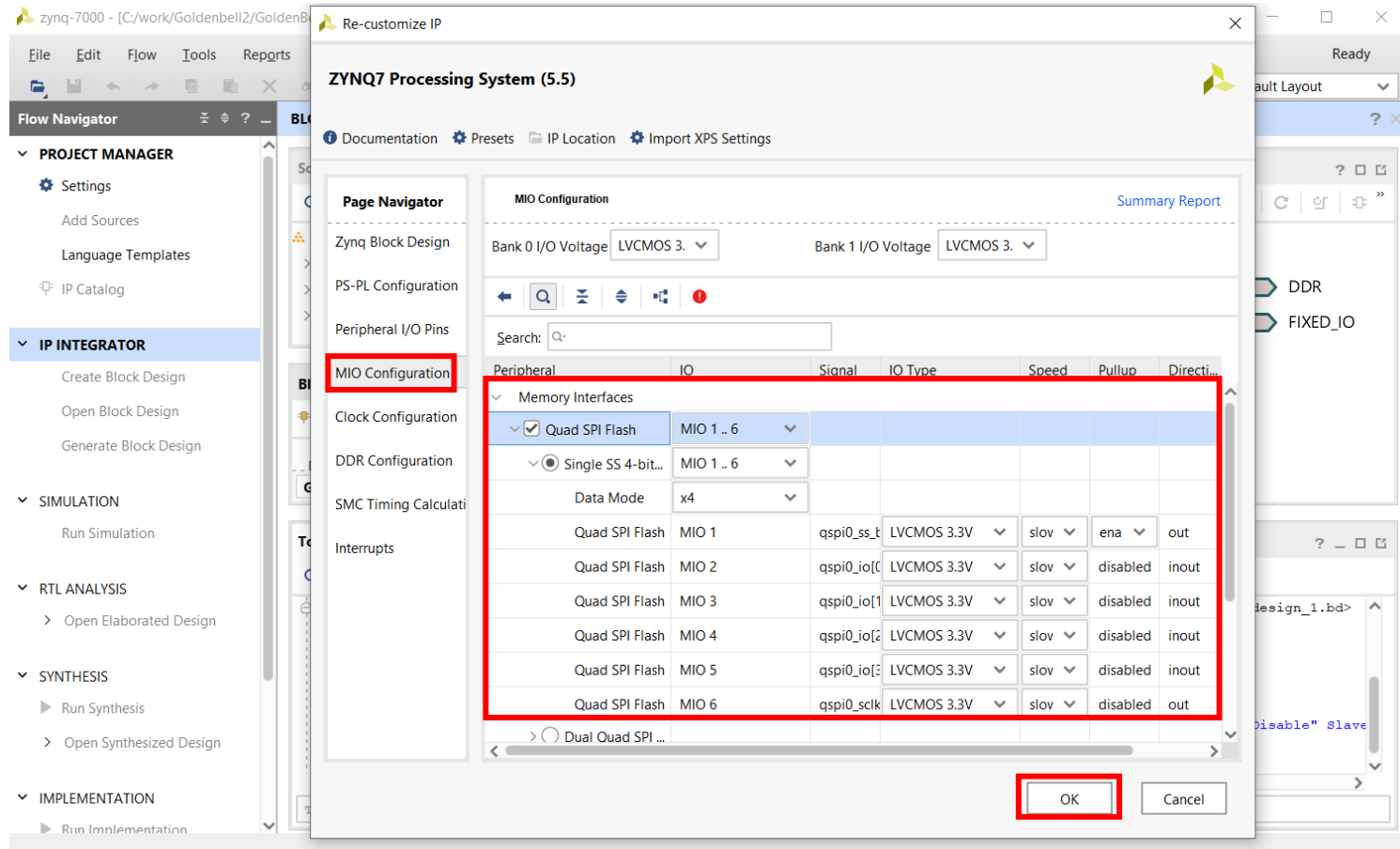
Search: Q

Name	Select	Description
tRASmin (ns)	37.5	Minimum number of memory clock cycles required by
tFAW (ns)	50	Determines the number of activates that can be perf
Training/Board Details		
User Input		
DRAM Training		
DQS to Clock Delay (ns)		
DQS0	0.103	DQS to Clock delay [0] (ns). The DQS path delay subtr
DQS1	0.103	DQS to Clock delay [1] (ns). The DQS path delay subtr
DQS2	0.215	DQS to Clock delay [2] (ns). The DQS path delay subtr
DQS3	0.215	DQS to Clock delay [3] (ns). The DQS path delay subtr
Board Delay (ns)		
Additive Latency (cycles)	0	Additive Latency (cycles). Increases the efficiency of t
Enable Advanced options		
Enable Advanced DDR QoS settings		

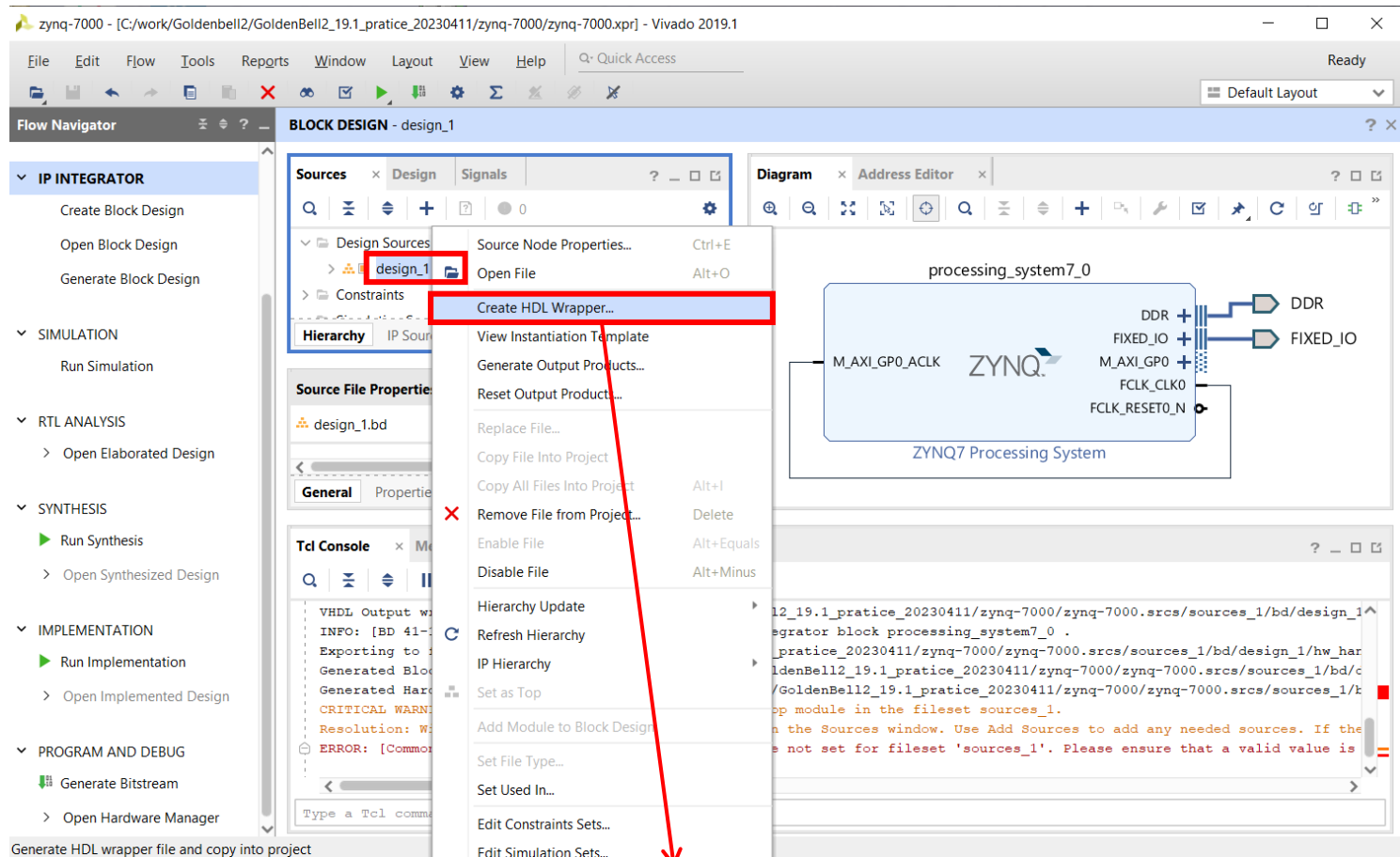
Summary Report

OK Cancel

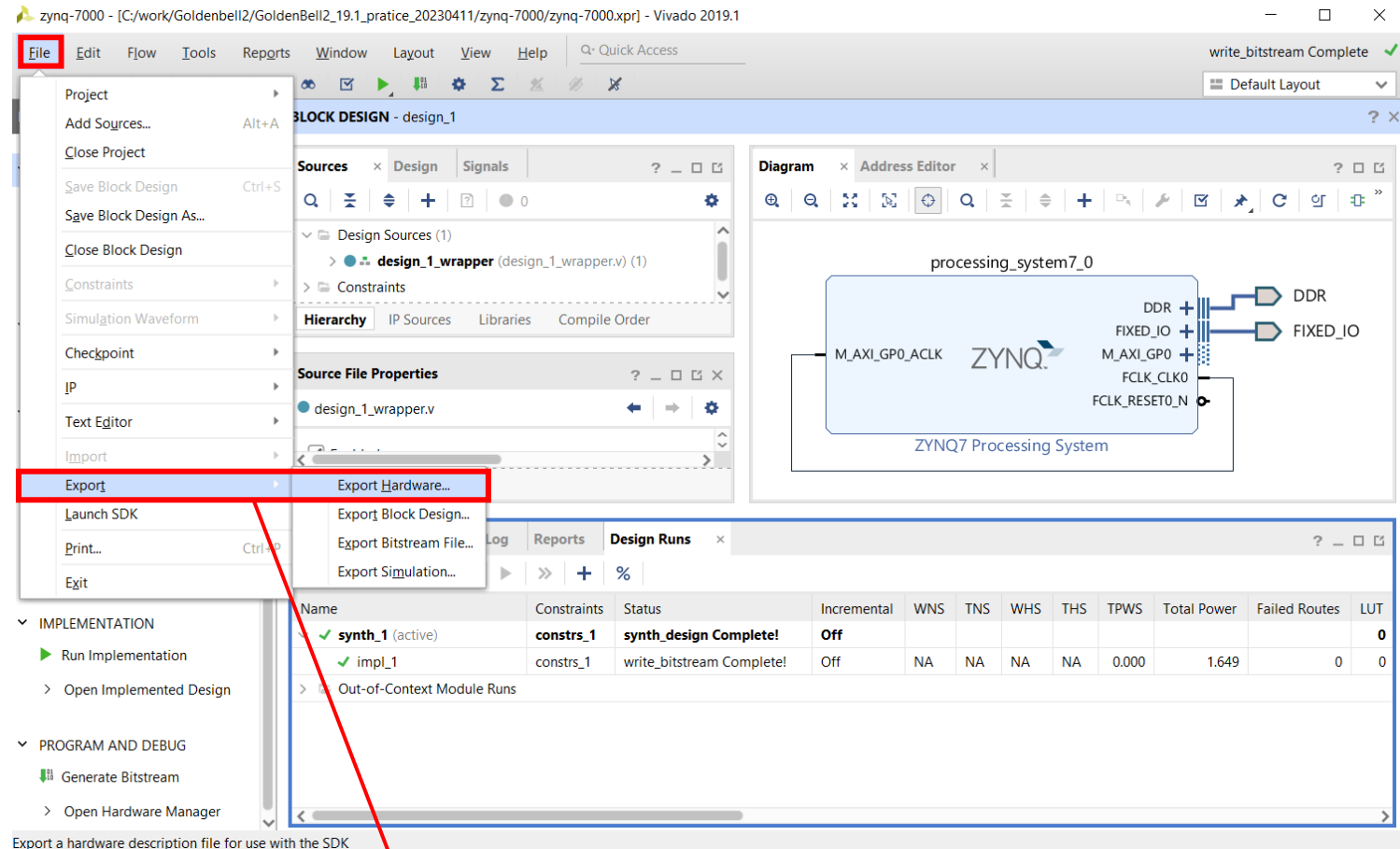
⑥ DQS0 ~ DQS3 Clock Delay 입력



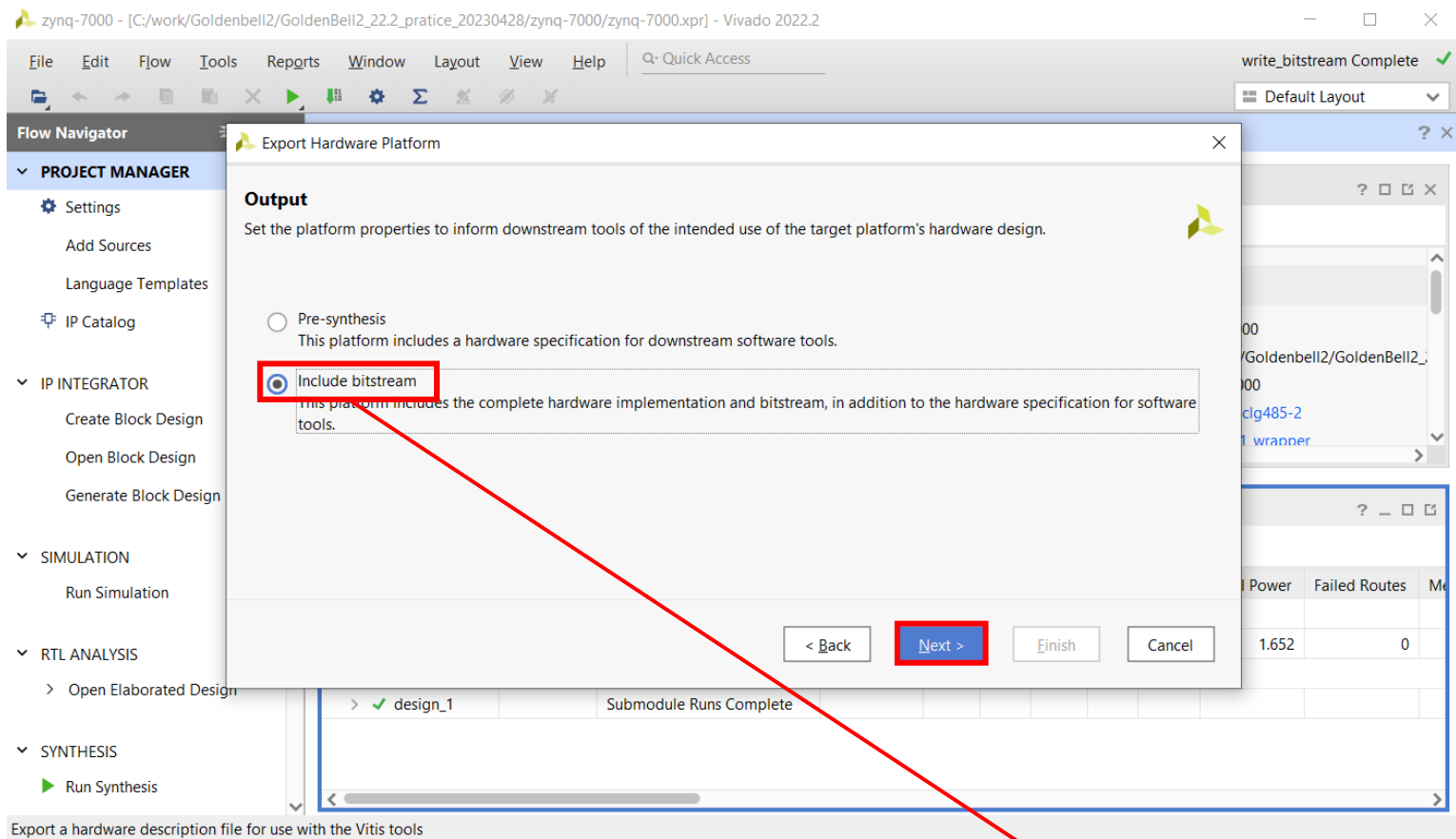
⑦ MIO Configuration 클릭, Quad SPI Flash 선택, OK 클릭



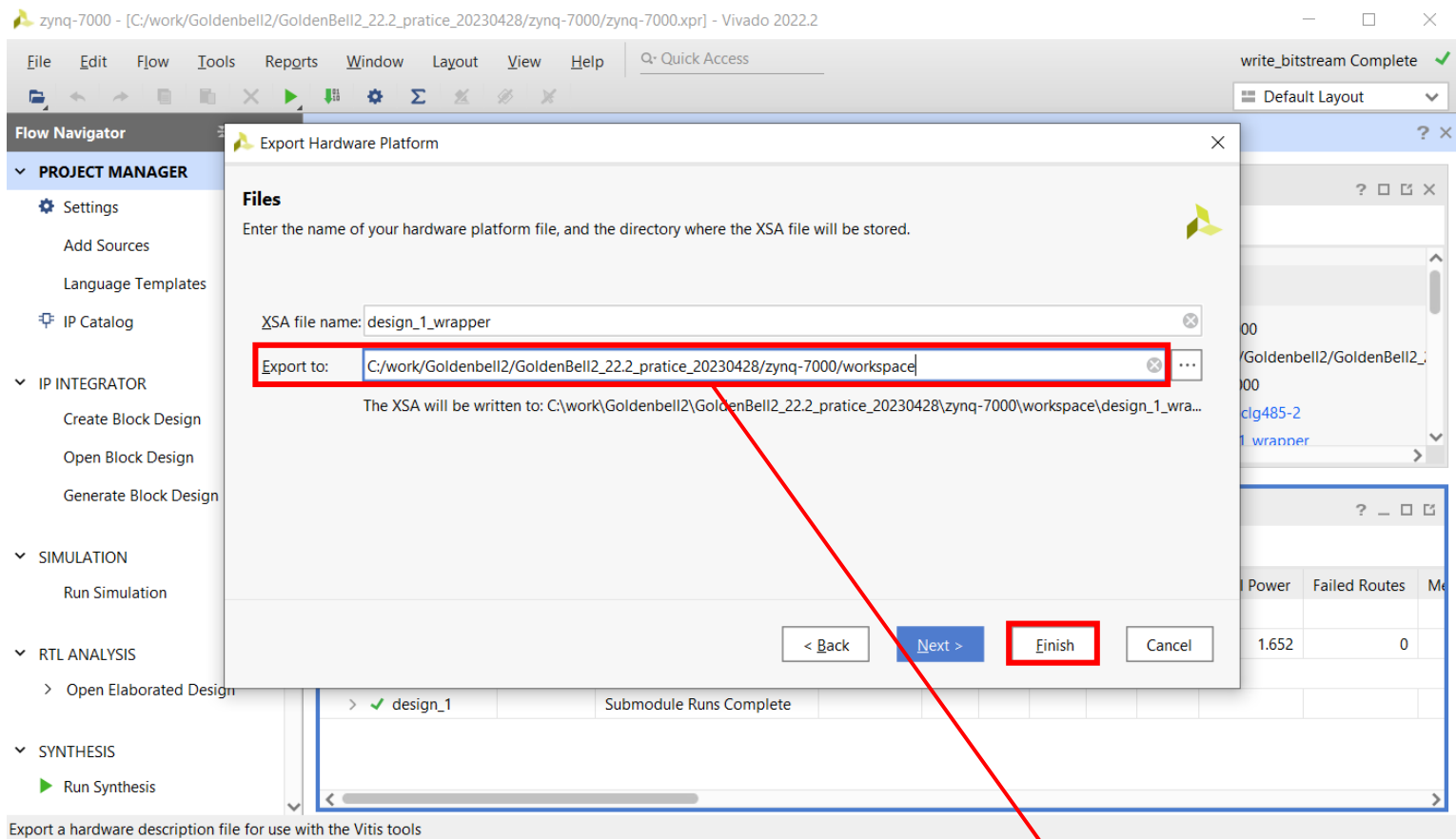
⑧ Sources / Design Sources / design_1에 우클릭, Create HDL Wrapper 클릭,
Diagram에서 FCLK_CLK0와 M_AXI_GPO_ACLK 연결 후 Generate Bitstream 클릭



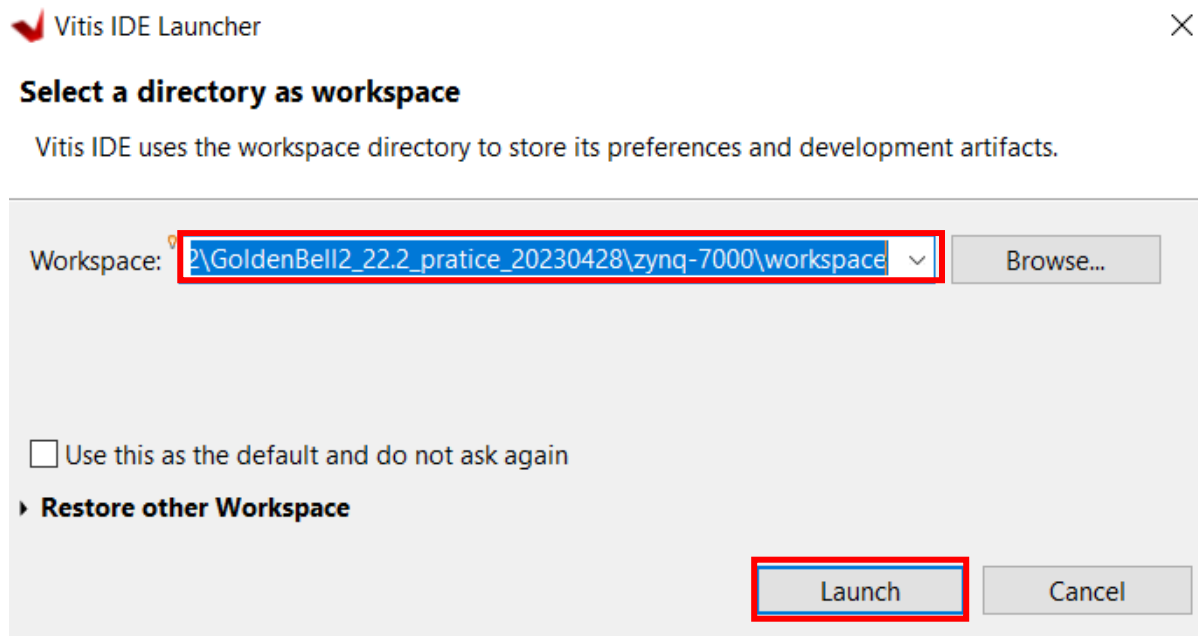
⑨ bit file 생성후 File / Export / Export Hardware 클릭



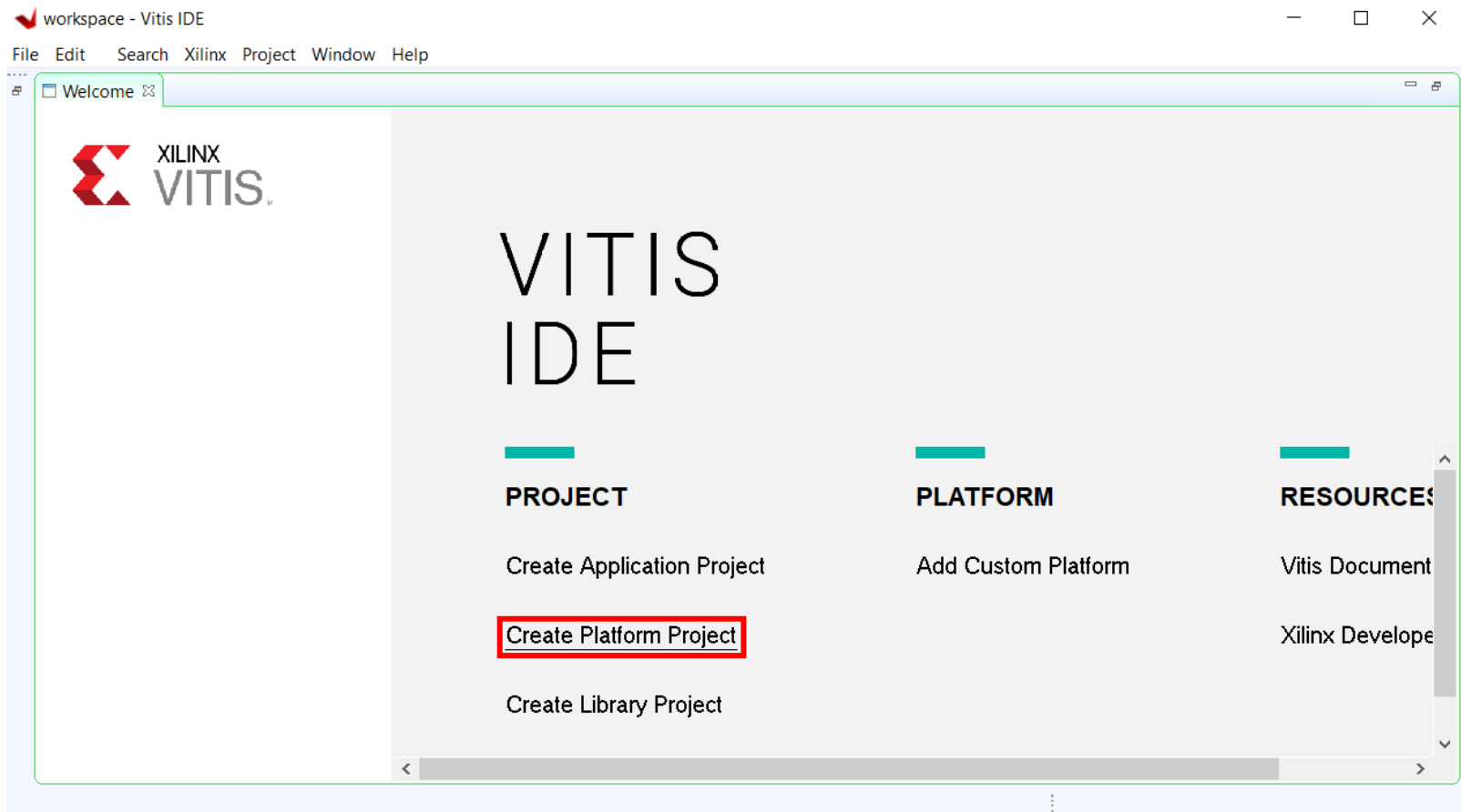
⑩ Include bitstream 체크후 Next 클릭



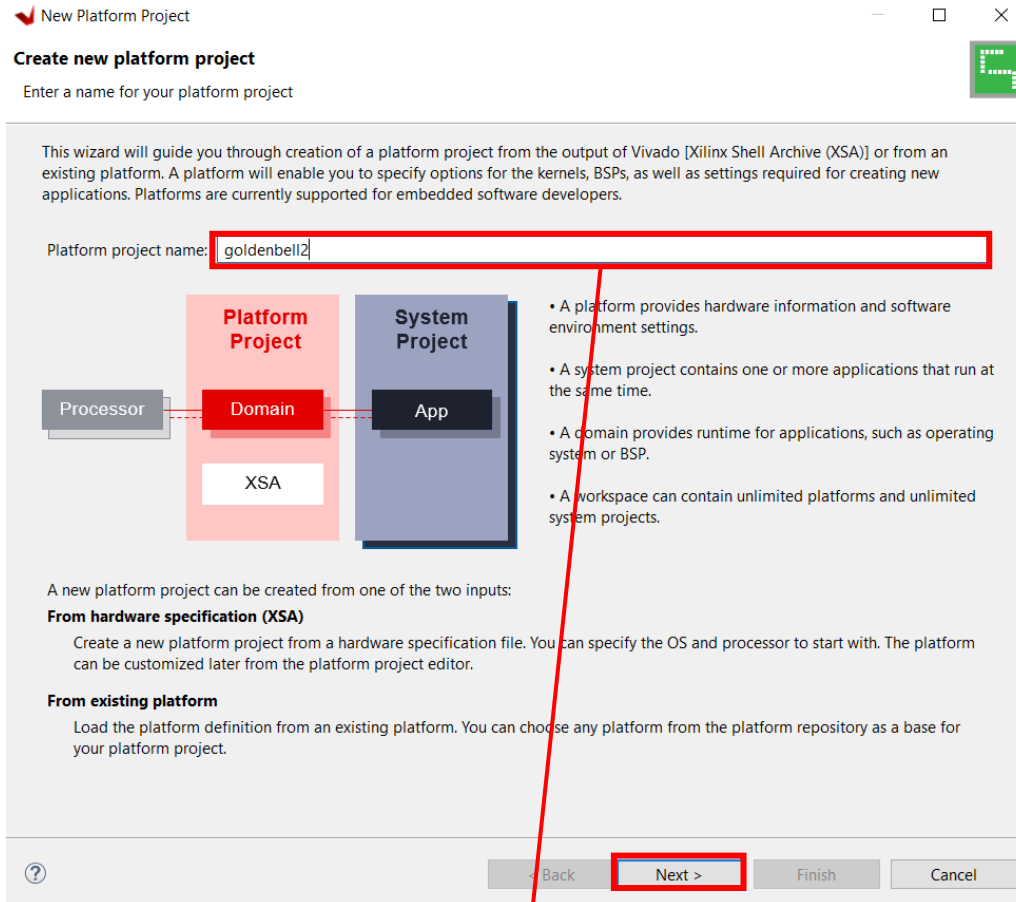
⑪ export할 위치를 명시하고 Finish 클릭



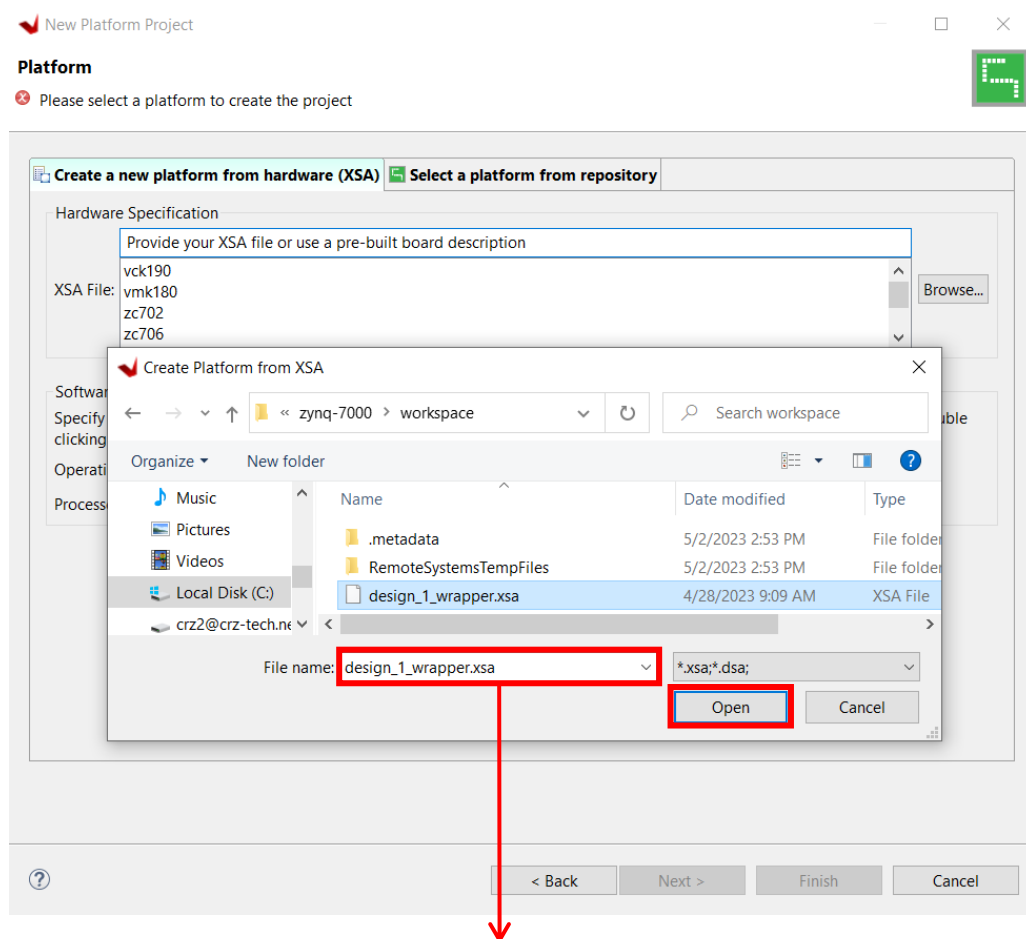
⑫ Vitis 실행 후 workspace 지정한 다음 Launch 클릭



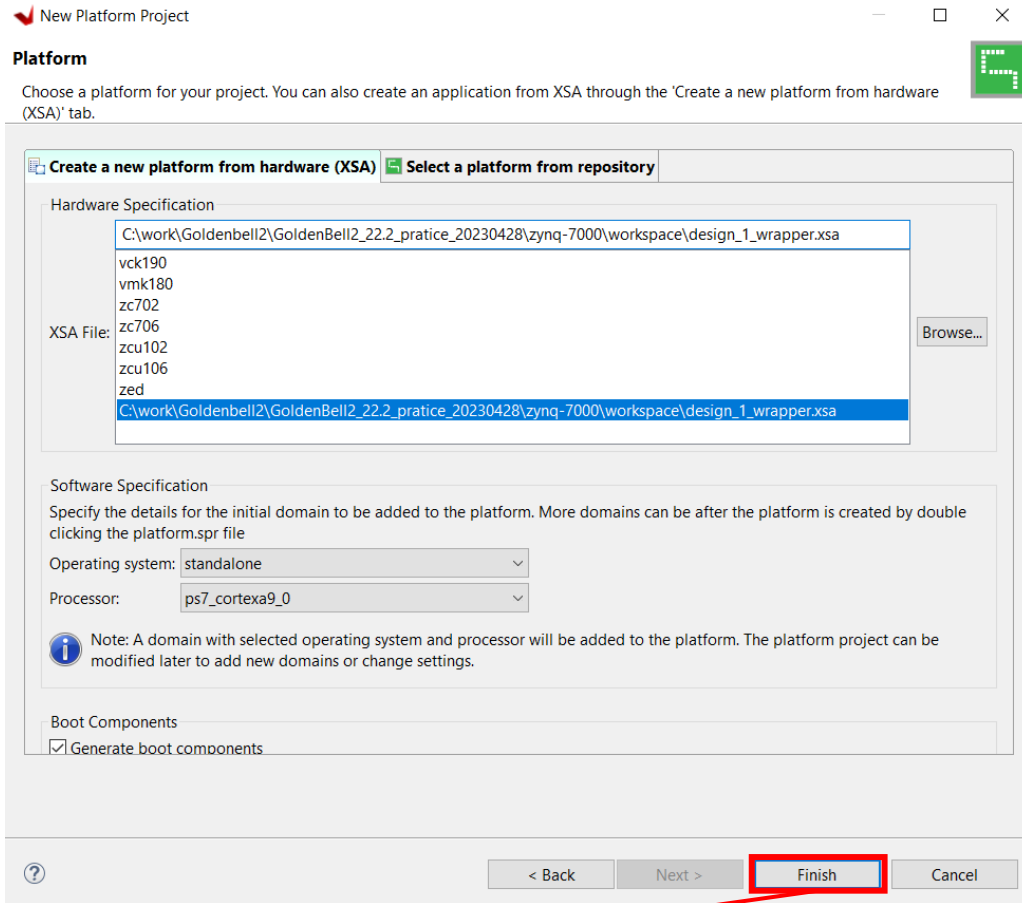
⑬ Create Platform Project 클릭



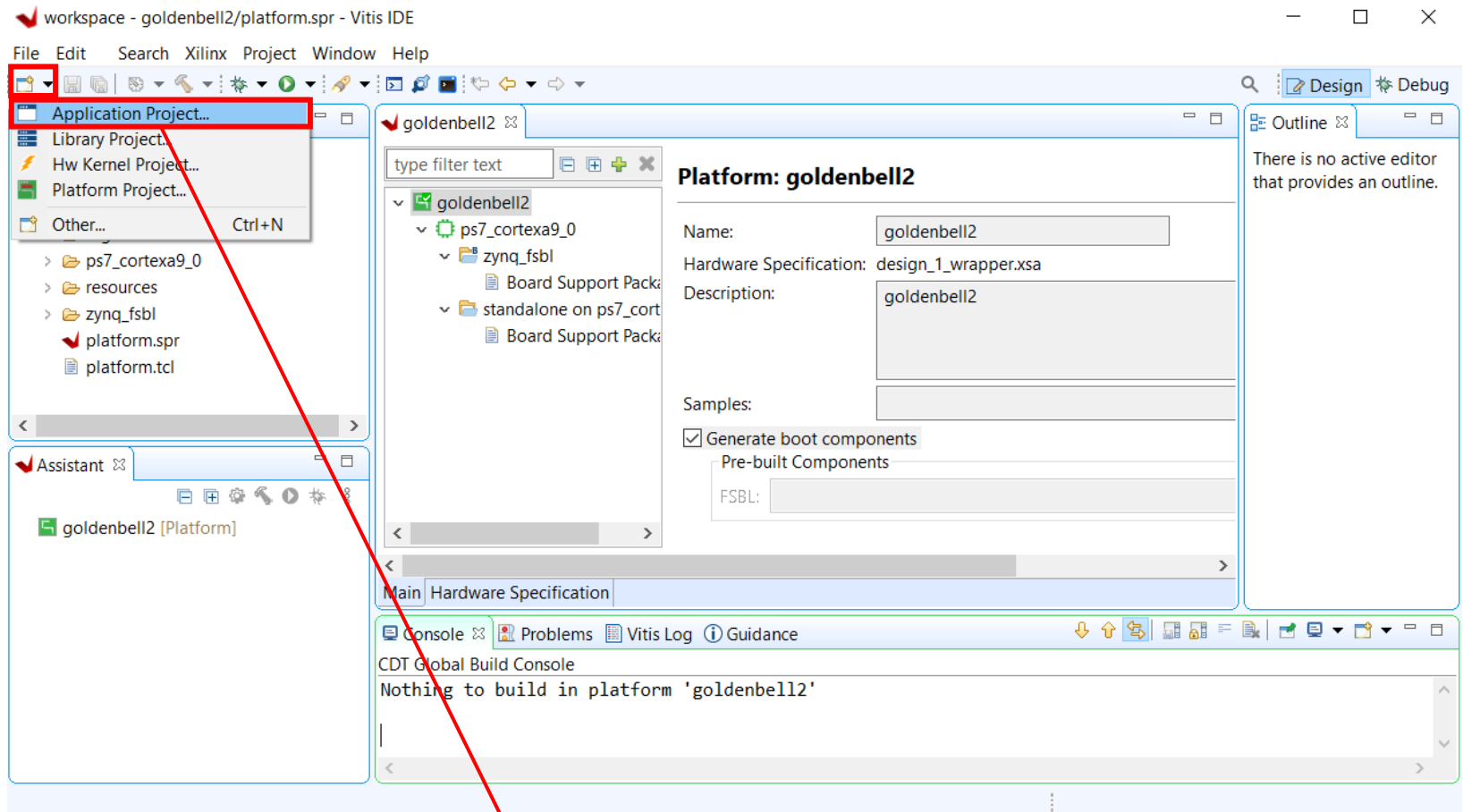
⑭ Platform Project name 입력 후, Next 클릭



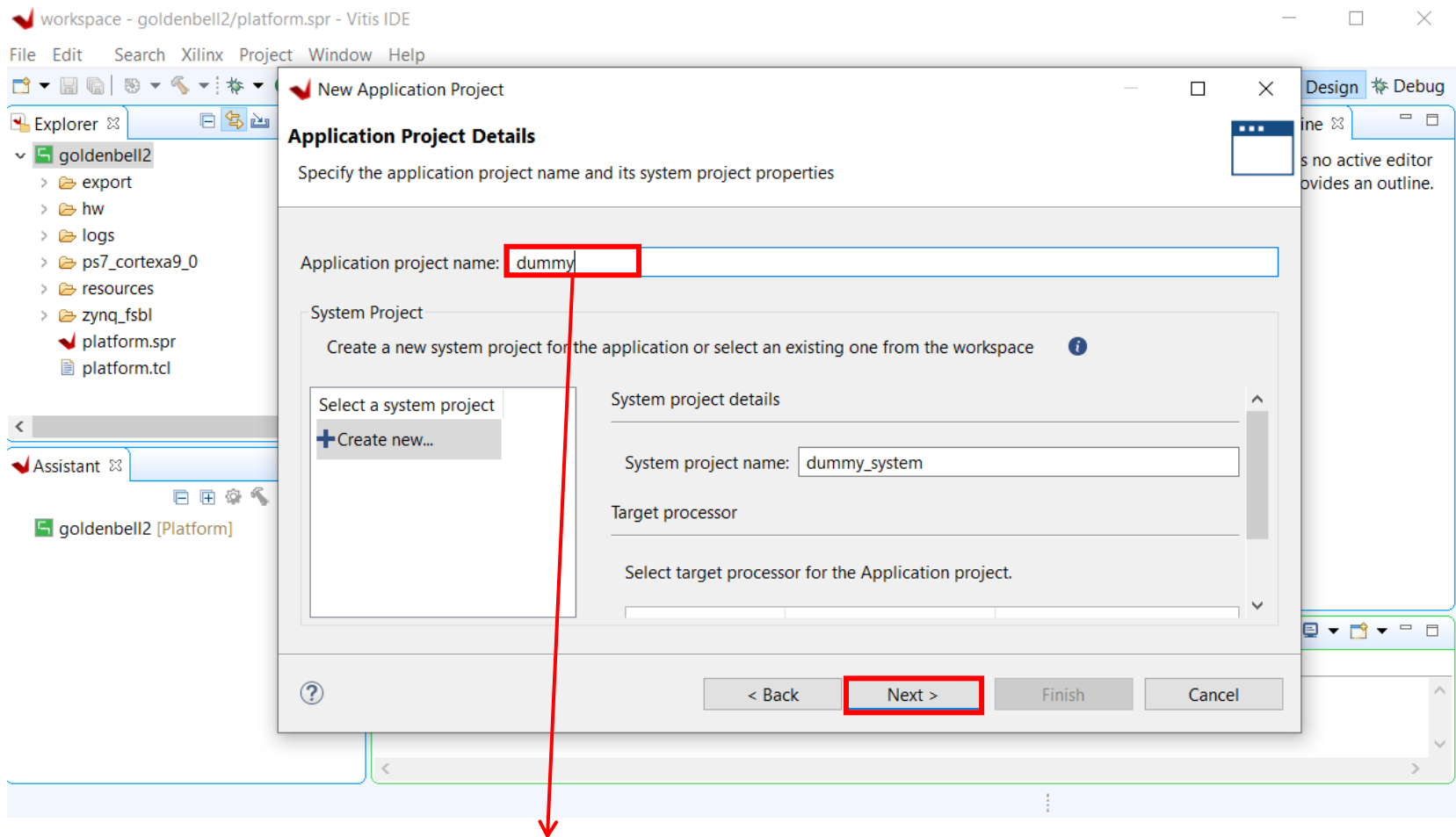
⑮ Vivado에서 export한 XSA 파일을 선택 후, Open 클릭



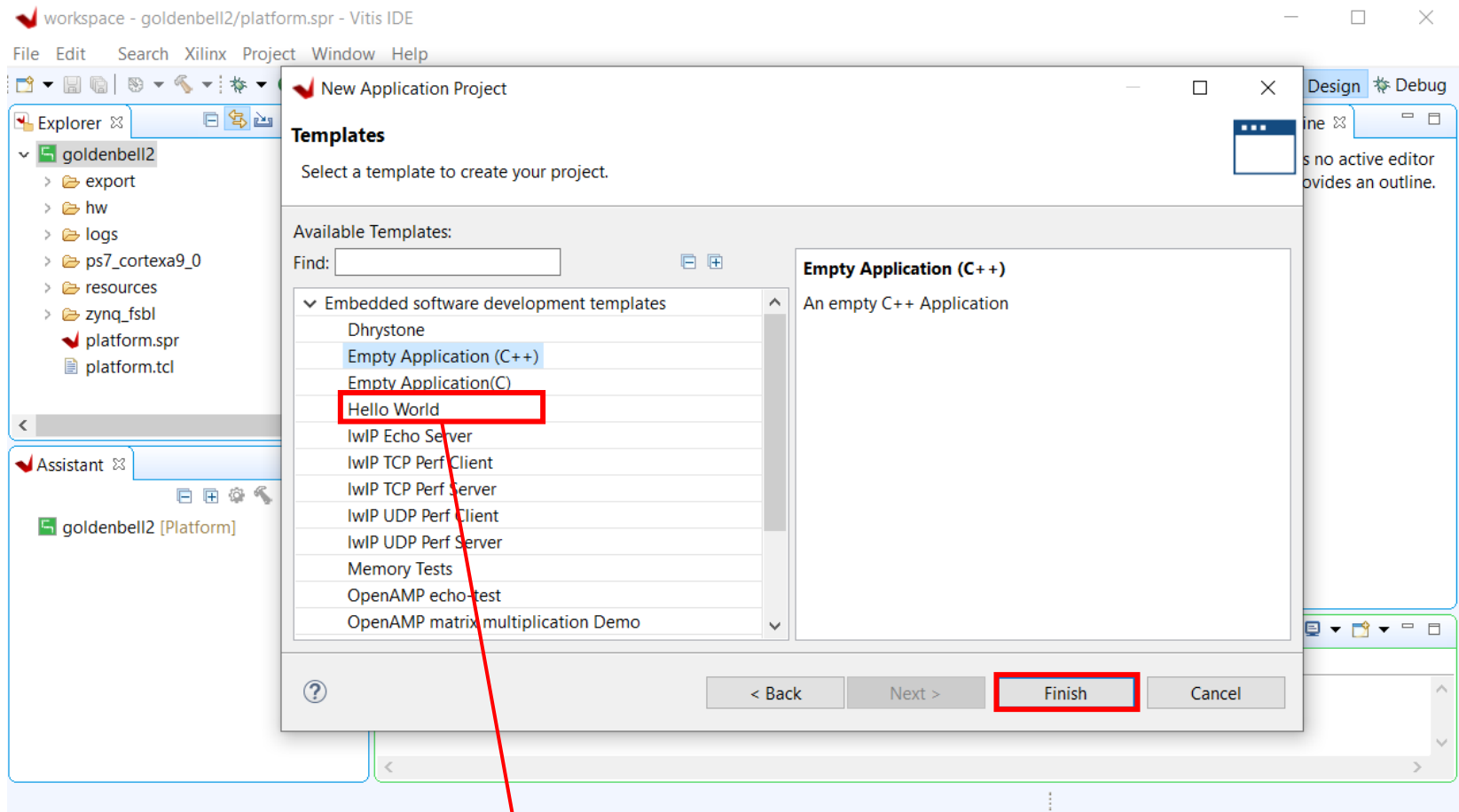
⑩ Vivado에서 export한 XSA 파일을 선택 후, Finish 클릭



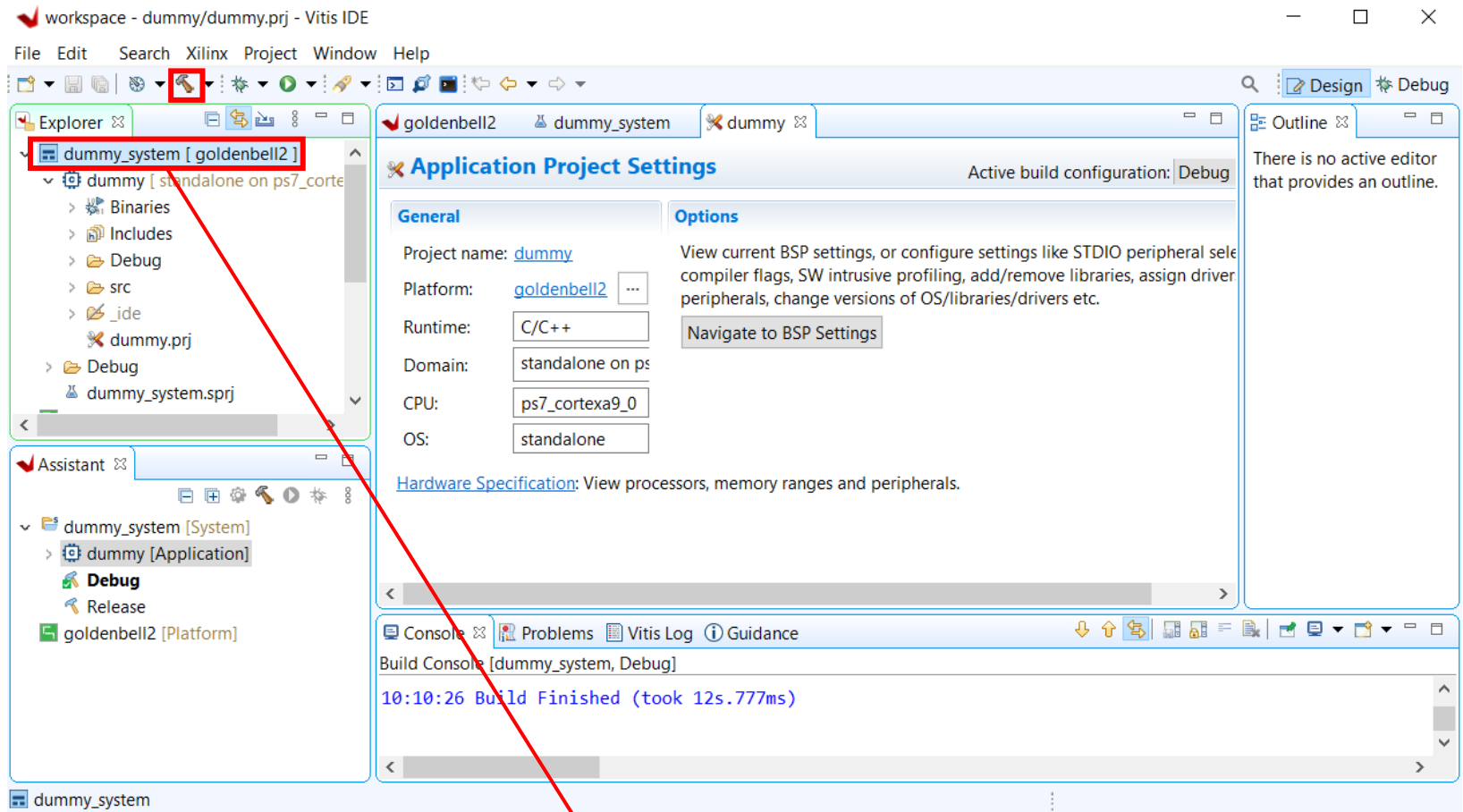
⑰ New -> Application Project 클릭



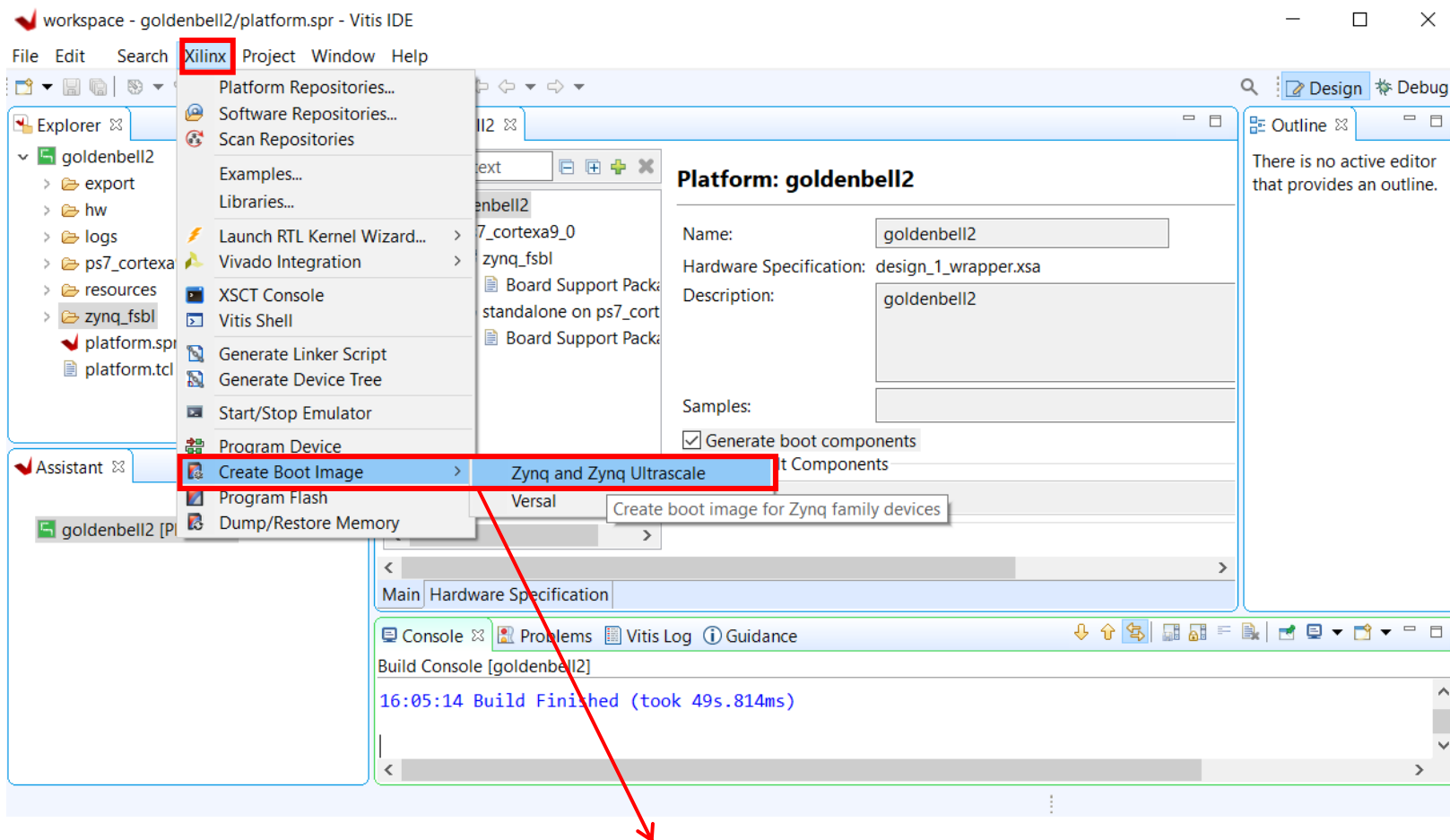
⑱ application name 입력 후, Next 클릭



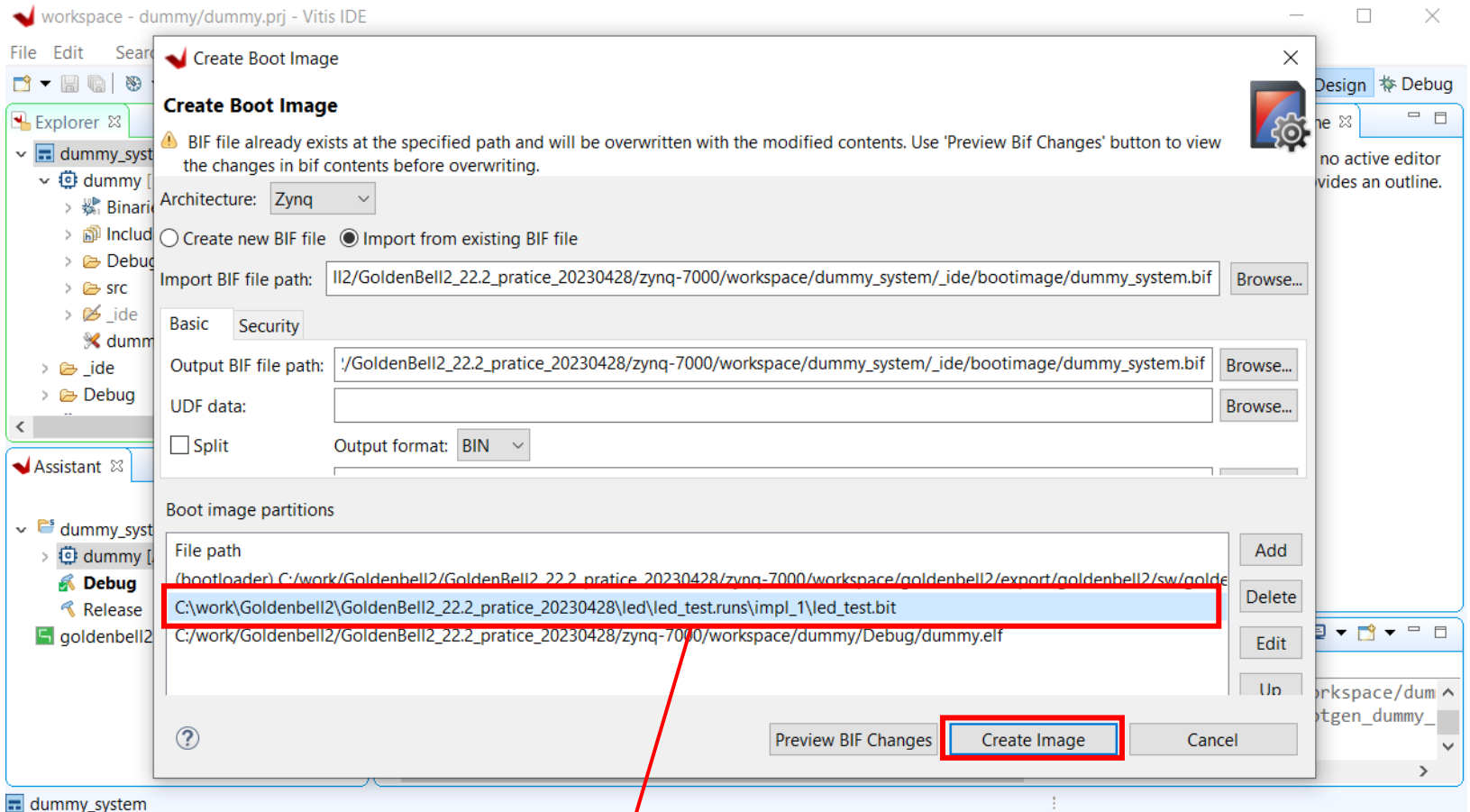
⑰ Hello World 선택 후, Finish 클릭



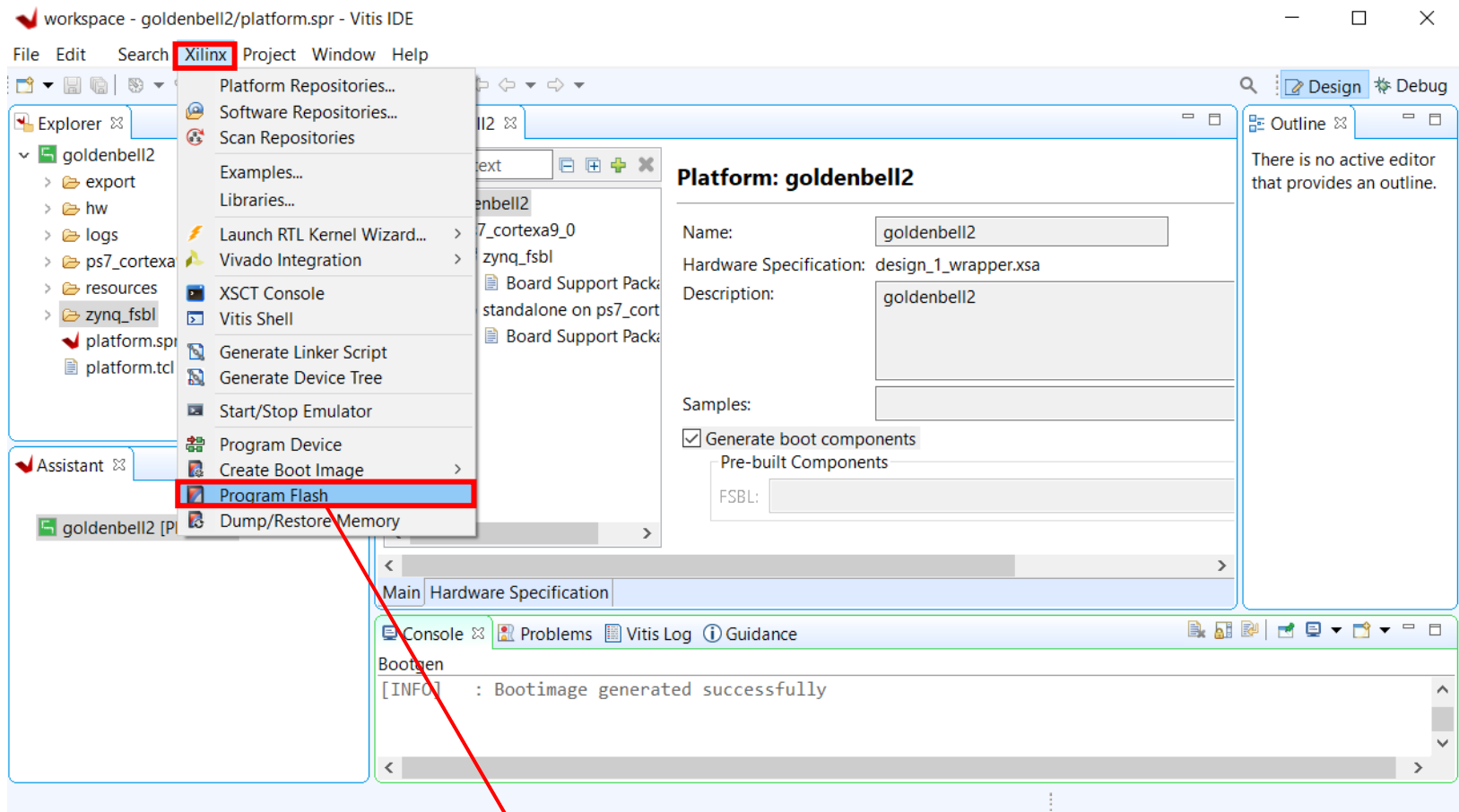
②① dummy_system 클릭 후, 빌드 아이콘 클릭



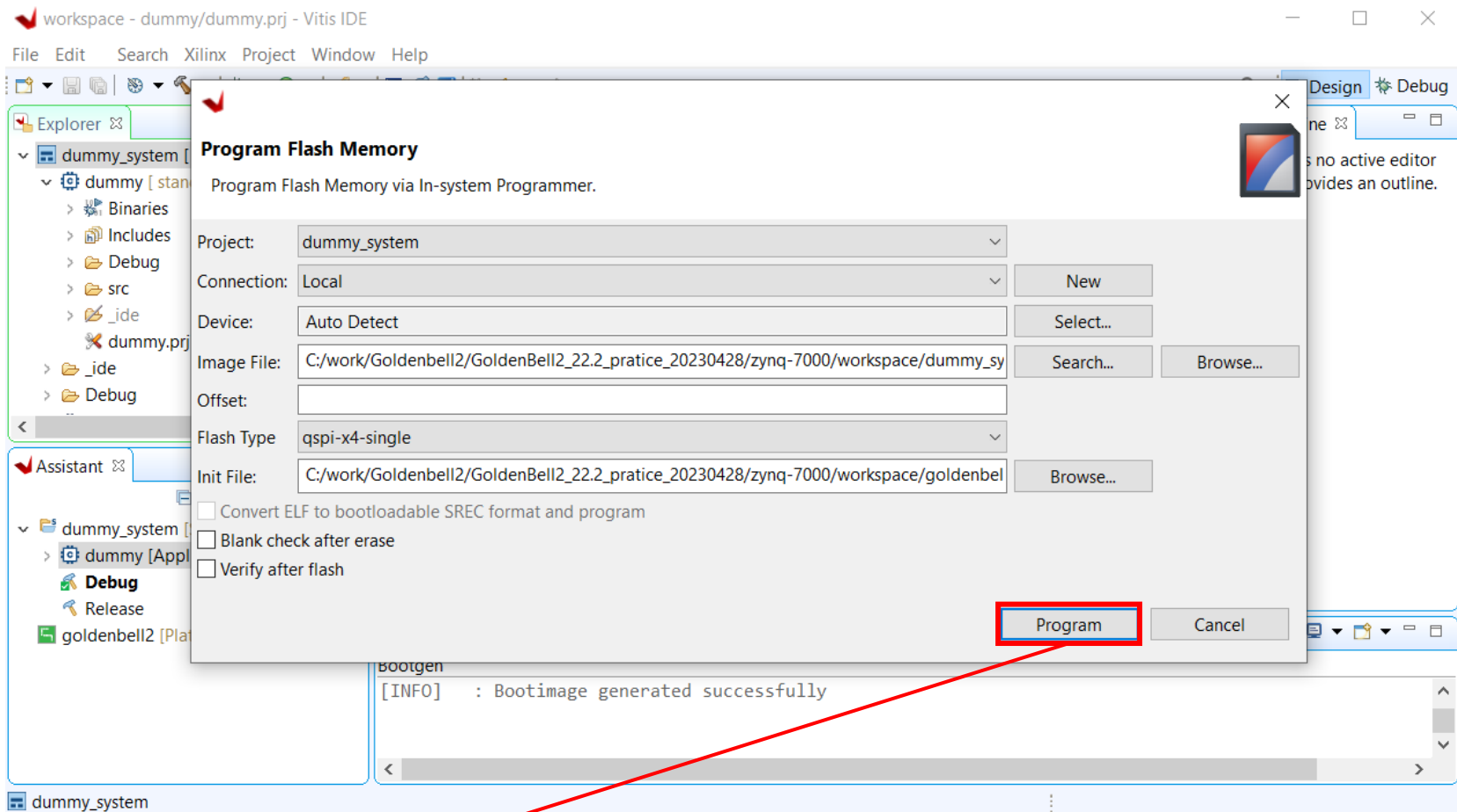
(21) Xilinx -> Create Boot Image -> Zynq and Zynq Ultrascale 클릭



(22) design_1_wrapper.bit를 삭제하고 led_test.bit 을 추가한 후, Create Image 클릭

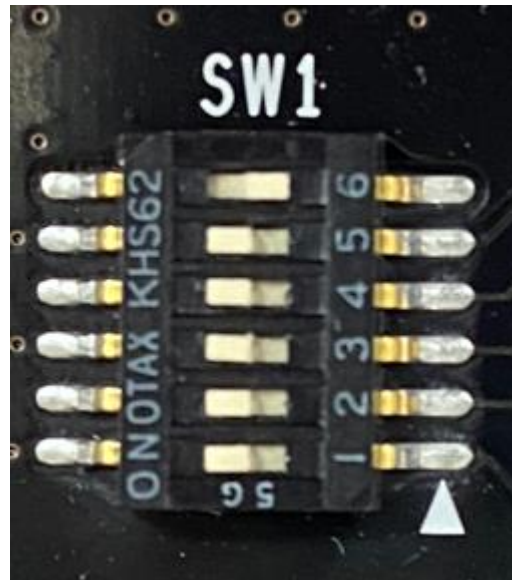


(23) Xilinx -> Program Flash 클릭



(24) Program 클릭

SPI Flash Boot Switch(SW1) 설정

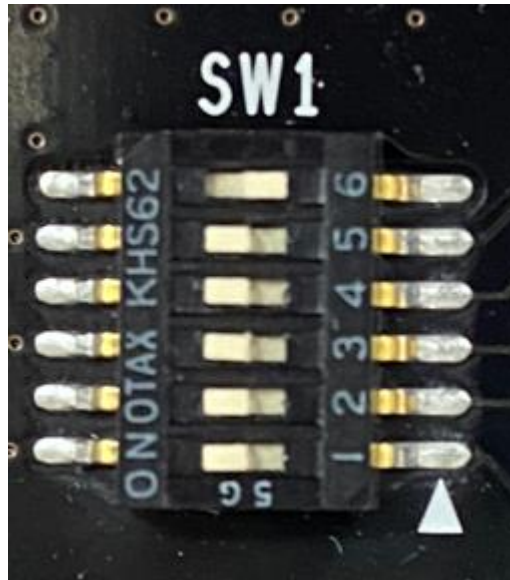


SPI Flash로 부팅 시 BOOT Switch 설정

1-2-3-4-5-6

ON-ON-ON-OFF-ON-OFF

SD card Boot Switch(SW1) 설정



BOOT.BIN을 SD card에 복사 후 BOOT Switch 설정
1-2-3-4-5-6
ON-ON-OFF-OFF-ON-OFF

첨부자료

- Example_test_project
 - LED 예제: led_test
 - FND 예제: fnd_test
 - KEY 예제: key_test
- Vivado 실습 소스 위치
 - Google Drive : [vivado source#1](#)
 - Github : [vivado source#2](#)

Linux 동작 실습

- Xilinx ZYNQ-7000 XC7Z015CLG485-2 상에서 Petalinux 동작 실습
- Host
 - Ubuntu 18.04 LTS

Petalinux 설치

- Petalinux 설치
 - Petalinux installer 2022.2 버전 다운로드
 - <https://www.xilinx.com/support/download/index.html/content/xilinx/en/downloadNav/embedded-design-tools.html>
- Petalinux Tool 인스톨
 - `$ sudo -u <user_name> mkdir -p /opt/pkg/petalinux-v2022.2`
 - `$ ./petalinux-v2022.2-final-installer.run /opt/pkg/petalinux-v2022.2`
- Petalinux 환경 설정
 - `source /opt/pkg/petalinux-v2022.2/settings.sh`

Petalinux 프로젝트 생성

- GoldenBell2 BSP 복사
 - [goldenbell2_20222_20230321.bsp](#)
- GoldenBell2 BSP 기반 프로젝트 생성
 - \$ petalinux-create -t project -s goldenbell2_20222_20230321.bsp -n goldenbell2
- 생성된 프로젝트 폴더로 이동
 - \$ cd goldenbell2
- HW description file의 경로를 설정
 - \$ petalinux-config --get-hw-description=./hardware/GoldenBell2_22.2_pcie_20230321/workspace

Petalinux 프로젝트 빌드

- SD 부트용 이미지 빌드
 - \$ cd project-spec/build
 - \$./build.sh
- 빌드 결과물 저장 디렉토리
 - project-spec/images/image

SD Card Fusing

- Host에 SD card 연결 후 부팅 SD card fusing
 - \$ cd project-spec/images
 - \$ sudo ./create-sdcard.sh
- SD card 부팅을 위한 BOOT Switch(SW1) 설정
 - 6-5-4-3-2-1
 - OFF-ON-OFF-OFF-ON-ON
- SD card 슬롯에 삽입 후 전원 인가
 - USB Type B to Type A를 J9로 연결
 - User id : petalinux

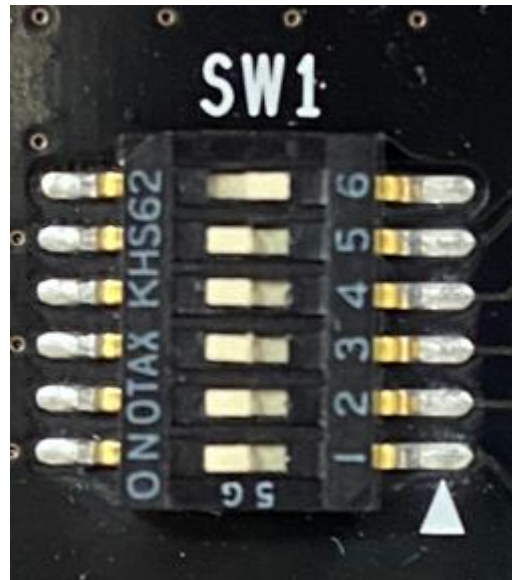
Peripheral 검증

- 링크 문서 참조 Peripheral 검증
 - [Petalinux_GoldenBell2_설정.docx](#)
 - 1Gbps Ethernet
 - 16MBytes QSPI flash
 - USB device
 - FND
 - LED
 - Key
 - PCIe

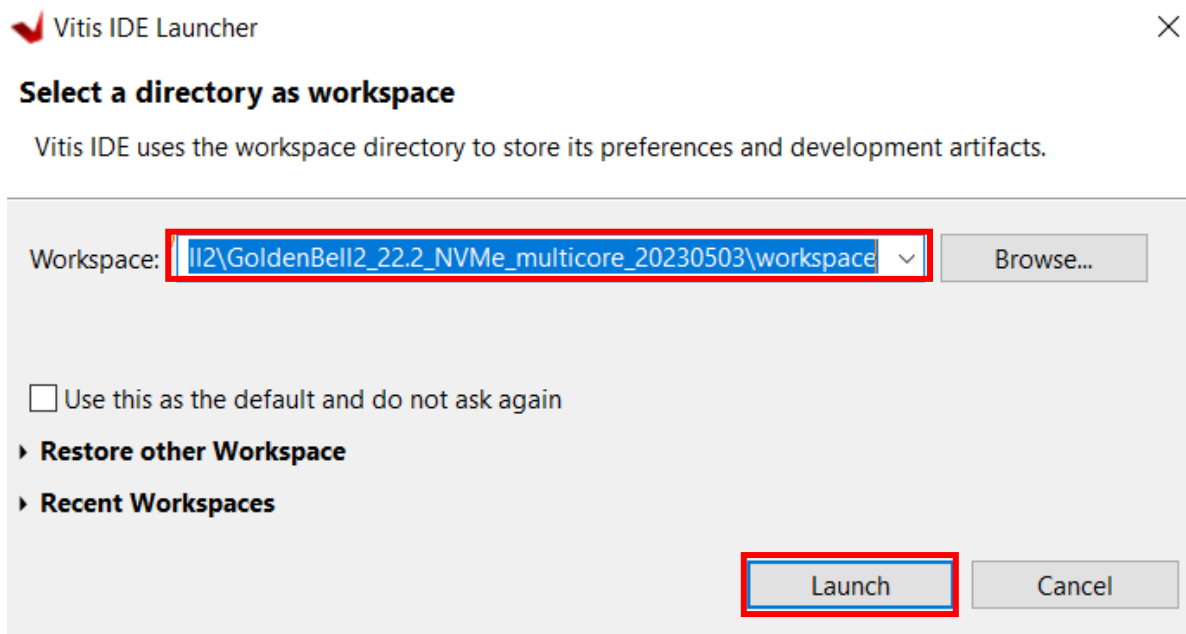
NVMe SSD 실습

- GoldenBell2 – PCIe Gen2 x4 lane 지원
- NVMe SSD 설정
 - GoldenBell2에 SD card를 삽입 후 SSD storage(16GB)로 이용
 - BIN 파일을 QSPI flash에 저장 후 QSPI BOOT
 - Host PC의 PCIe 슬롯에 GoldenBell2 보드 장착
- Vivado 실습 소스 위치
 - Google Drive : [vivado source#1](#)
 - Github : [vivado source#2](#)

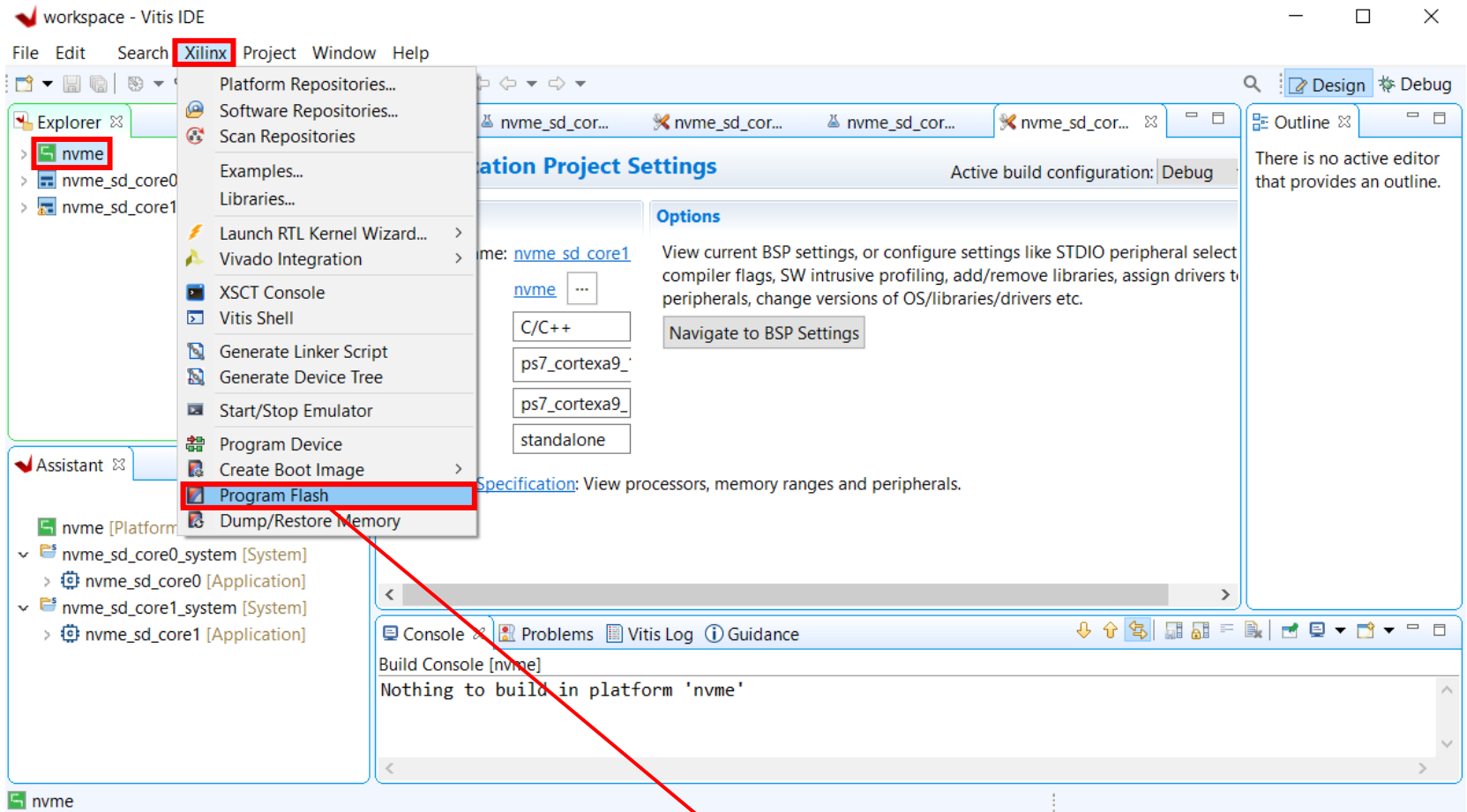
JTAG Boot Switch(SW1) 설정



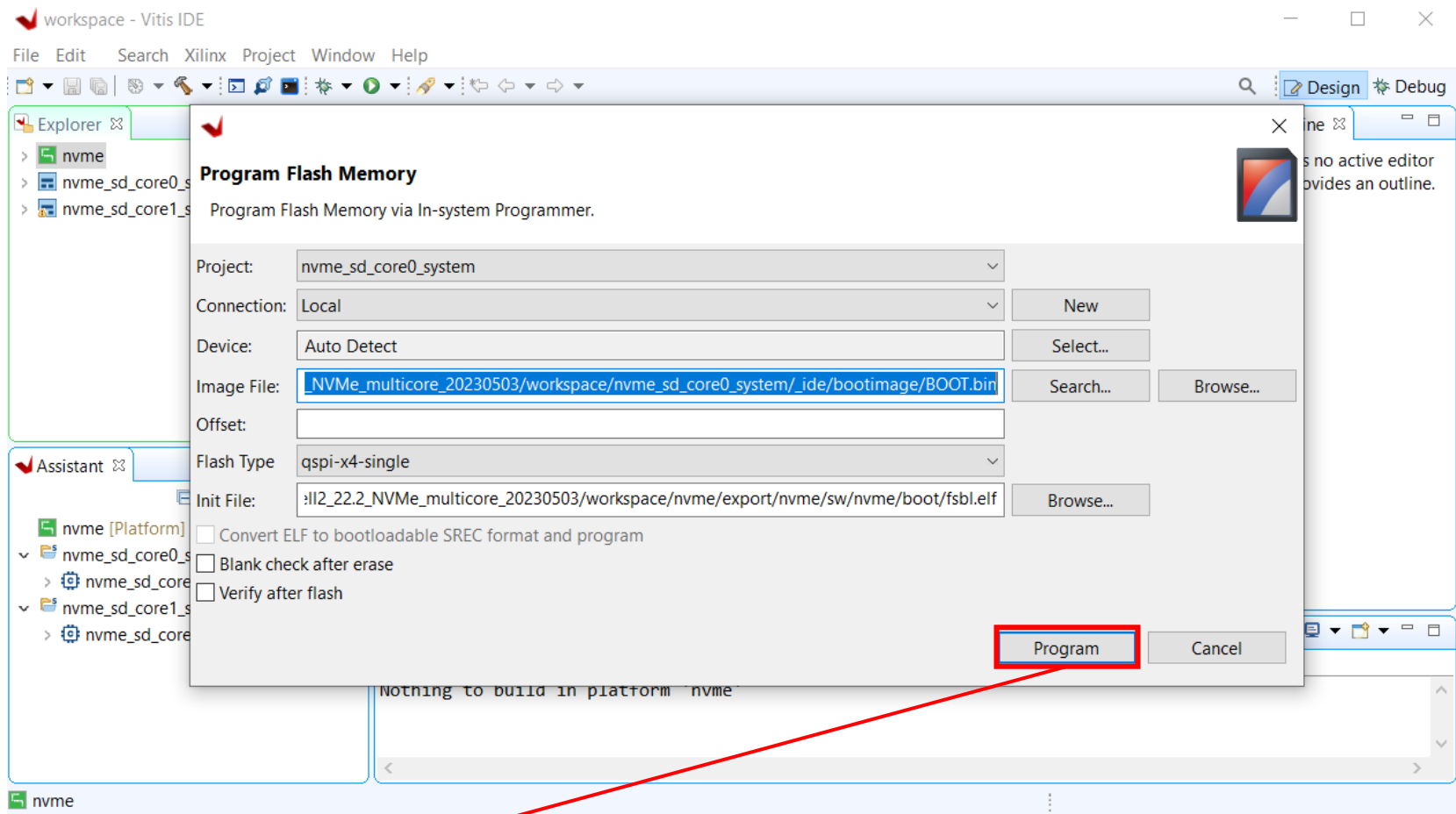
1-2-3-4-5-6
ON-ON-ON-ON-ON-OFF



(1) Vitis 실행 후 실습 소스의 workspace 지정한 다음 Launch 클릭

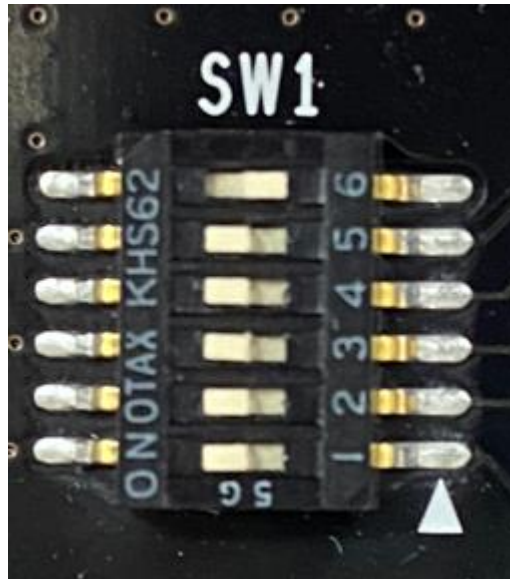


(2) nvme platform project 클릭 후, Xilinx -> Program Flash 클릭



(3) Program 클릭

SPI Flash Boot Switch(SW1) 설정



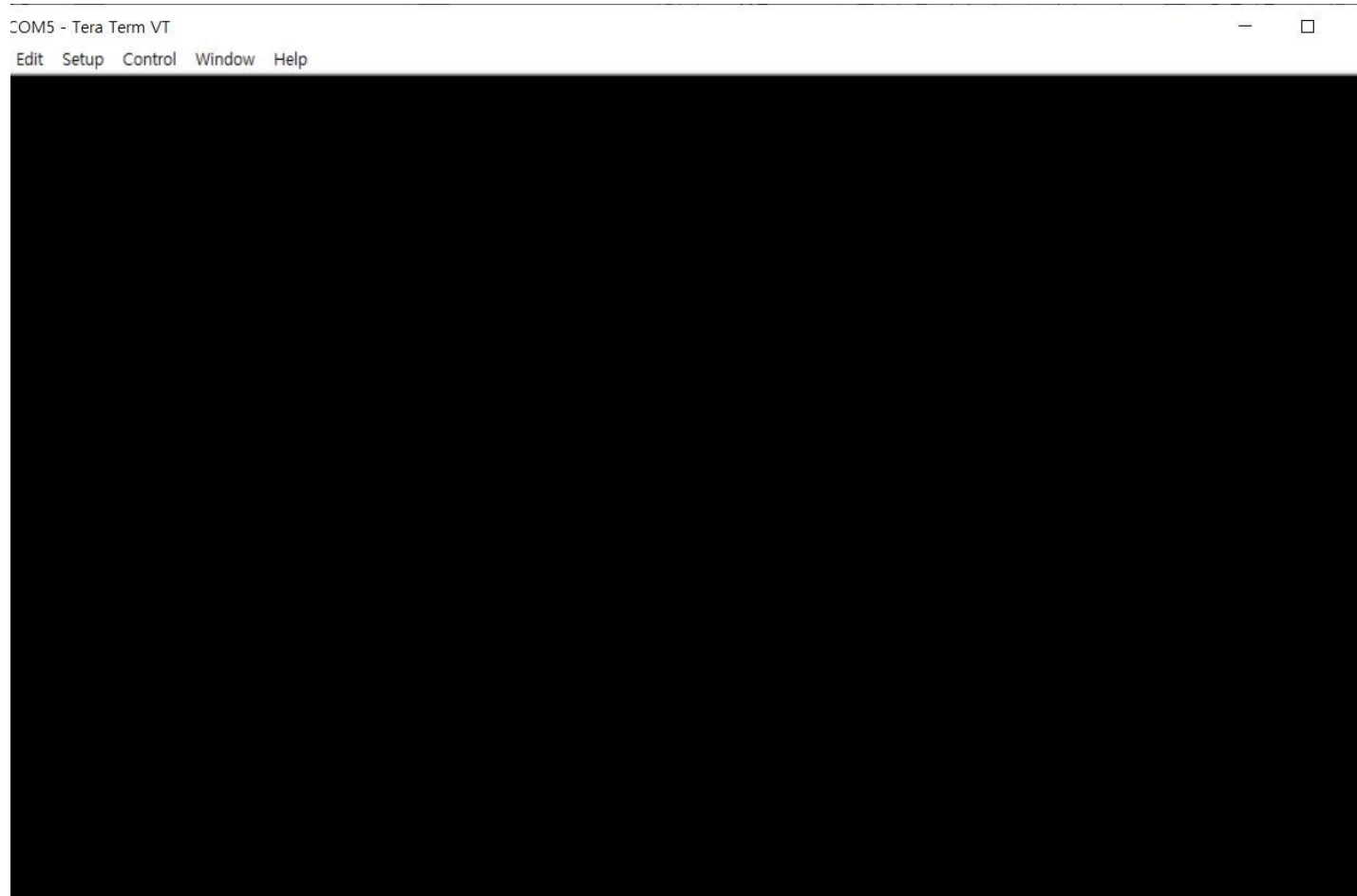
SPI Flash로 부팅 시 BOOT Switch 설정

1-2-3-4-5-6

ON-ON-ON-OFF-ON-OFF

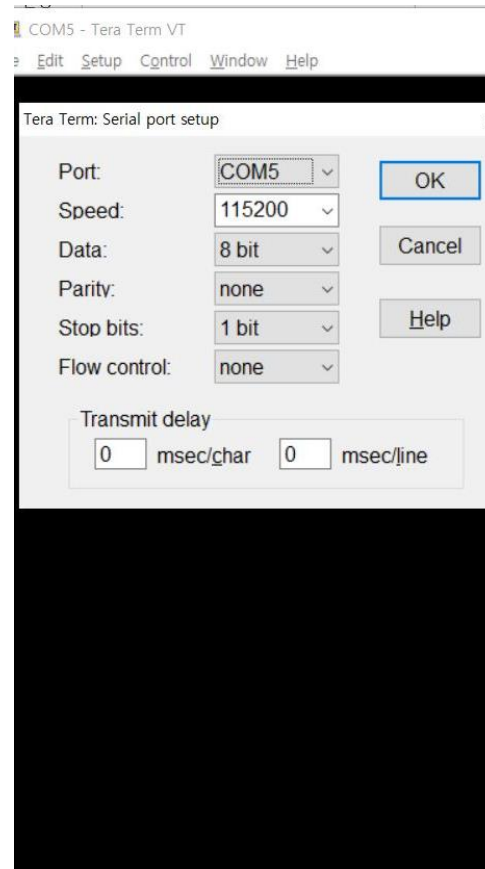
Configure UART

- Open terminal program.



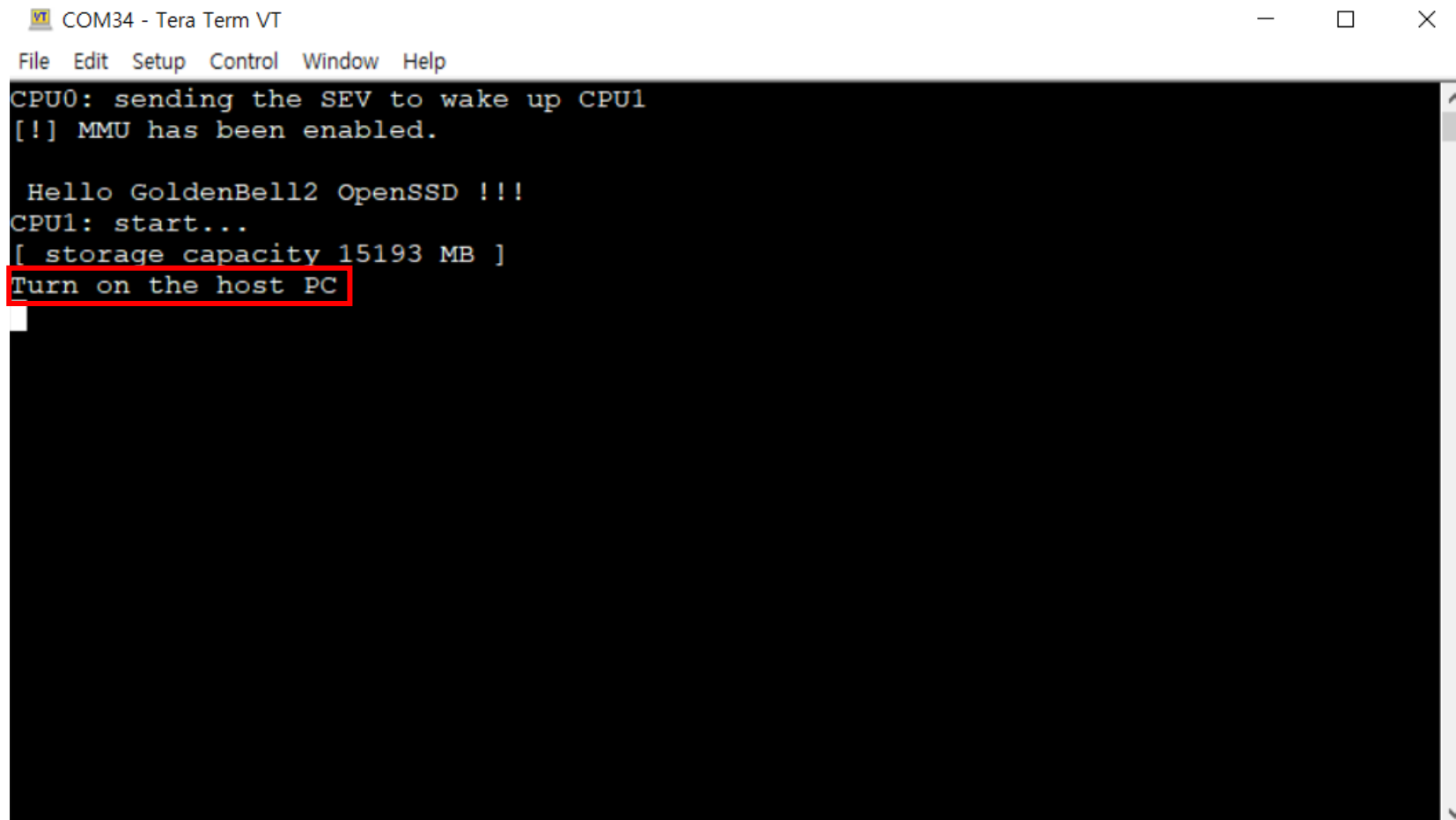
Configure UART

- On Terminal program, set “Port” and “Baud Rate” to 115200, respectively



Configure UART

- Power on GoldenBell2. Check boot message.



```
COM34 - Tera Term VT
File Edit Setup Control Window Help
CPU0: sending the SEV to wake up CPU1
[!] MMU has been enabled.

Hello GoldenBell2 OpenSSD !!!
CPU1: start...
[ storage capacity 15193 MB ]
Turn on the host PC
```

Power on Host PC

- GoldenBell2 보드를 Host PC에 장착
- Host PC 전원 인가

UART Messages While Host Computer is Booting up

- NVMe SSD initialization steps are on going

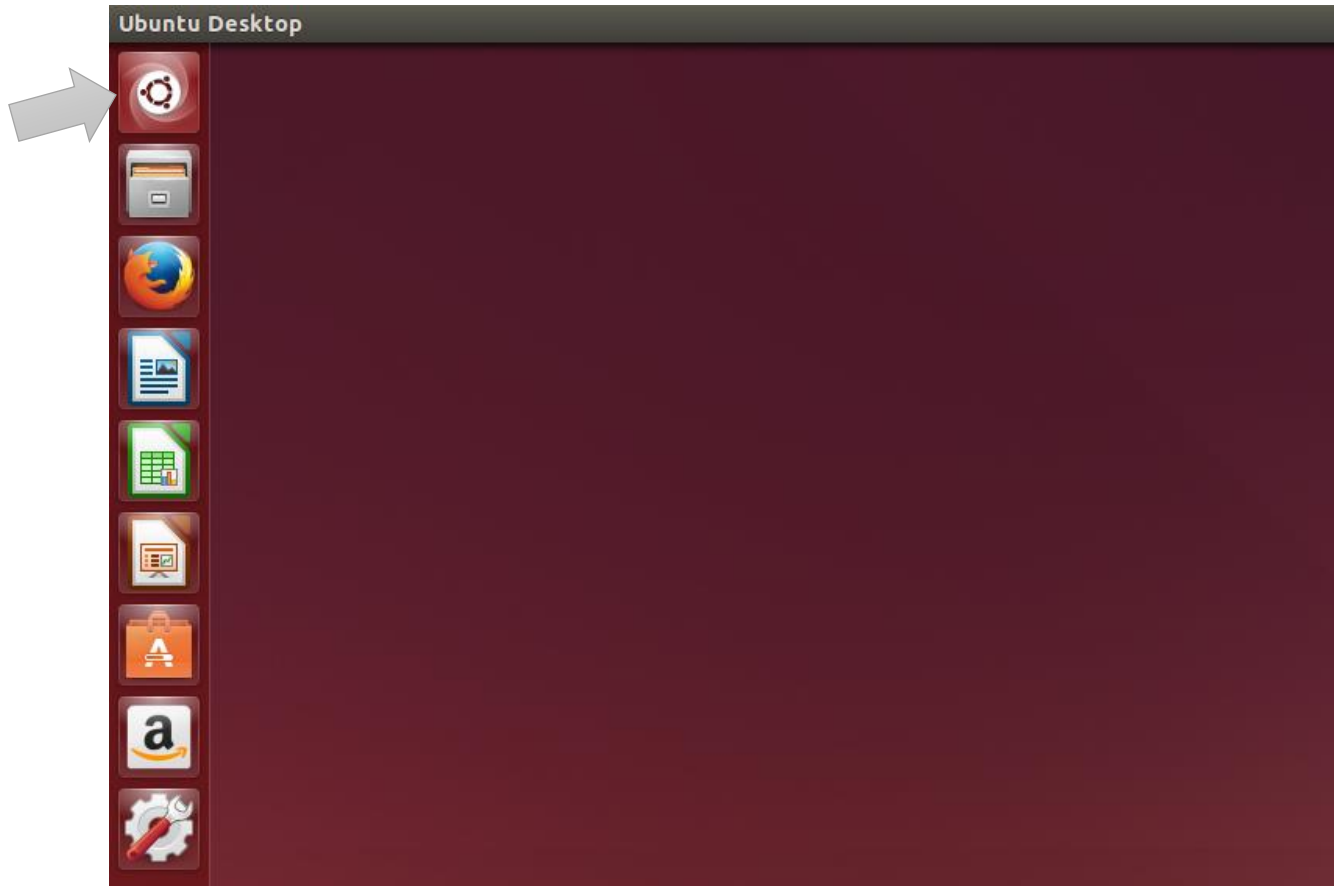
```
FTL reset complete!!!  
Turn on the host PC  
PCIe Link: 1  
PCIe Bus Master: 1  
PCIe Bus Master: 0  
PCIe Bus Master: 1  
NVME CC.EN: 1  
  
NVMe ready!!!  
Create IO CQ, DW11: 0x00000001, DW10: 0x00010001  
Create IO SQ, DW11: 0x00010001, DW10: 0x00010001  
PCIe IRQ Disable: 0  
PCIe IRQ Disable: 1  
PCIe IRQ Disable: 0  
PCIe MSI Enable: 1, 0x0  
NVME CC.EN: 0  
  
NVMe disable!!!
```

Operating GoldenBell2 OpenSSD(Linux)

1. Check device recognition
2. Create a partition
3. Check the created partition
4. Format the partition
5. Create a mount point
6. Mount the partition
7. Check the mounted partition

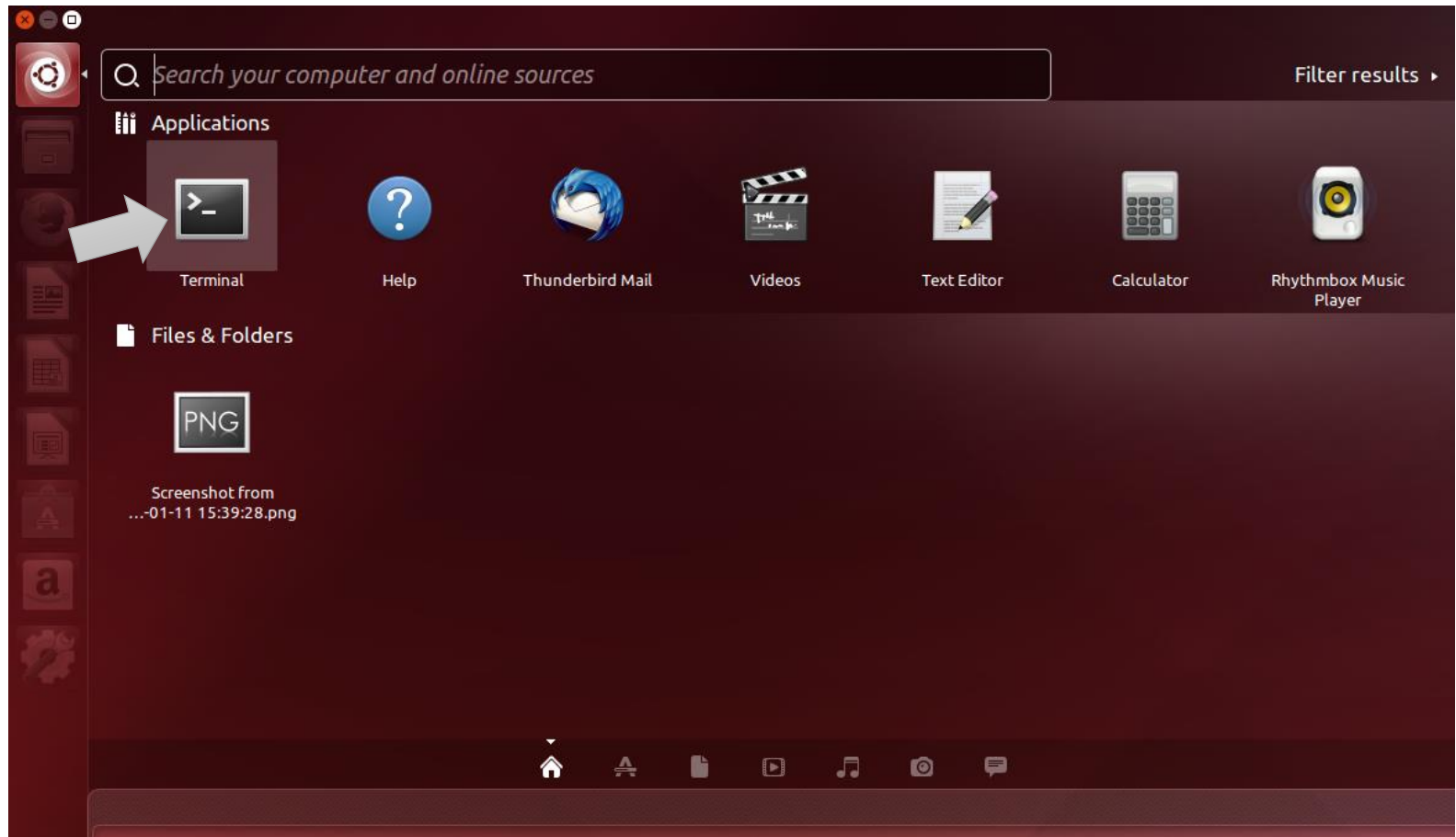
Open a Terminal (1 / 2)

- Click the pointed icon



Open a Terminal (2 / 2)

- Click the terminal icon



Check Device Recognition (1 / 2)

- Types “lspci” -> press ENTER -> check “Non-Volatile memory controller: Xilinx Corporation Device 7028” on the PCI device list

```
hgchoe@ubuntu:~$ lspci
00:00.0 Host bridge: Intel Corporation Xeon E3-1200 v6/7th Gen Core Processor Host Bridge/DRAM Registers (rev 05)
00:01.0 PCI bridge: Intel Corporation Xeon E3-1200 v5/E3-1500 v5/6th Gen Core Processor PCIe Controller (x16) (rev 05)
00:02.0 VGA compatible controller: Intel Corporation HD Graphics 630 (rev 04)
00:14.0 USB controller: Intel Corporation 200 Series/Z370 Chipset Family USB 3.0 xHCI Controller
00:14.2 Signal processing controller: Intel Corporation 200 Series PCH Thermal Subsystem
00:16.0 Communication controller: Intel Corporation 200 Series PCH CSME HECI #1
00:17.0 SATA controller: Intel Corporation 200 Series PCH SATA controller [AHCI mode]
00:1c.0 PCI bridge: Intel Corporation 200 Series PCH PCI Express Root Port #5 (rev f0)
00:1f.0 ISA bridge: Intel Corporation 200 Series PCH LPC Controller (H270)
00:1f.2 Memory controller: Intel Corporation 200 Series/Z370 Chipset Family Power Management Controller
00:1f.3 Audio device: Intel Corporation 200 Series PCH HD Audio
00:1f.4 SMBus: Intel Corporation 200 Series/Z370 Chipset Family SMBus Controller
01:00.0 Non-Volatile memory controller: Xilinx Corporation Device 7028
02:00.0 Ethernet controller: Realtek Semiconductor Co., Ltd. RTL8111/8168/8411 PCI Express Gigabit Ethernet Controller (rev 15)
hgchoe@ubuntu:~$
```

Check Device Recognition (2 / 2)

- Types “ls /dev” -> press ENTER -> check “nvme0n1” on the device list

```
hgchoe@ubuntu:~$ ls /dev
autofs          loop-control    loop5            pts
block           fd              loop6            random
bsg             full            loop7            rfkill
btrfs-control   fuse            loop8            rtc
bus             hidraw0         loop9            rtc0
cdrom           hidraw1         mapper           sda
cdrw            hidraw2         mcelog           sda1
char            hpet            mei0             sda2
console         hugepages       mem              sda3
core            hwrng           mqueue           sda4
cpu_dma_latency i2c-0           net              sda5
cuse            i2c-1           ng0n1            sdb
disk            i2c-2           null             sdb1
dma_heap        i2c-3           nvme0            sdb2
dri             i2c-4           nvme0n1          sdb3
drm_dp_aux0     initctl         nvram            sg0
drm_dp_aux1     input           port             sg1
dvd            kmsg           ppp              sg2
dvdrw          kvm            psaux            shm
ecryptfs        log            ptmx             snapshot
hgchoe@ubuntu:~$
```

Create a Partition

- Type “sudo fdisk /dev/nvme0n1”, press ENTER -> type your password, press ENTER -> type “n”, press ENTER->press ENTER->press ENTER, press ENTER -> type “w”, press ENTER

```
hgchoe@ubuntu:~$ sudo fdisk /dev/nvme0n1

Welcome to fdisk (util-linux 2.31.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (256-3816959, default 256):
Last sector, +sectors or +size{K,M,G,T,P} (256-3816959, default 3816959):

Created a new partition 1 of type 'Linux' and of size 14.6 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

hgchoe@ubuntu:~$
```

Check the Created Partition

- Types “ls /dev” -> press ENTER -> check “nvme0n1p1” on the device list

```
hgchoe@ubuntu:~$ ls /dev
autofs      i2c-4      mem        sr0         tty34      tty8        ttyS9
block       initctl    mqueue     stderr      tty35      tty9        udmabuf
bsg         input      net        stdin       tty36      ttyprintk   uhid
btrfs-control kmsg      ng0n1      stdout      tty37      ttyS0       uinput
bus         kvm        null       tpm0        tty38      ttyS1       urandom
cdrom       log        nvme0      tty         tty39      ttyS10      usb
cdrw        loop0      nvme0n1    tty0        tty4       ttyS11      userio
char        loop1      nvme0n1p1  tty1        tty40      ttyS12      vcs
console     loop10     nvram      tty10       tty41      ttyS13      vcs1
core        loop11     port       tty11       tty42      ttyS14      vcs2
cpu_dma_latency loop12     ppp        tty12       tty43      ttyS15      vcs3
cuse        loop13     psaux      tty13       tty44      ttyS16      vcs4
disk        loop14     ptmx       tty14       tty45      ttyS17      vcs5
dma_heap    loop15     pts        tty15       tty46      ttyS18      vcs6
dri         loop16     random     tty16       tty47      ttyS19      vcsa
drm_dp_aux0 loop17     rfkill     tty17       tty48      ttyS2       vcsa1
drm_dp_aux1 loop18     rtc        tty18       tty49      ttyS20      vcsa2
```


Format the Partition

- Type “`sudo mkfs -t ext4 /dev/nvme0n1p1`”, press ENTER

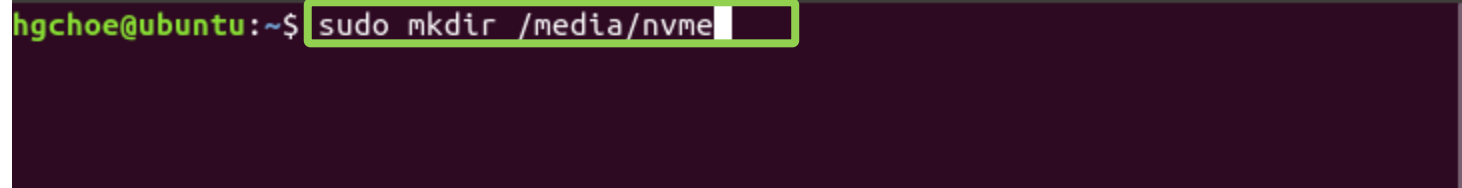
```
hgchoe@ubuntu:~$ sudo mkfs -t ext4 /dev/nvme0n1p1
mke2fs 1.44.1 (24-Mar-2018)
Creating filesystem with 60365568 4k blocks and 15097856 inodes
Filesystem UUID: 4c5f1175-5c68-459d-b516-558811bc19ca
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
    4096000, 7962624, 11239424, 20480000, 23887872

Allocating group tables: done
Writing inode tables: done
Creating journal (262144 blocks): done
Writing superblocks and filesystem accounting information: done

hgchoe@ubuntu:~$
```

Create a Mount Point

- Type “sudo mkdir /media/nvme”, press ENTER

A terminal window with a dark purple background. The prompt 'hgchoe@ubuntu:~\$' is in green. The command 'sudo mkdir /media/nvme' is being typed in white, with a white cursor at the end. The command is highlighted by a green rectangular box.

```
hgchoe@ubuntu:~$ sudo mkdir /media/nvme
```

Mount the Partition

- Type “sudo mount /dev/nvme0n1p1 /media/nvme”, press ENTER

```
hgchoe@ubuntu:~$ sudo mount /dev/nvme0n1p1 /media/nvme
hgchoe@ubuntu:~$
```

Check the Mounted Partition (1 / 2)

- Type “lsblk”, press ENTER -> check the mounted partition on the block device list

```
hgchoe@ubuntu:~$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
loop0        7:0      0    1.5M  1 loop /snap/gnome-system-monitor/181
loop1        7:1      0    73M   1 loop /snap/core22/607
loop2        7:2      0    2.6M   1 loop /snap/gnome-system-monitor/178
loop3        7:3      0   81.3M  1 loop /snap/gtk-common-themes/1534
loop4        7:4      0  218.4M  1 loop /snap/gnome-3-34-1804/90
loop5        7:5      0     2M   1 loop /snap/gnome-calculator/926
loop6        7:6      0   55.6M  1 loop /snap/core18/2714
loop7        7:7      0  218.4M  1 loop /snap/gnome-3-34-1804/93
loop8        7:8      0   63.3M  1 loop /snap/core20/1852
loop9        7:9      0   53.2M  1 loop /snap/snapd/18933
loop10       7:10     0     4K   1 loop /snap/bare/5
loop11       7:11     0  349.7M  1 loop /snap/gnome-3-38-2004/137
loop12       7:12     0    508K   1 loop /snap/gnome-characters/785
loop13       7:13     0  460.6M  1 loop /snap/gnome-42-2204/99
loop14       7:14     0   63.3M  1 loop /snap/core20/1828
loop16       7:16     0  260.7M  1 loop /snap/kde-frameworks-5-core18/32
loop17       7:17     0  349.7M  1 loop /snap/gnome-3-38-2004/140
loop18       7:18     0  107.5M  1 loop /snap/kdiskmark/88
loop19       7:19     0  460.6M  1 loop /snap/gnome-42-2204/87
loop20       7:20     0  107.5M  1 loop /snap/kdiskmark/83
loop21       7:21     0     2M   1 loop /snap/gnome-calculator/934
loop22       7:22     0   49.9M  1 loop /snap/snapd/18596
loop23       7:23     0   91.7M  1 loop /snap/gtk-common-themes/1535
loop24       7:24     0   55.6M  1 loop /snap/core18/2721
loop25       7:25     0    556K  1 loop /snap/gnome-logs/112
loop26       7:26     0  289.8M  1 loop /snap/kde-frameworks-5-core18/35
loop27       7:27     0   696K   1 loop /snap/gnome-logs/115
loop28       7:28     0   476K   1 loop /snap/gnome-characters/781
loop29       7:29     0    73M   1 loop /snap/core22/617
sda          8:0      0  953.9G  0 disk
├─sda1        8:1      0   499M   0 part
├─sda2        8:2      0   100M   0 part /boot/efi
├─sda3        8:3      0    16M   0 part
├─sda4        8:4      0  952.7G  0 part
└─sda5        8:5      0   568M   0 part
sdb          8:16     0  931.5G  0 disk
├─sdb1        8:17     0   512M   0 part
├─sdb2        8:18     0  899.1G  0 part /
└─sdb3        8:19     0   31.9G  0 part
scd          11:0     1   1024M  0 rom
nvme0n1      259:0    0   14.6G  0 disk
└─nvme0n1p1  259:2    0   14.6G  0 part /media/nvme
hgchoe@ubuntu:~$
```

Check the Mounted Partition (2 / 2)

- Type “df -h”, press ENTER -> check the mounted partition on the storage list

```
hgchoe@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            16G   0    16G   0% /dev
tmpfs           3.2G  3.6M  3.2G   1% /run
/dev/sdb2       884G  343G  497G  41% /
tmpfs           16G   0    16G   0% /dev/shm
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
tmpfs           16G   0    16G   0% /sys/fs/cgroup
/dev/loop1       73M   73M   0 100% /snap/core22/607
/dev/loop2       2.7M  2.7M   0 100% /snap/gnome-system-monitor/178
/dev/loop0       1.5M  1.5M   0 100% /snap/gnome-system-monitor/181
/dev/loop4       219M  219M   0 100% /snap/gnome-3-34-1804/90
/dev/loop3       82M   82M   0 100% /snap/gtk-common-themes/1534
/dev/loop5       2.2M  2.2M   0 100% /snap/gnome-calculator/926
/dev/loop6       56M   56M   0 100% /snap/core18/2714
/dev/loop7       219M  219M   0 100% /snap/gnome-3-34-1804/93
/dev/loop12      512K  512K   0 100% /snap/gnome-characters/785
/dev/loop9       54M   54M   0 100% /snap/snapd/18933
/dev/loop13      461M  461M   0 100% /snap/gnome-42-2204/99
/dev/loop10      128K  128K   0 100% /snap/bare/5
/dev/loop8       64M   64M   0 100% /snap/core20/1852
/dev/loop11      350M  350M   0 100% /snap/gnome-3-38-2004/137
/dev/loop14       64M   64M   0 100% /snap/core20/1828
/dev/loop21       2.2M  2.2M   0 100% /snap/gnome-calculator/934
/dev/loop16      261M  261M   0 100% /snap/kde-frameworks-5-core18/32
/dev/loop20      108M  108M   0 100% /snap/kdiskmark/83
/dev/loop17      350M  350M   0 100% /snap/gnome-3-38-2004/140
/dev/loop23       92M   92M   0 100% /snap/gtk-common-themes/1535
/dev/loop19      461M  461M   0 100% /snap/gnome-42-2204/87
/dev/loop22       50M   50M   0 100% /snap/snapd/18596
/dev/loop25      640K  640K   0 100% /snap/gnome-logs/112
/dev/loop24       56M   56M   0 100% /snap/core18/2721
/dev/loop26      290M  290M   0 100% /snap/kde-frameworks-5-core18/35
/dev/loop28      512K  512K   0 100% /snap/gnome-characters/781
/dev/loop27      768K  768K   0 100% /snap/gnome-logs/115
/dev/loop18      108M  108M   0 100% /snap/kdiskmark/88
/dev/sda2        96M   53M   44M  56% /boot/efi
tmpfs           3.2G  16K   3.2G   1% /run/user/121
tmpfs           3.2G  1.9M   3.2G   1% /run/user/1000
/dev/loop29       73M   73M   0 100% /snap/core22/617
/dev/nvme0n1p1   15G   24K   14G   1% /media/nvme
```

Operating Daisy+ OpenSSD (Windows)

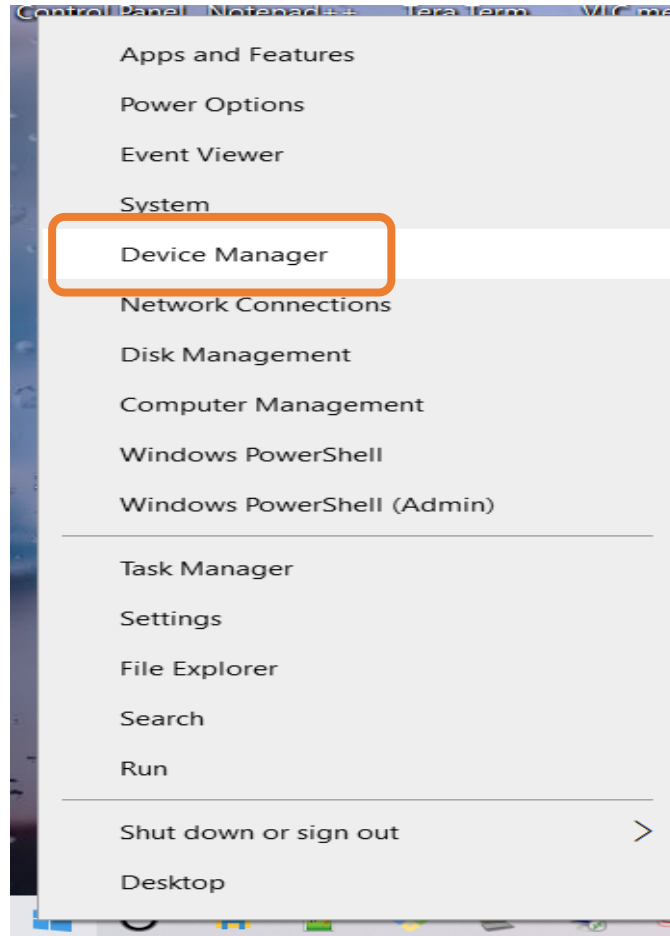
1. Check device recognition

2. Create a partition

3. Format the partition

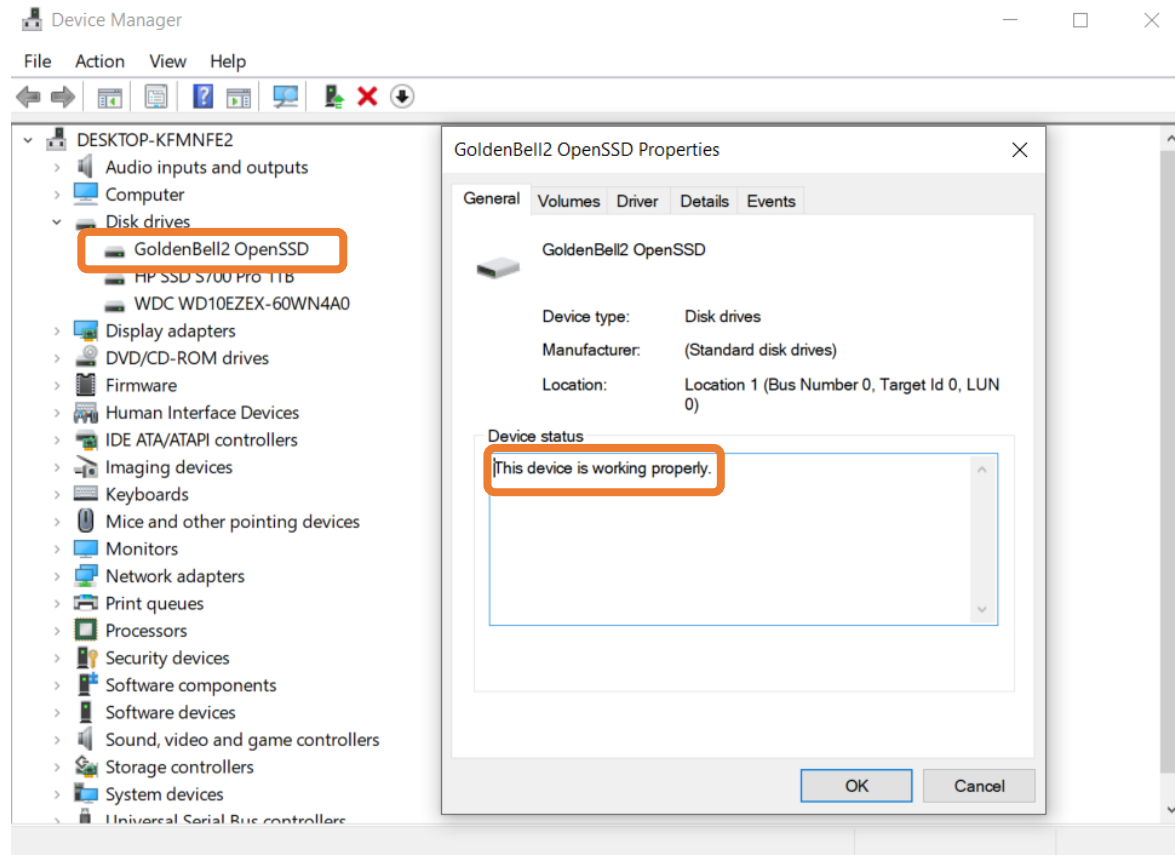
Check Device Recognition (1 / 2)

- Right-click “Start” menu → click “Device Manager”



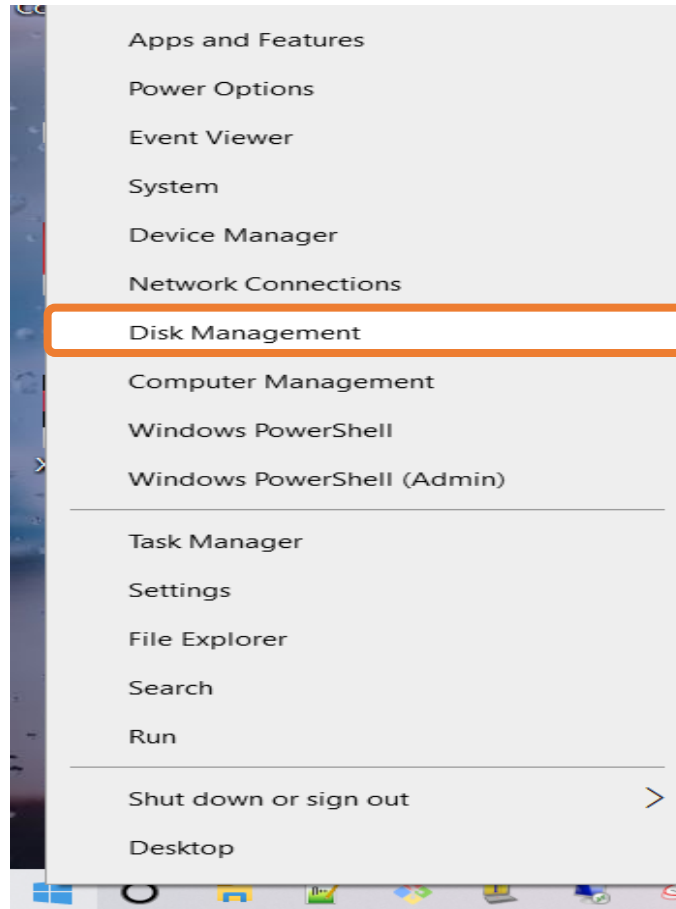
Check Device Recognition (2 / 2)

- Disk drives → double-click “DaisyPlus OpenSSD”



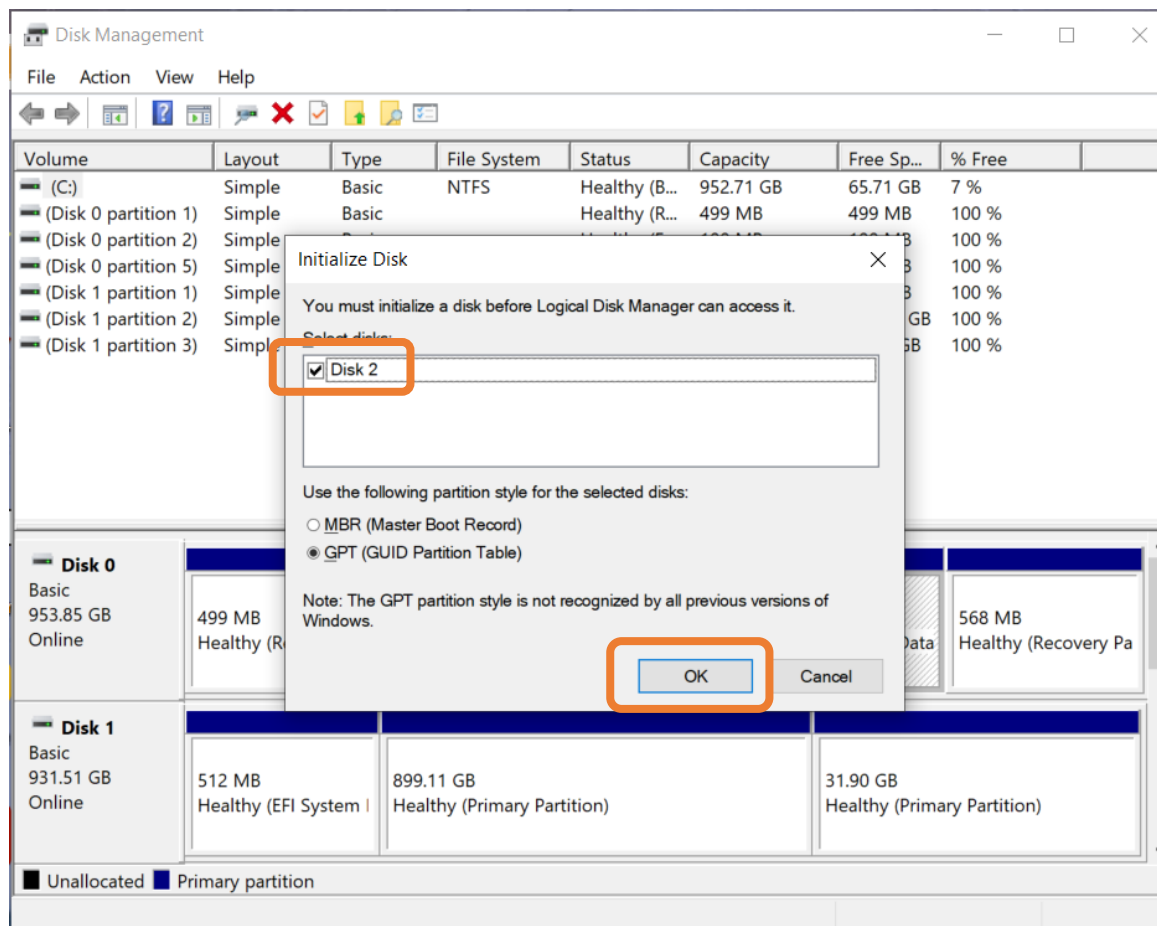
Create a Partition (1 / 4)

- Right-click “Start” menu → click “Disk Management”



Create a Partition (2 / 4)

- Click “OK” to confirm disk initialization



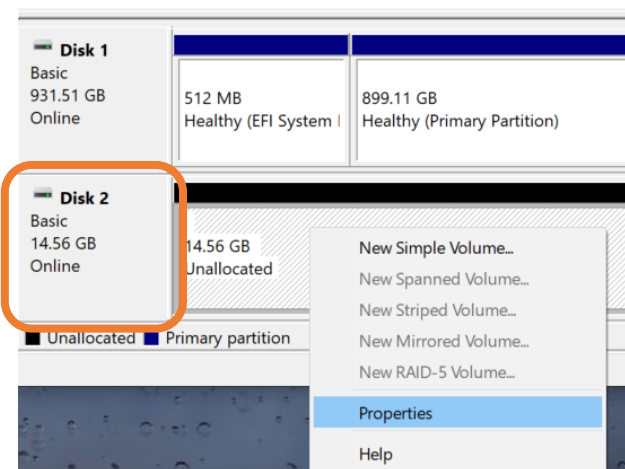
Create a Partition (3 / 4)

- Click right mouse button on “Disk 2” which was shown in 2nd step → click “Properties”

Disk Management

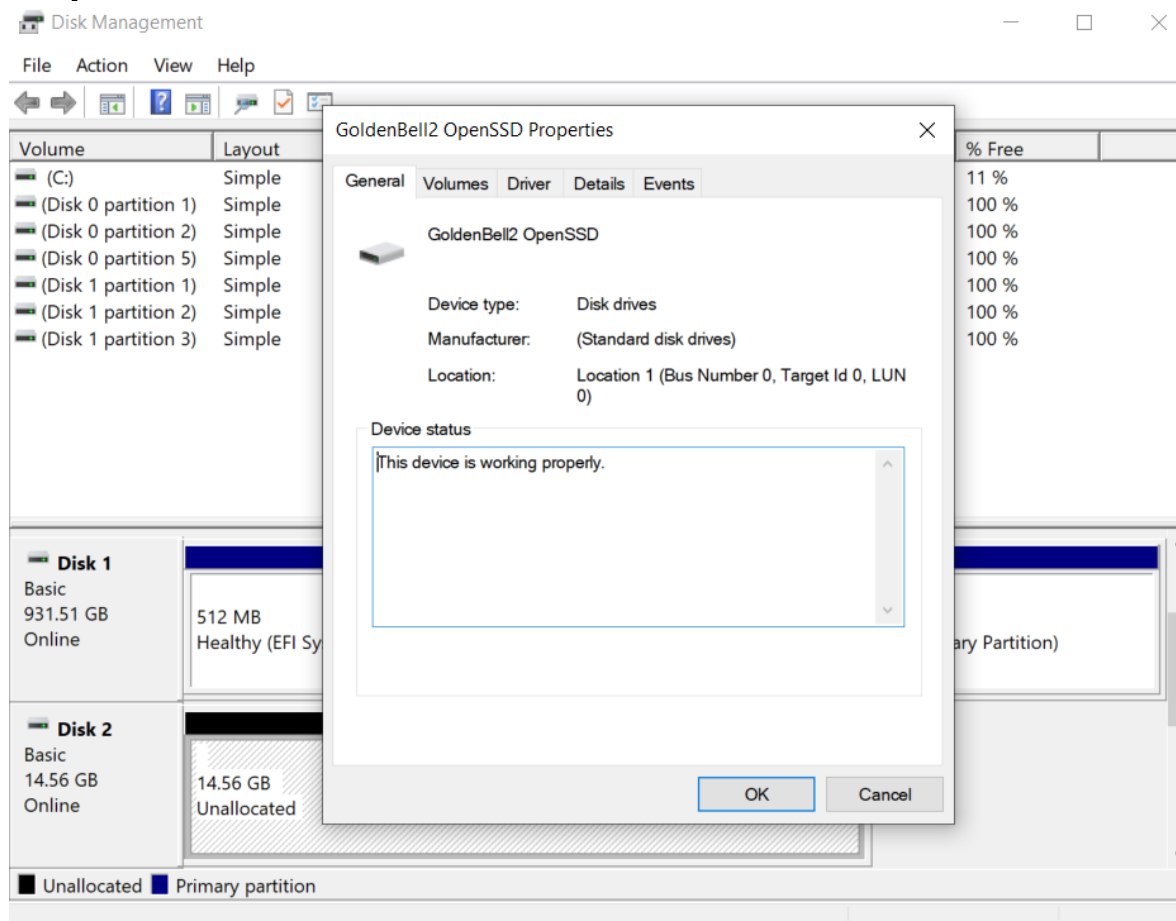
File Action View Help

Volume	Layout	Type	File System	Status
(C:)	Simple	Basic	NTFS	Healthy (B...
(Disk 0 partition 1)	Simple	Basic		Healthy (R...
(Disk 0 partition 2)	Simple	Basic		Healthy (E...
(Disk 0 partition 5)	Simple	Basic		Healthy (R...
(Disk 1 partition 1)	Simple	Basic		Healthy (E...
(Disk 1 partition 2)	Simple	Basic		Healthy (P...
(Disk 1 partition 3)	Simple	Basic		Healthy (P...



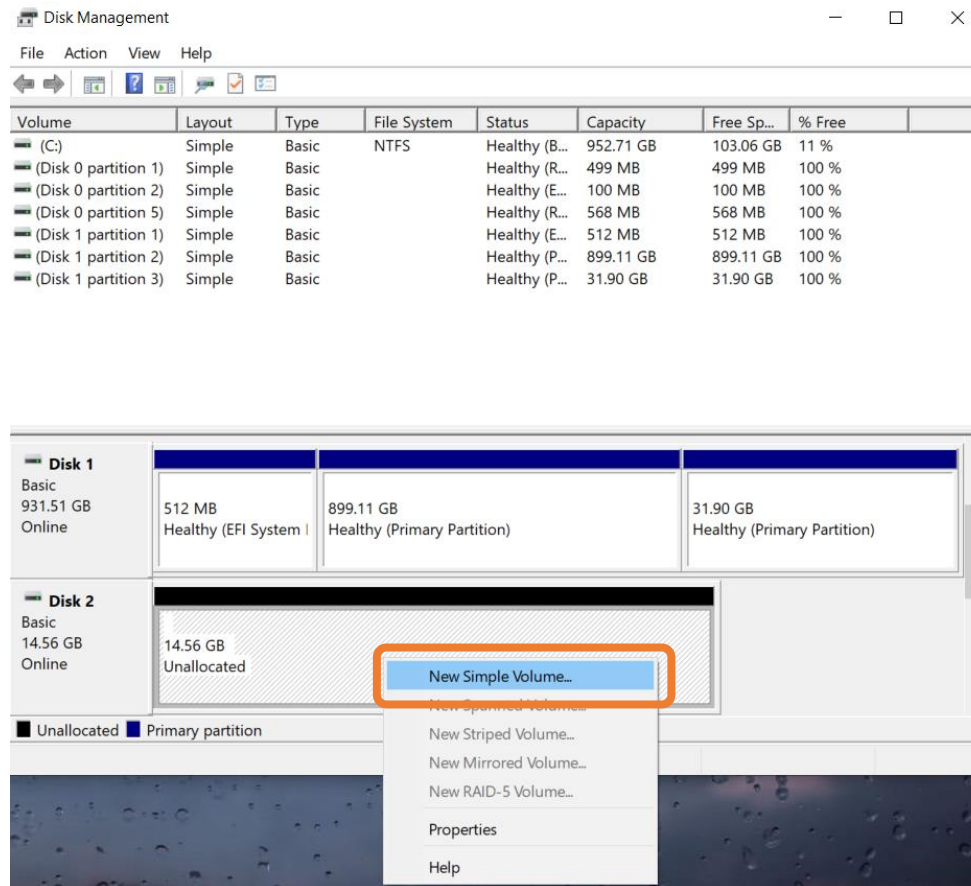
Create a Partition (4 / 4)

- Make sure that the “Disk 2” is GoldenBell2 OpenSSD before you proceed to the next step



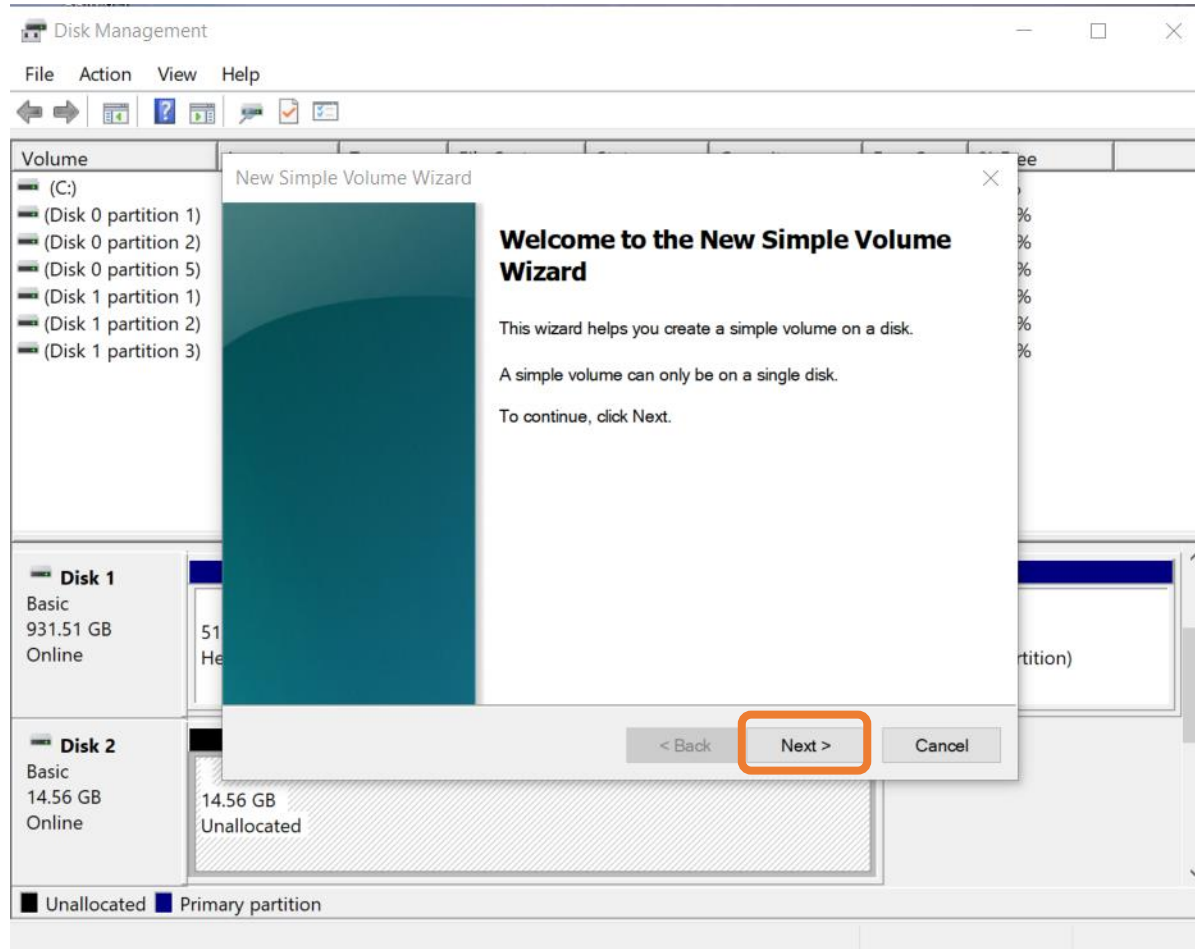
Format the Partition (1 / 8)

- Click right mouse button on the right part of “Disk 2” → click “New Simple Volume”



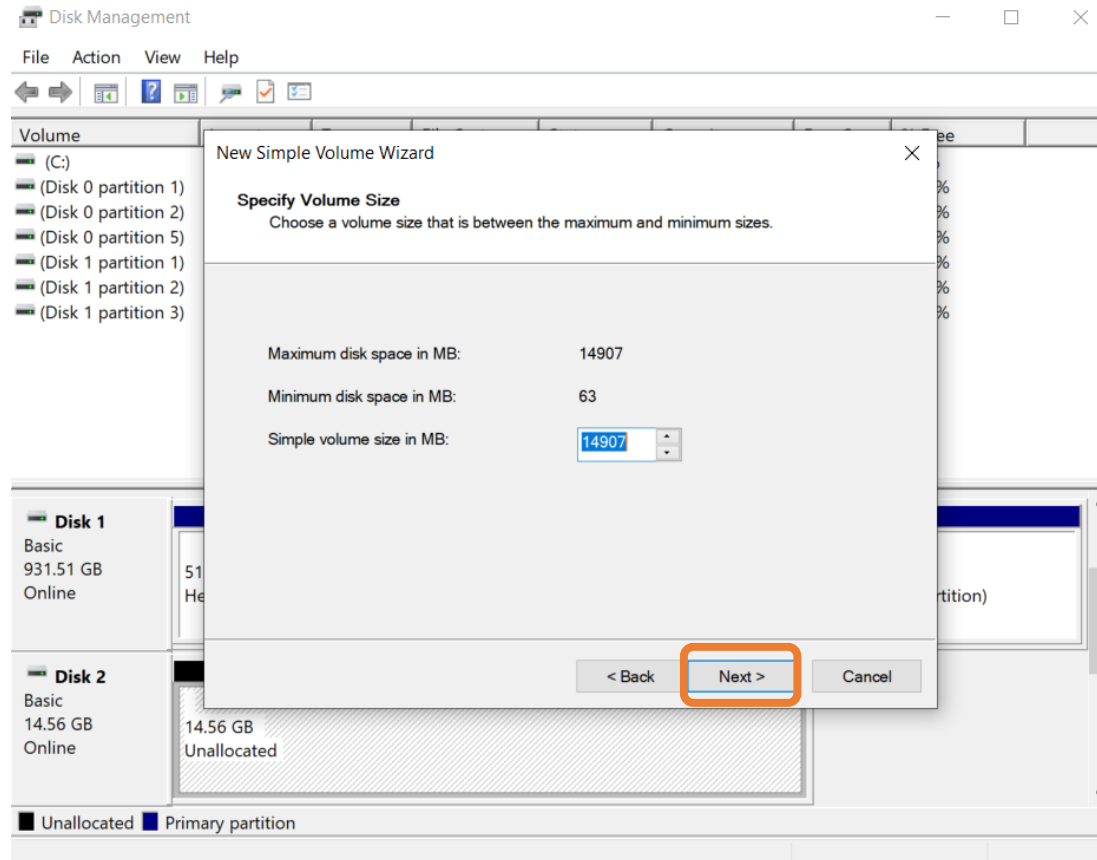
Format the Partition (2 / 8)

- Click “Next”



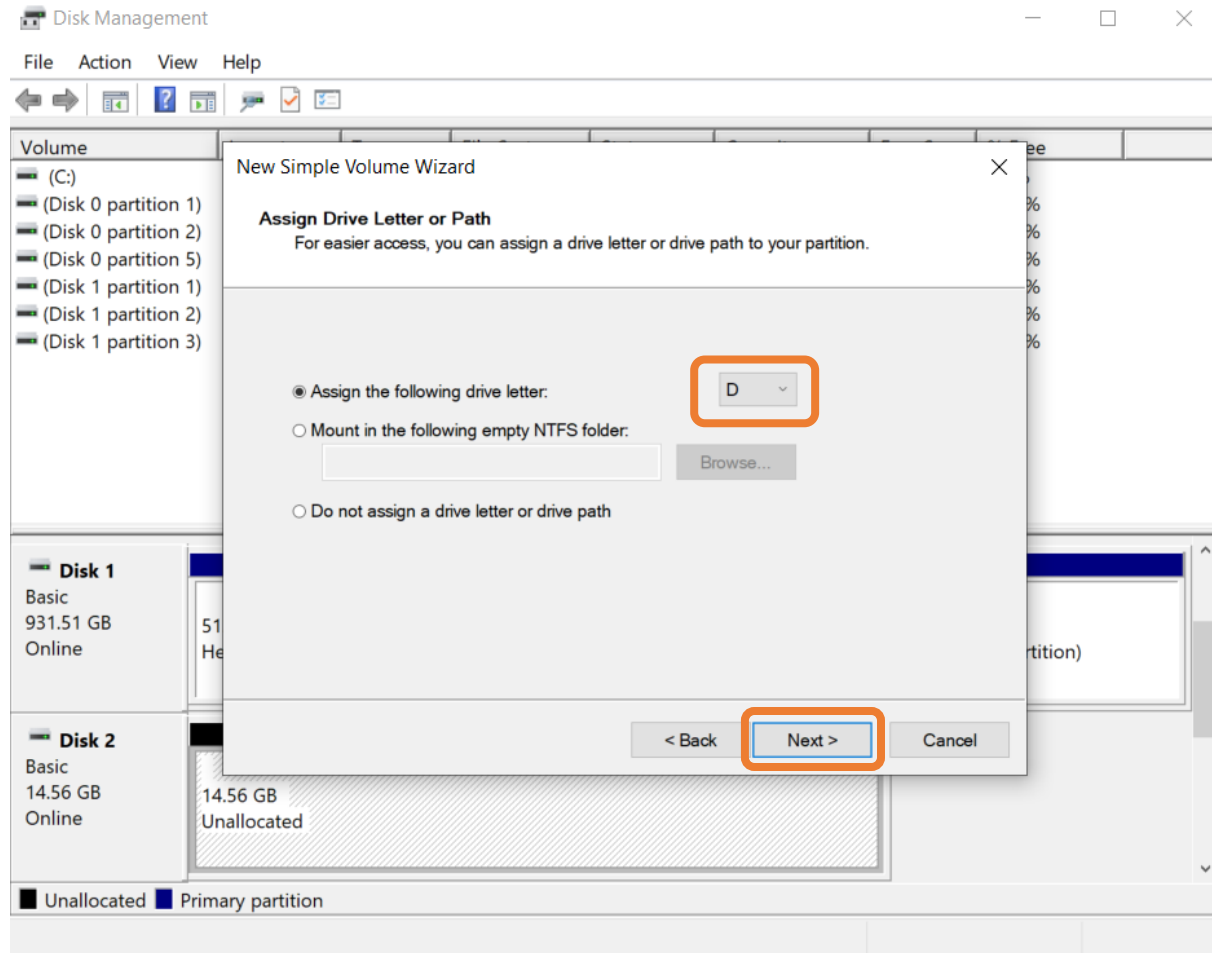
Format the Partition (3 / 8)

- Click “Next”



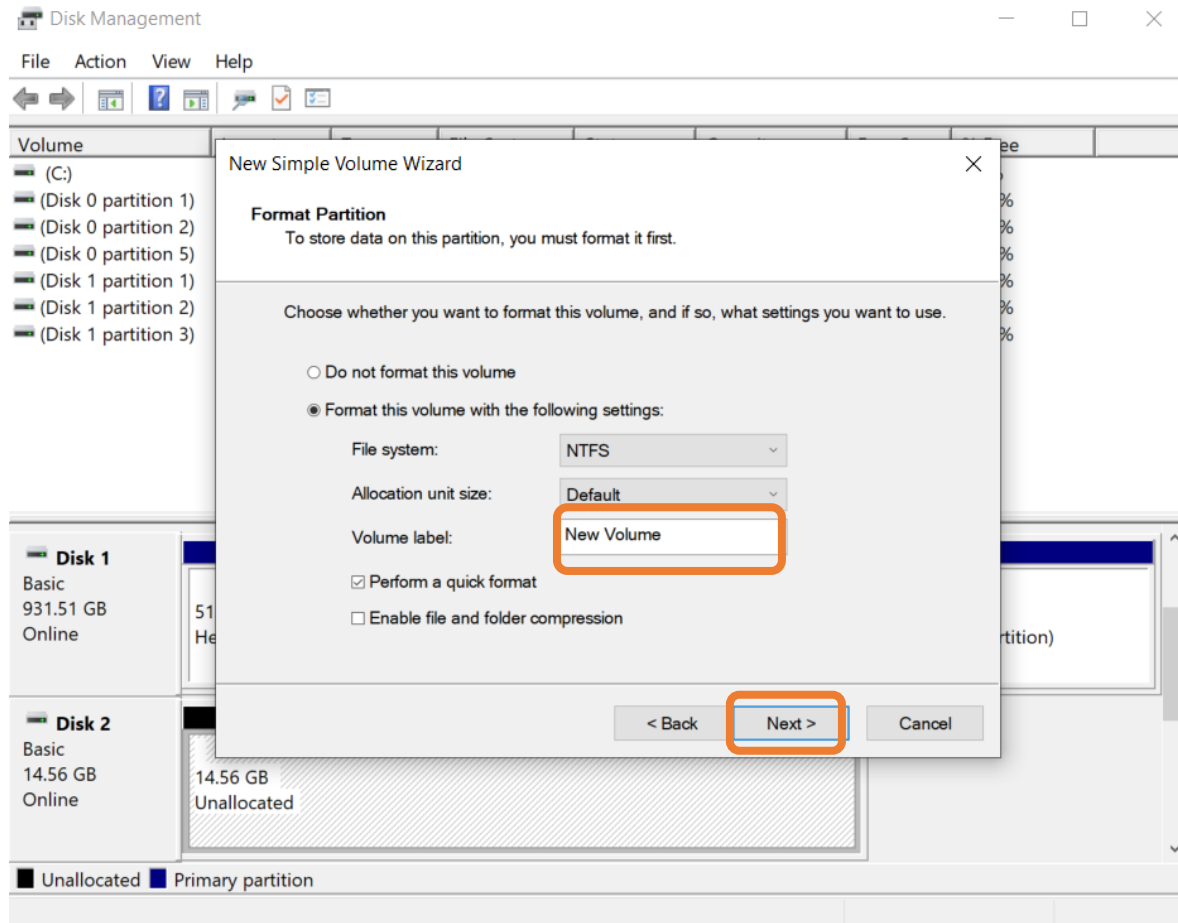
Format the Partition (4 / 8)

- Select desired drive letter → Click “Next”



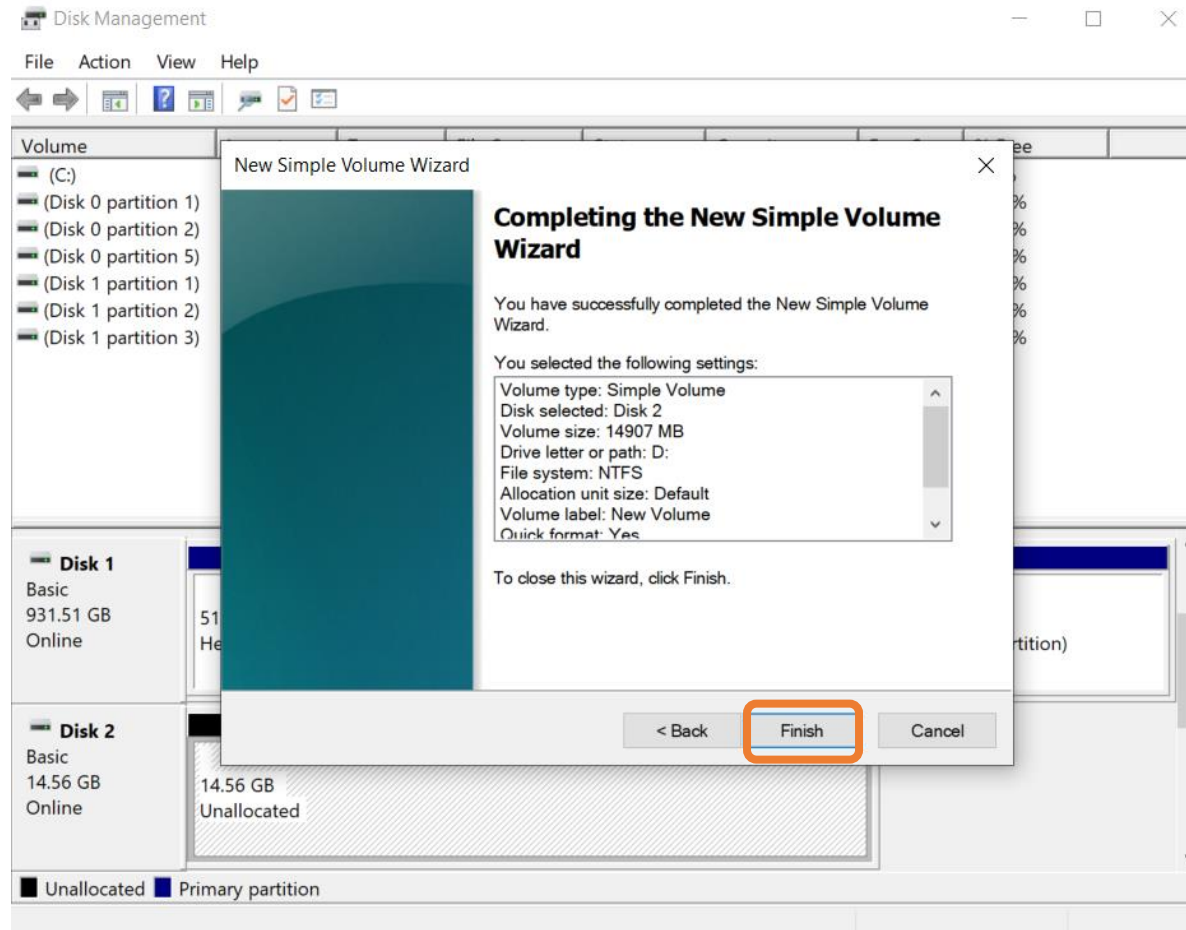
Format the Partition (5 / 8)

- Type desired volume label → Click “Next”



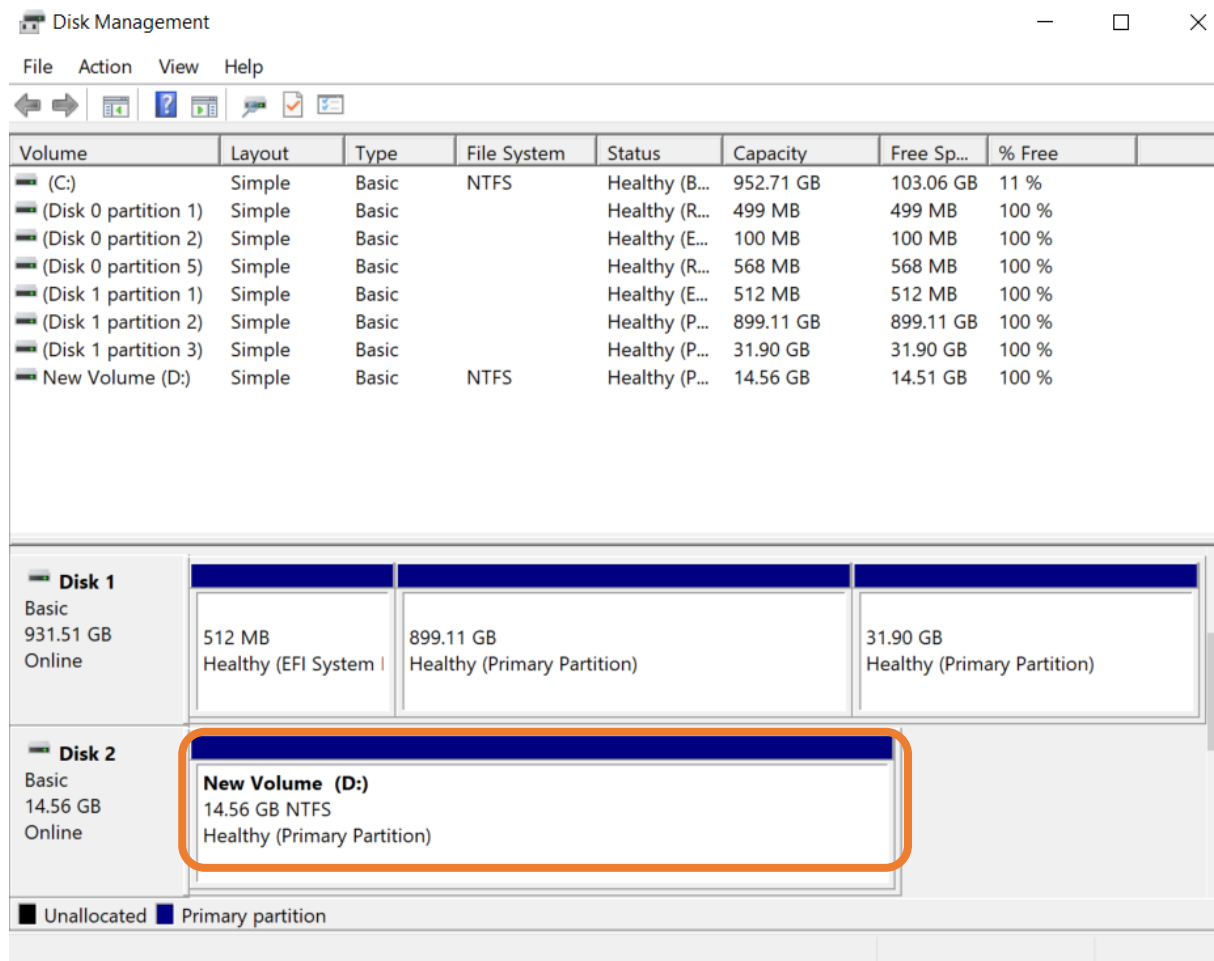
Format the Partition (6 / 8)

- Click “Finish”



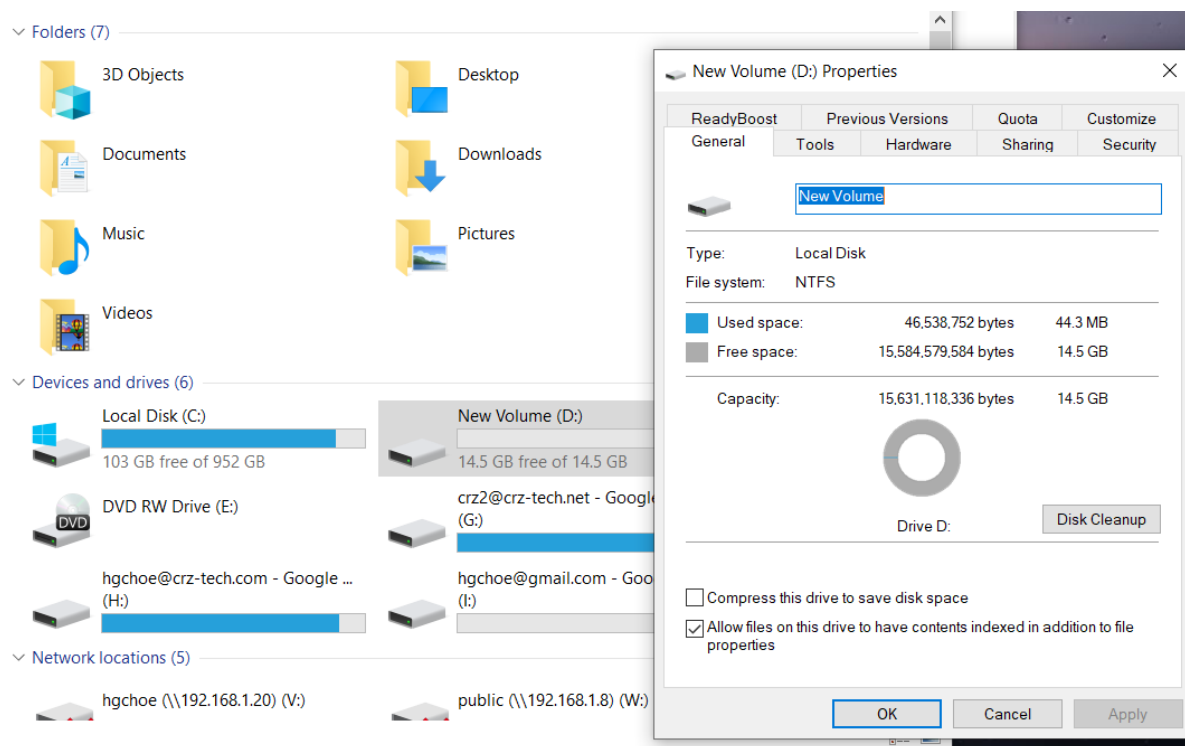
Format the Partition (7 / 8)

- Formatting is now finished



Format the Partition (8 / 8)

- Now you can find the formatted Daisy+ OpenSSD at “This PC”



Thank You